Lecture

Internet Trends and Web Basics

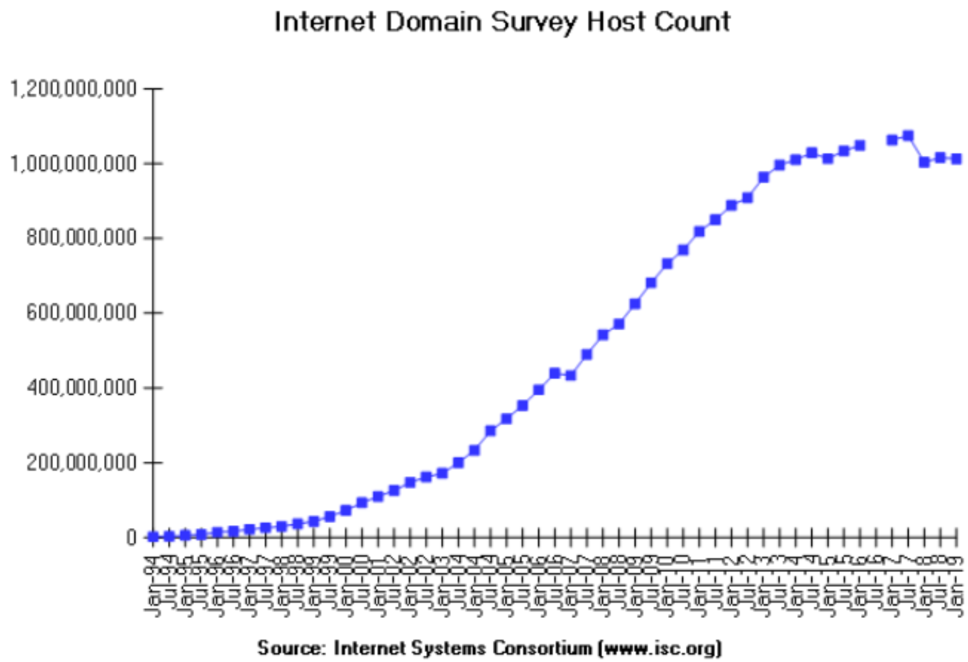
The Internet and the WWW are Different

- The *Internet* is a global digital infrastructure that connects hundreds of millions of computers and people
- The *World Wide Web* is a mechanism that unifies the retrieval and display of a subset of data on the Internet
- An *intranet* is a local/global information structure that connects an organization internally. Intranets today often make use of Web technologies
- An *extranet* is a private network that uses the public telecommunication system to securely share part of a business's information or operations with suppliers, vendors, partners, customers, or other businesses.

Recent Trends in Internet Development

- Growth in number of users connected
- Growth in Smartphone use, particularly iOS and Android
- Growth in digital data, especially photos and video
- Growth in Social Media
- Growth in Internet use from Mobile over desktop/laptop
- Growth in tablet usage over desktops/laptops
- Decreased dominance of Microsoft Windows
- Growth in use of the cloud

How Big is the Internet (historical)
https://www.isc.org/network/survey



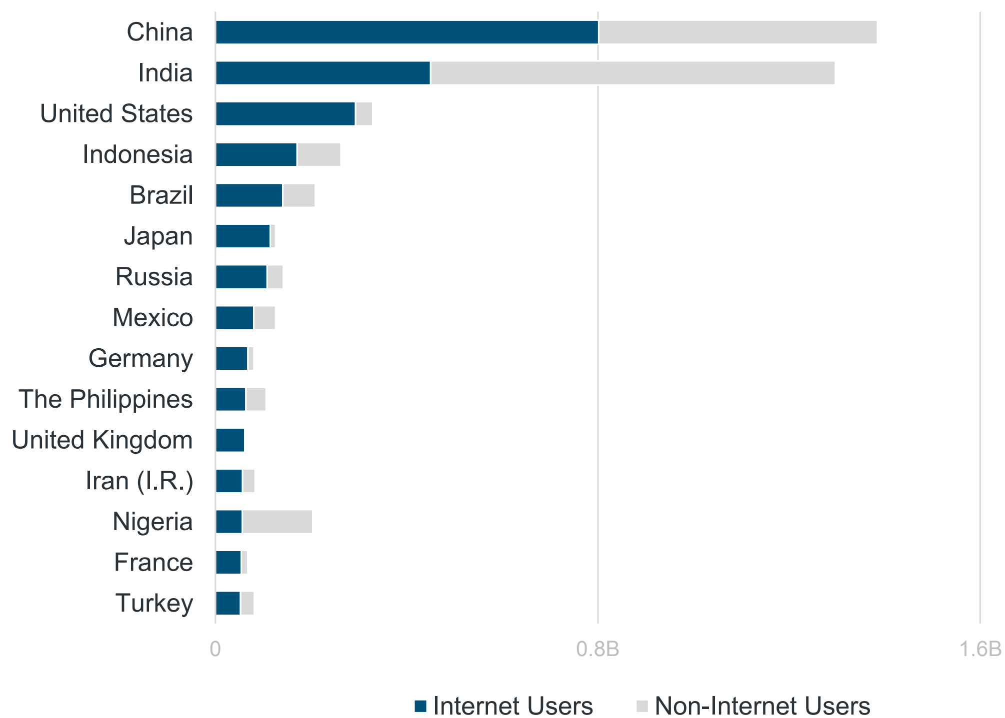
hosts were doubling every 18 months, but growth has slowed
See the survey background at: http:///www.isc.org/network/survey

It counts the number of IP addresses that have been assigned a name. The survey queries the domain name system for the name assigned to every possible IP address. But rather than sending a query to every one of the 4.3 billion possible IP addresses, the survey starts with a list of all network numbers that have been delegated within the IN-ADDR.ARPA domain. This survey has been discontinued.

Date	HostCount
Jan 19	1,012,695,272
Jul 18	1,015,787,389
Jan 18	1,003,604,363
Jul 17	1,074,971,748
Jan 17	1,062,660,523
Jan 16	1,048,766,623
Jul 15	1,033,836,245
Jan 15	1,012,706,608
Jul 14	1,028,544,414
Jan 14	1,010,251,829
Jul 13	996,230,757
Jan 13	963,518,598
Jul 12	908,585,739
Jan 12	888,239,420
Jul 11	849,869,781
Jan 11	818,374,269
Jul 10	768,913,036
Jan 10	732,740,444
Jul 09	681,064,561
Jan 09	625,226,456
Jul 08	570,937,778
Jan 08	541,677,360
Jul 07	489,774,269
Jan 07	433,193,199
Jul 06	439,286,364
Jan 06	394,991,609
Jul 05	353,284,187
Jan 05	317,646,084
Jul 04	285,139,107
Jan 04	233,101,481
Jan 03	171,638,297
Jul 02	162,128,493
Jan 02	147,344,723
Jul 01	125,888,197
Jan 01	109,574,429
Jul 00	93,047,785
Jan 00	72,398,092
Jul 99	56,218,000
Jan 99	43,230,000
Jul 98	36,739,000
Jan 98	29,670,000
Jul 97	19,540,000
Jan 97	16,146,000
Jul 96	12,881,000
Jan 96	9,472,000
Jul 95	6,642,000
Jan 95	4,852,000
Jul 94	3,212,000
Jan 94	2,217,000
Jul 93	1,776,000
Jan 93	1,313,000

Global Internet Users = China @ 21% of Total...India @ 12%...USA @ 8%

Internet Users – Top Countries, 2018



BOND
Internet Trends
2019

Internet user data is as of mid-year. Source: United Nations / International Telecommunications Union, USA Census Bureau. Pew Research (USA), China Internet Network Information Center (China), Islamic Republic News Agency / InternetWorldStats / Bond estimates (Iran), Bond estimates based on IAMAI data (India), & APJII (Indonesia).

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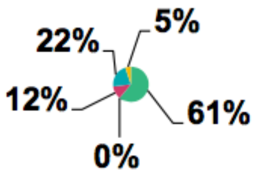
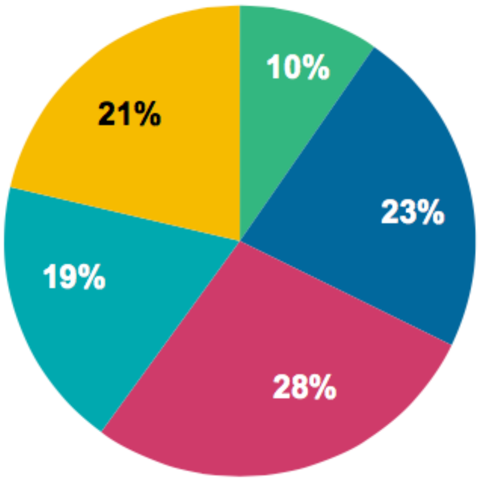
The following slides are based upon a presentation by Mary Meeker of Bond and formerly of Kleiner Perkins Caufield and Byers, see https://www.bondcap.com/#internettrends

Internet Users – 1995 → 2014…
<1% to 39% Population Penetration Globally

1995
35MM+ Internet Users
0.6% Population Penetration



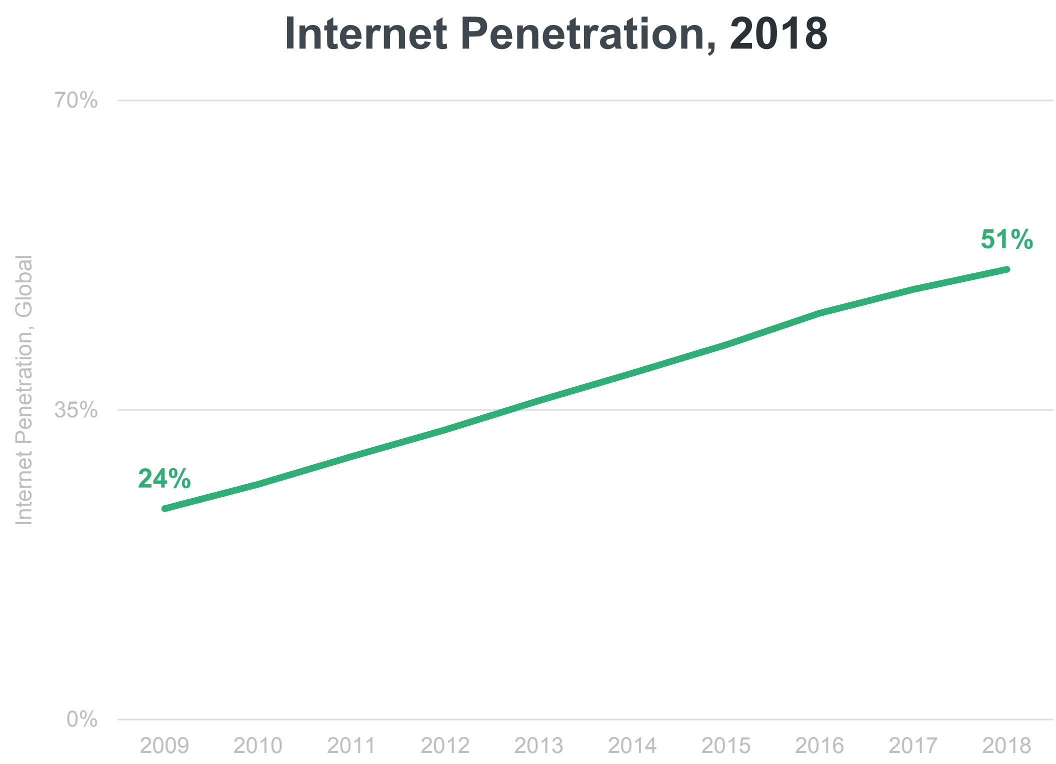
2014
2.8B Internet Users
39% Population Penetration



■ USA ■ China ■ Asia (ex. China) ■ Europe ■ Rest of World

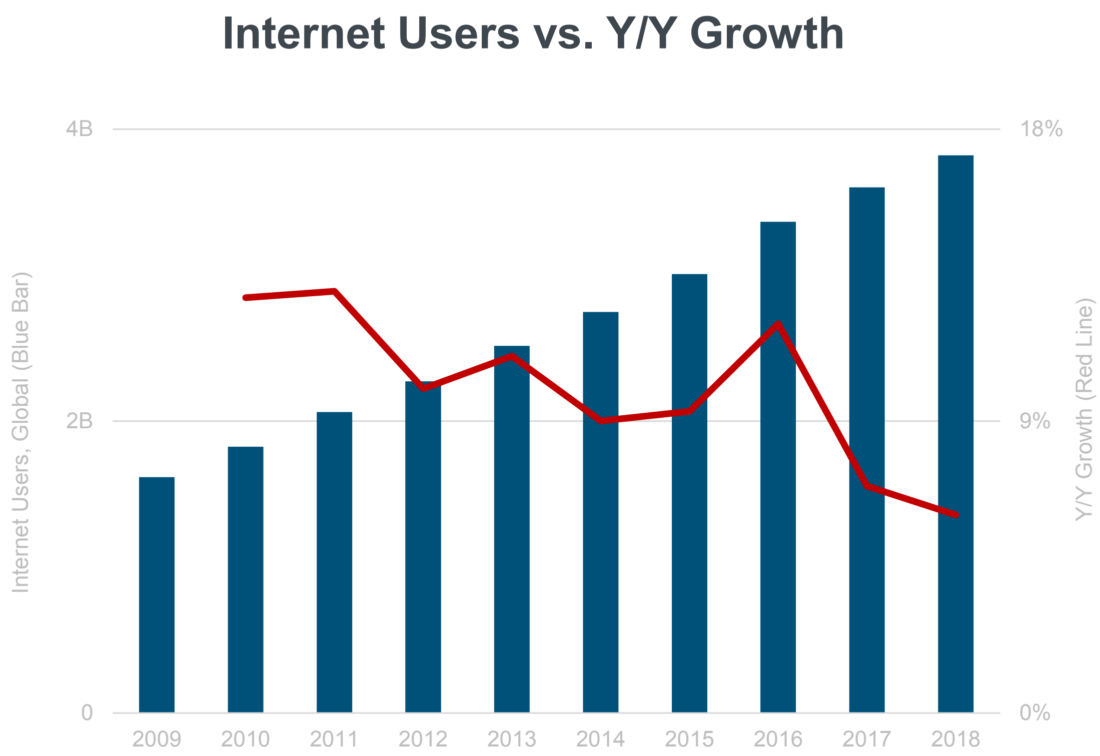
@KPCB Source: Euromonitor, ITU, US Census.

Global Internet Users = 3.8B >50% of Population



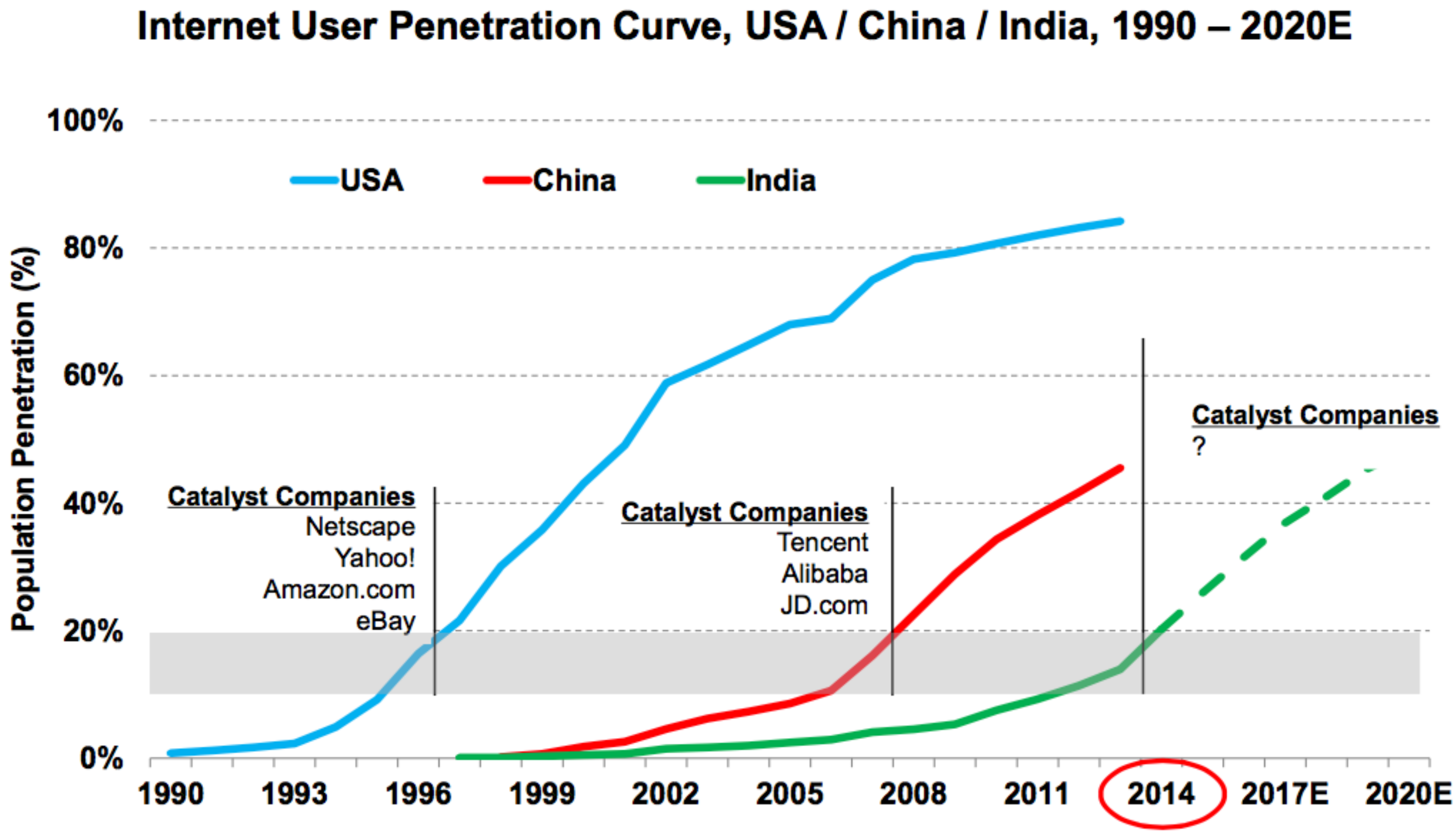
Internet user data is as of mid-year. Source: United Nations / International Telecommunications Union, USA Census Bureau. Pew Research (USA), China Internet Network Information Center (China), Islamic Republic News Agency / InternetWorldStats / Bond estimates (Iran), Bond estimates based on IAMAI data (India), & APJII (Indonesia).

Global Internet User Growth = Solid But Slowing +6% vs. +7% Y/Y



Internet user data is as of mid-year. Source: United Nations / International Telecommunications Union, USA Census Bureau. Pew Research (USA), China Internet Network Information Center (China), Islamic Republic News Agency / InternetWorldStats / Bond estimates (Iran), Bond estimates based on IAMAI data (India), & APJII (Indonesia).

India = Appears to Be @ Internet Penetration Growth Inflection



Source: World Bank, Hillhouse Capital forecast for India beyond 2014.

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Hillhouse Capital

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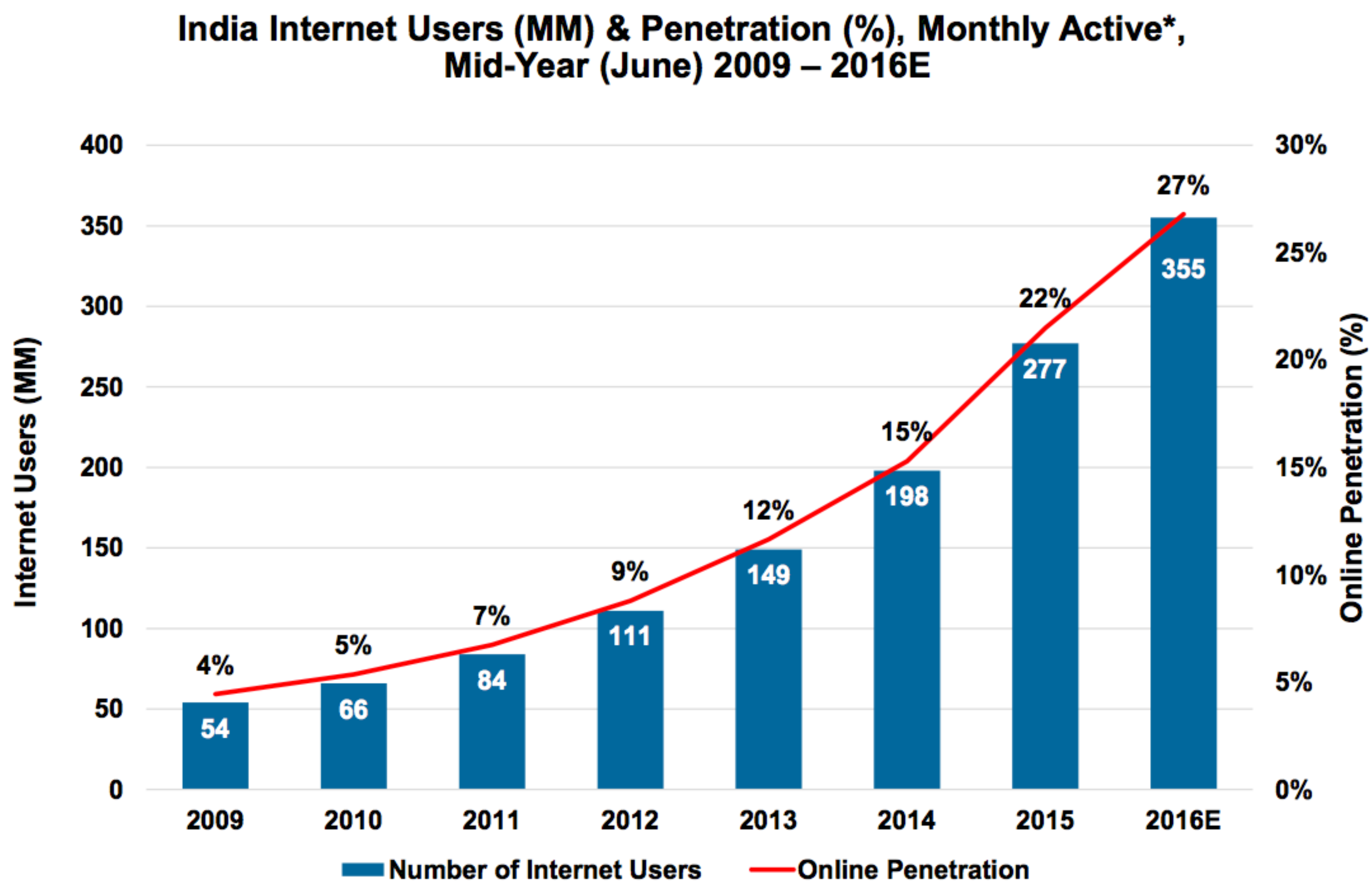
India Internet Users = +28% (2016-June) vs. 40% Y/Y Growth @ 27% Penetration   355MM Users   #2 Behind China



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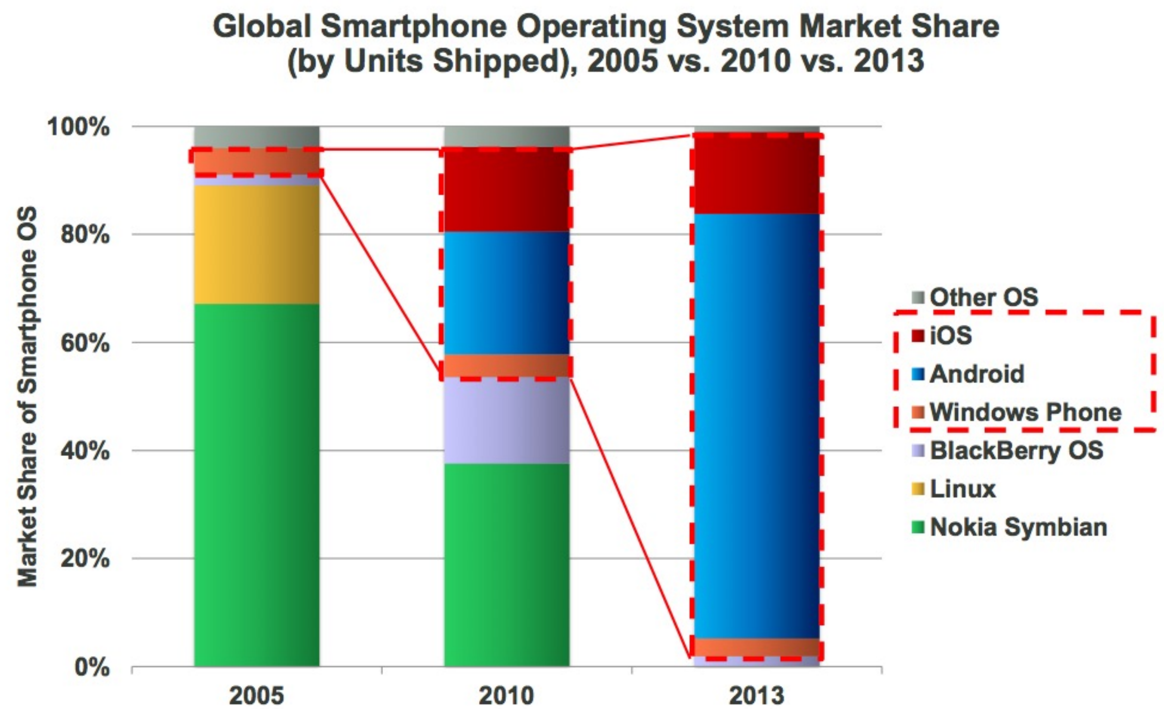
Source: IAMAI. UN Population Division, Worldometer, KPCB estimates based on IAMAI data. Uses mid-year figures.
*Note that "Monthly Active Users" are distinct from "Ever" users, which IAMAI defines as anyone who has ever accessed the internet. Owing to increasing activity levels, the number of "Monthly Active Users" may grow faster than "Ever" users.

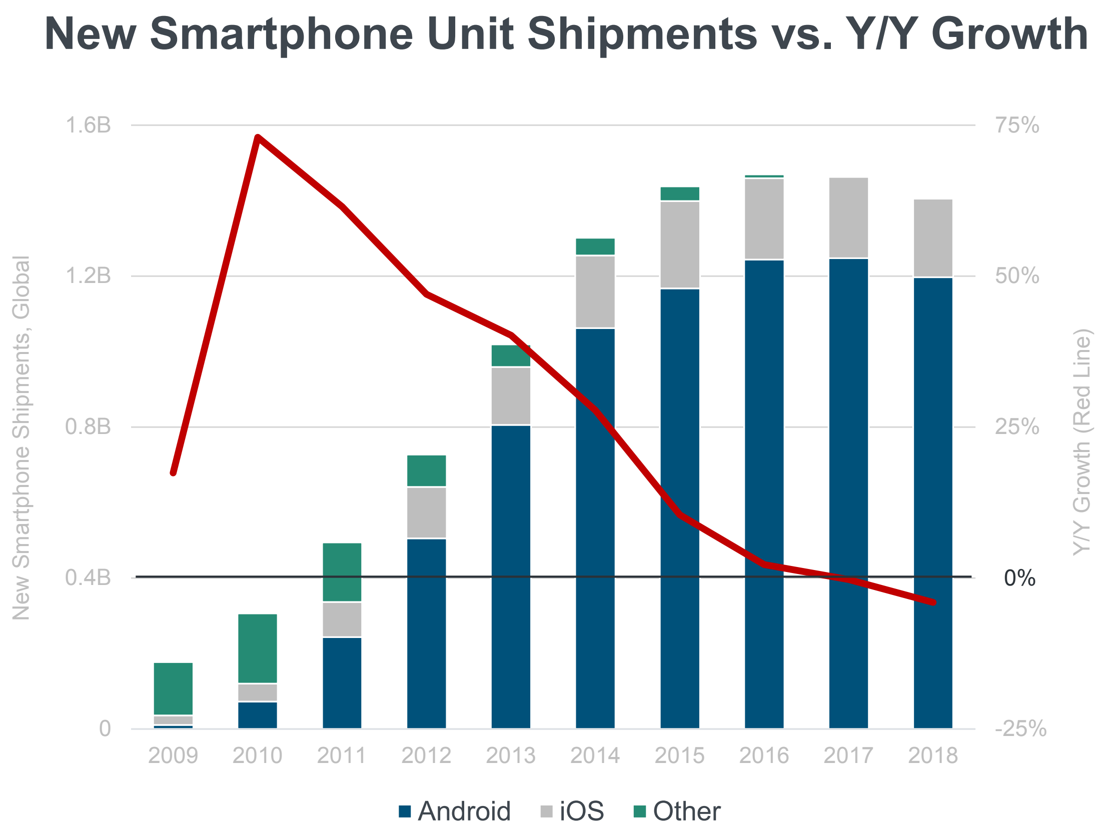
Global Smartphone Operating Systems 'Made in USA'... 97% Share from 5% Eight Years Ago

Examining smartphone operating systems, over the past seven years, iOS and Android have made major gains with Nokia disappearing and Linux a very small piece of the pie



@KPCB Source: 2005 & 2010 data per Gartner, 2013 data per IDC. 10

Global New Smartphone Unit Shipments = Declining -4% vs. 0% Y/Y

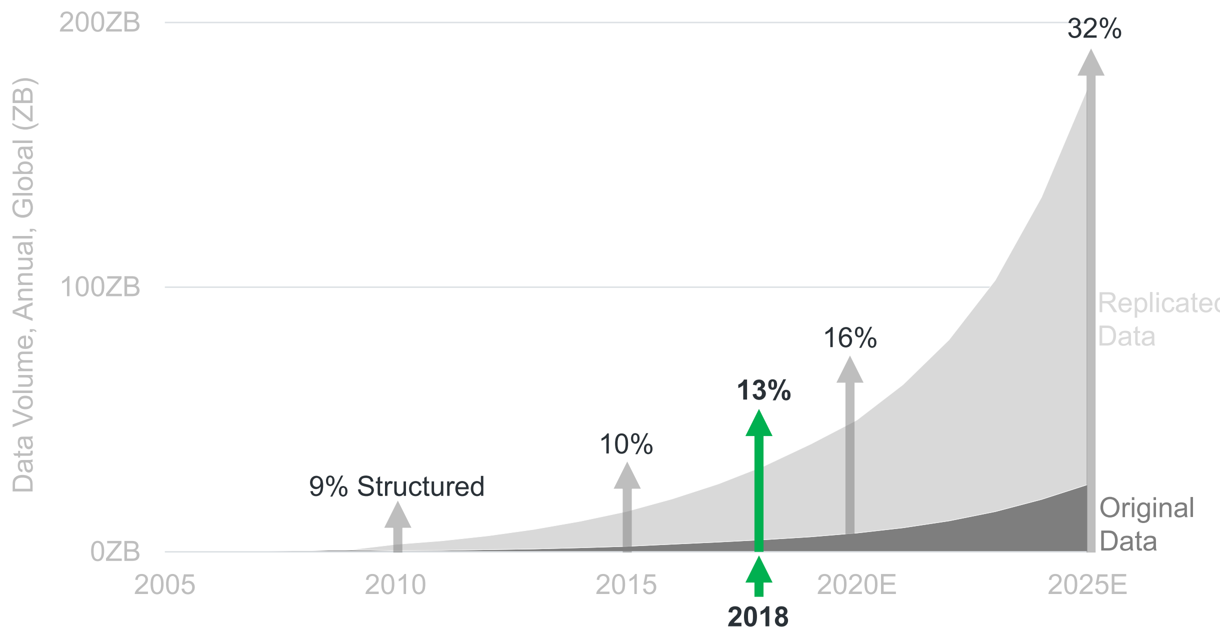


Source: Katy Huberty @ Morgan Stanley (3/19), IDC.

Data Volume = Extraordinary Growth…
~13% Structured / Tagged & Rising Rapidly

There has been exponential growth in online information;
1 Zettabyte = 1,024 Exabytes
1 Exabyte = 1,024 Petabytes
1 Petabyte = 1,024 Terabytes
1 Terabyte = 1,024 Gigabytes
or
1 Zettabyte = 1,000,000,000,000 gigabytes

New Data Captured / Created / Replicated, per IDC



Source: IDC 'Digitization of the World From Edge to Core White Paper' developed in collaboration with Seagate (11/18), IDC DataSphere. Note: 1 petabyte = 1MM gigabytes, 1 zeta byte = 1MM petabytes of new data created / captured each year. The grey area in the graph represents data generated, not stored. Structured data indicates data that has been organized so that it is easily searchable & includes metadata & machine-to-machine (M2M) data. Replicated data = data that is a copy of the original..

BOND Internet Trends 2019

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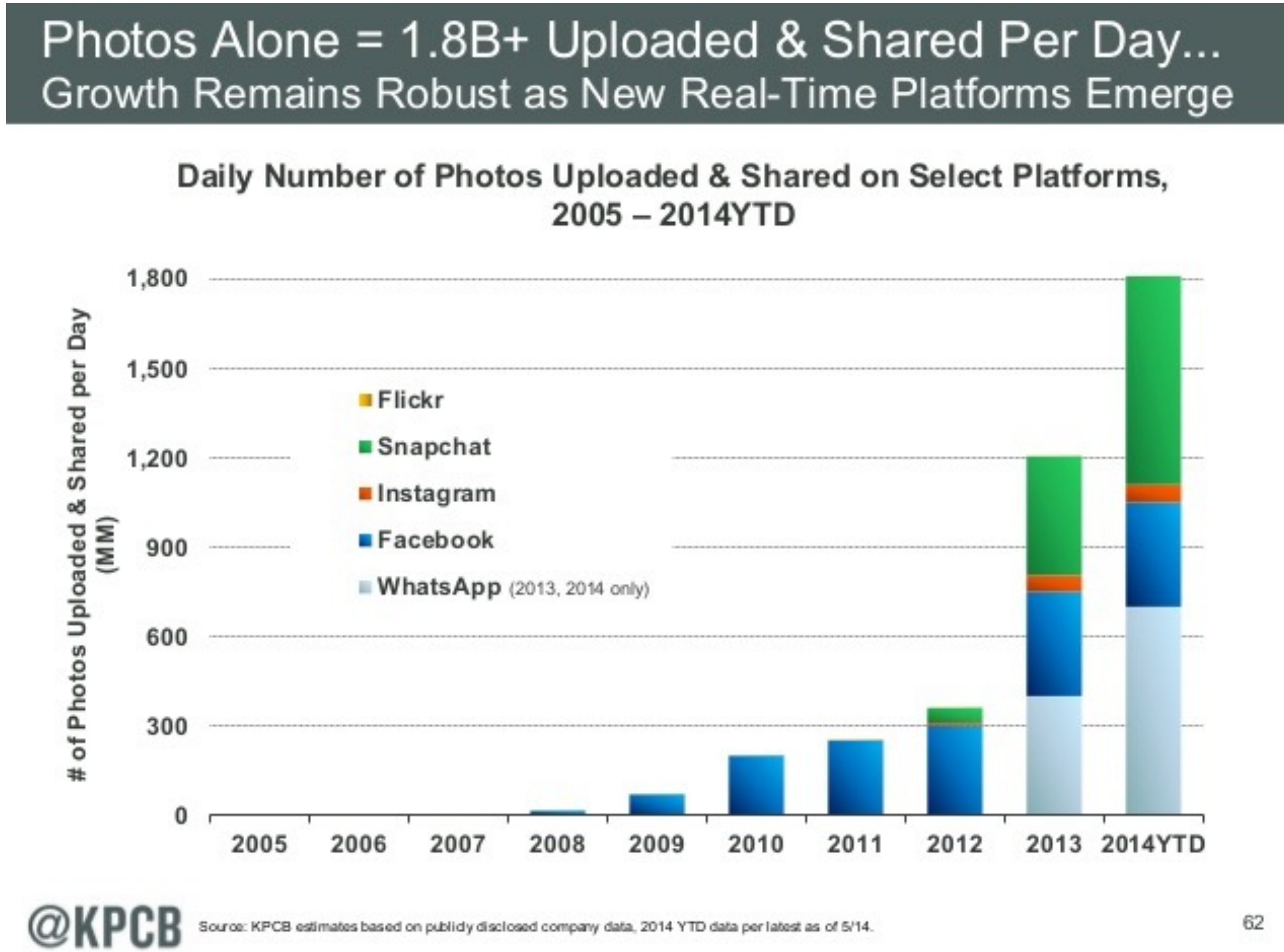
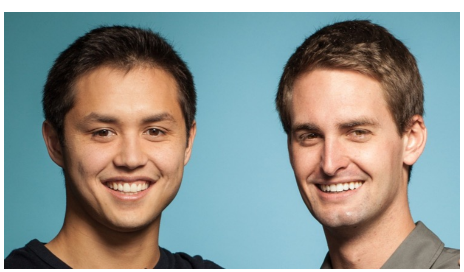
500 million photos are uploaded every day and that number is doubling every year

Instagram was recently (2010) purchased by Facebook for $1 billion

Snapchat is a photo messaging application developed by two Stanford students (IPO March 2017, $17B valuation);

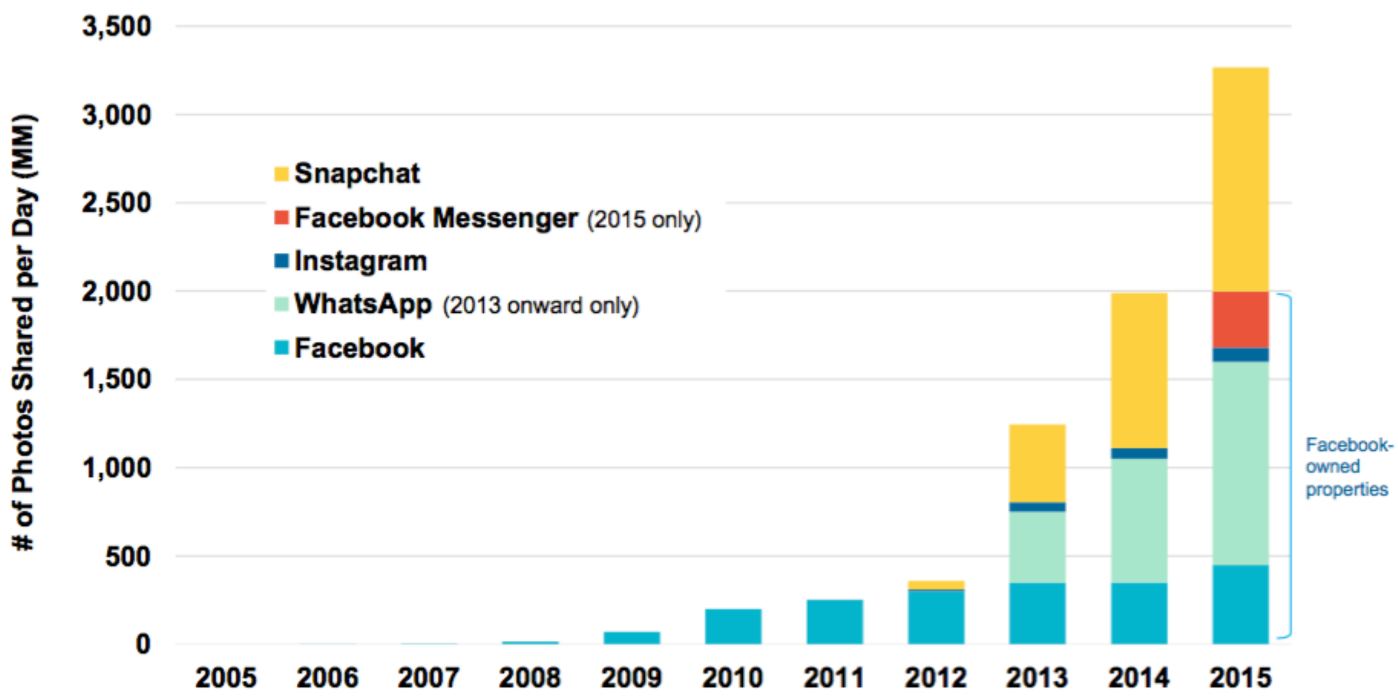




bobby Murphy - Evan Spiegel

Image Growth Remains Strong

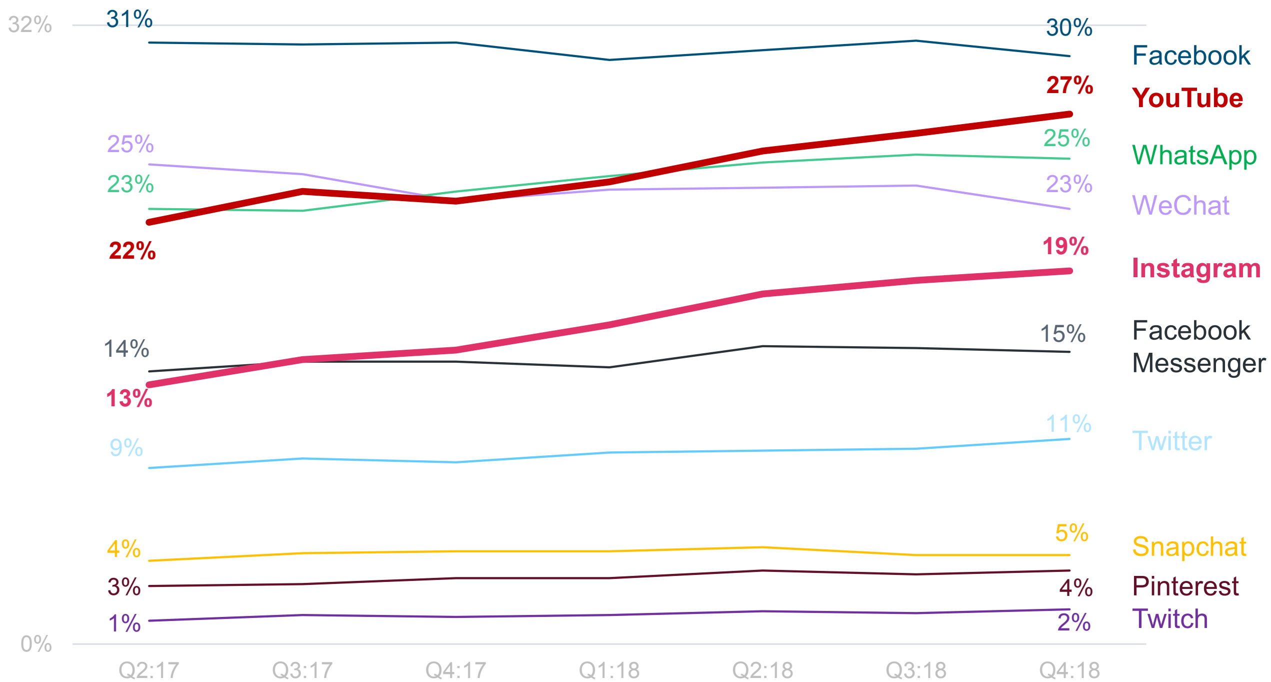
Daily Number of Photos Shared on Select Platforms, Global, 2005 – 2015



Source: Snapchat, Company disclosed information, KPCB estimates
Note: Snapchat data includes images and video. Snapchat stories are a compilation of images and video. WhatsApp data estimated based on average of photos shared disclosed in Q1:15 and Q1:16. Instagram data per Instagram press release. Messenger data per Facebook (~9.5B photos per month). Facebook shares ~2B photos per day across Facebook, Instagram, Messenger, and WhatsApp (2015).

Online Platform Time = YouTube + Instagram Gaining Most

% Internet Users Using Select Platforms > 1x per Day, Global*



BOND Internet Trends 2019

Source: Global Web Index Survey of 50K+ Global Internet users aged 16-64 (3/2017-12/2018). Note: WeChat is displayed as "Weixin" in China. *Countries surveyed include major Internet populations – Argentina, Australia, Austria, Belgium, Brazil, Canada, China, Colombia, Denmark, Egypt, France, Germany, Hong Kong, India, Indonesia, Ireland, Italy, Japan, Malaysia, Mexico, Netherlands, New Zealand, Philippines, Poland, Portugal, Romania, Russia, Saudi Arabia, Singapore, South Africa, South Korea, Spain, Sweden, Switzerland, Taiwan, Thailand, Turkey, UK, USA, United Arab Emirates, Vietnam. Toutiao estimate = ~4% of Internet users using Douyin in 2018….Not included owing to lack of trend data & regulatory action in period.

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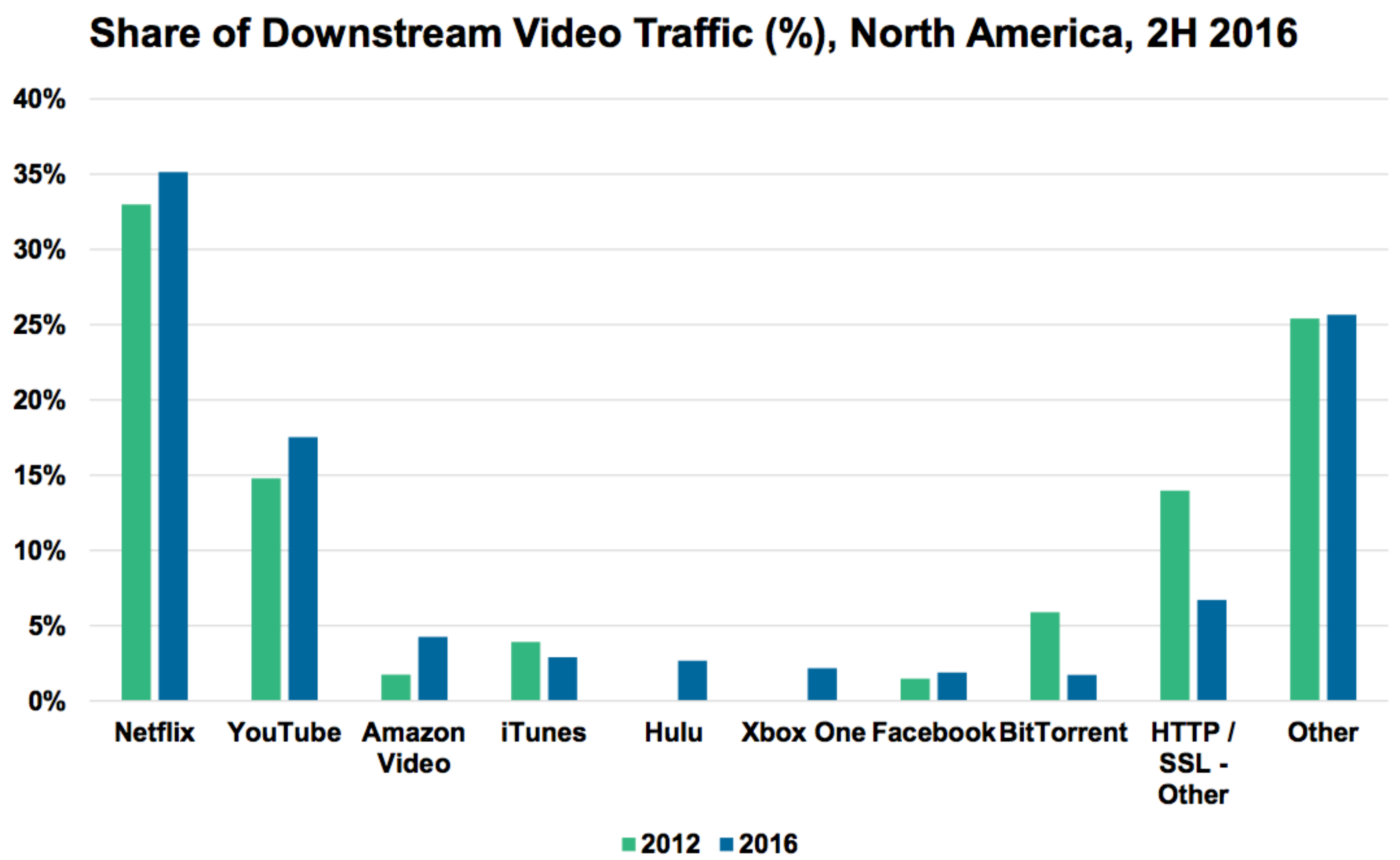
Netflix / YouTube =
Fixed-Access Video Traffic Share Leaders



Source: Sandvine Global Internet Phenomena Report (2H 2012 and 2016)

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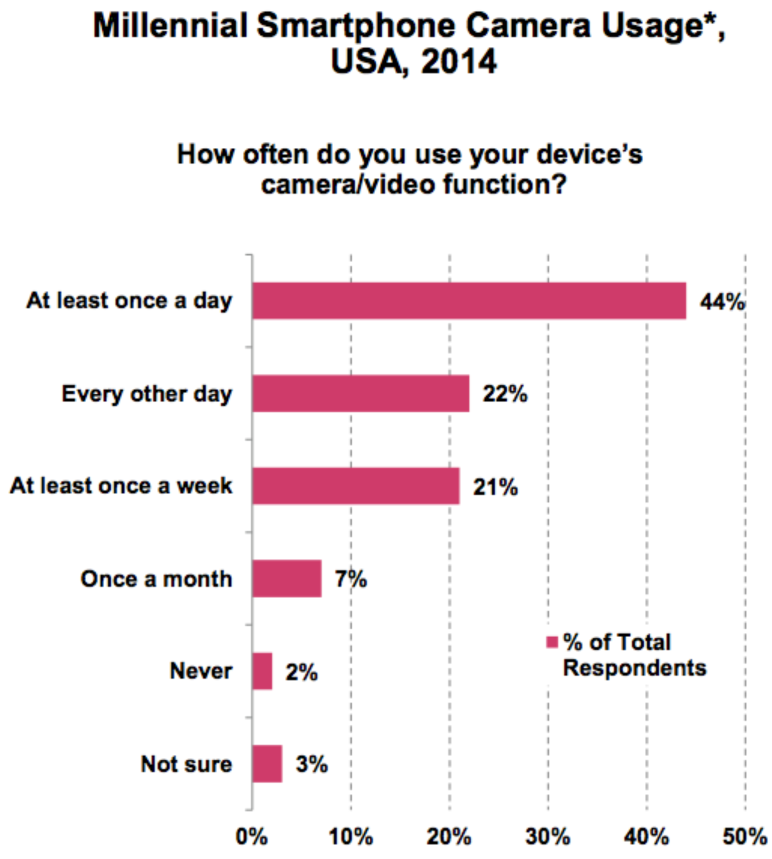
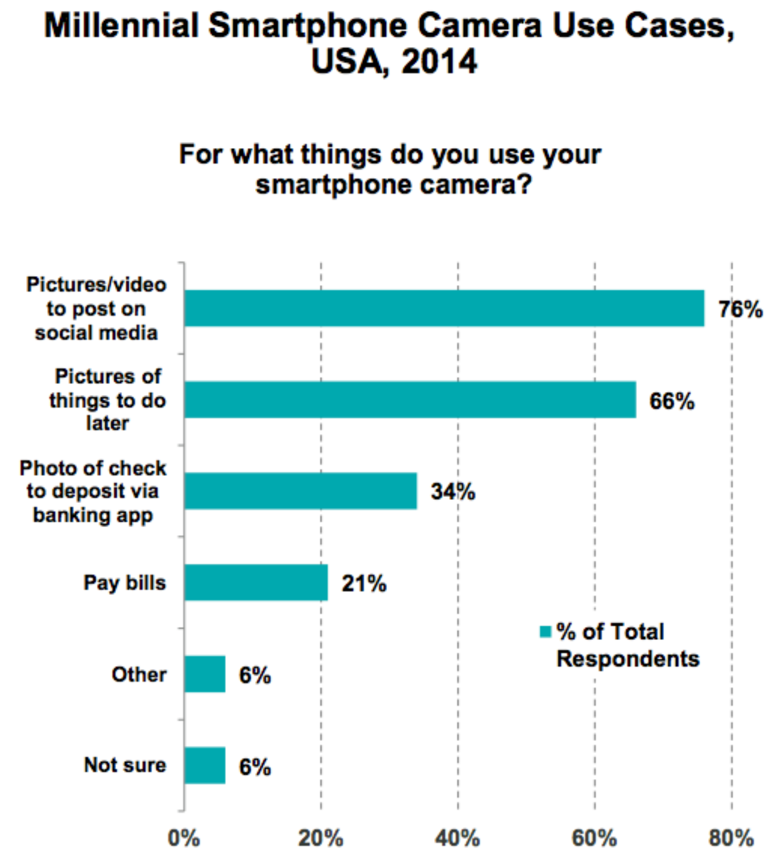
Millennials Love Their Smartphone Cameras...
44% Use Camera / Video Function Daily...76% Post on Social Media





@KPCB

Source: Zogby Analytics.
*18-24 year olds.
Note: Zogby Analytics was commissioned by Mitek Systems, Inc. to conduct an online survey of 1,019 millennials who have a smartphone. For the purposes of this survey, "millennials" are defined as adults between the ages of 18-34. All interviews were completed May 30 through June 6, 2014.

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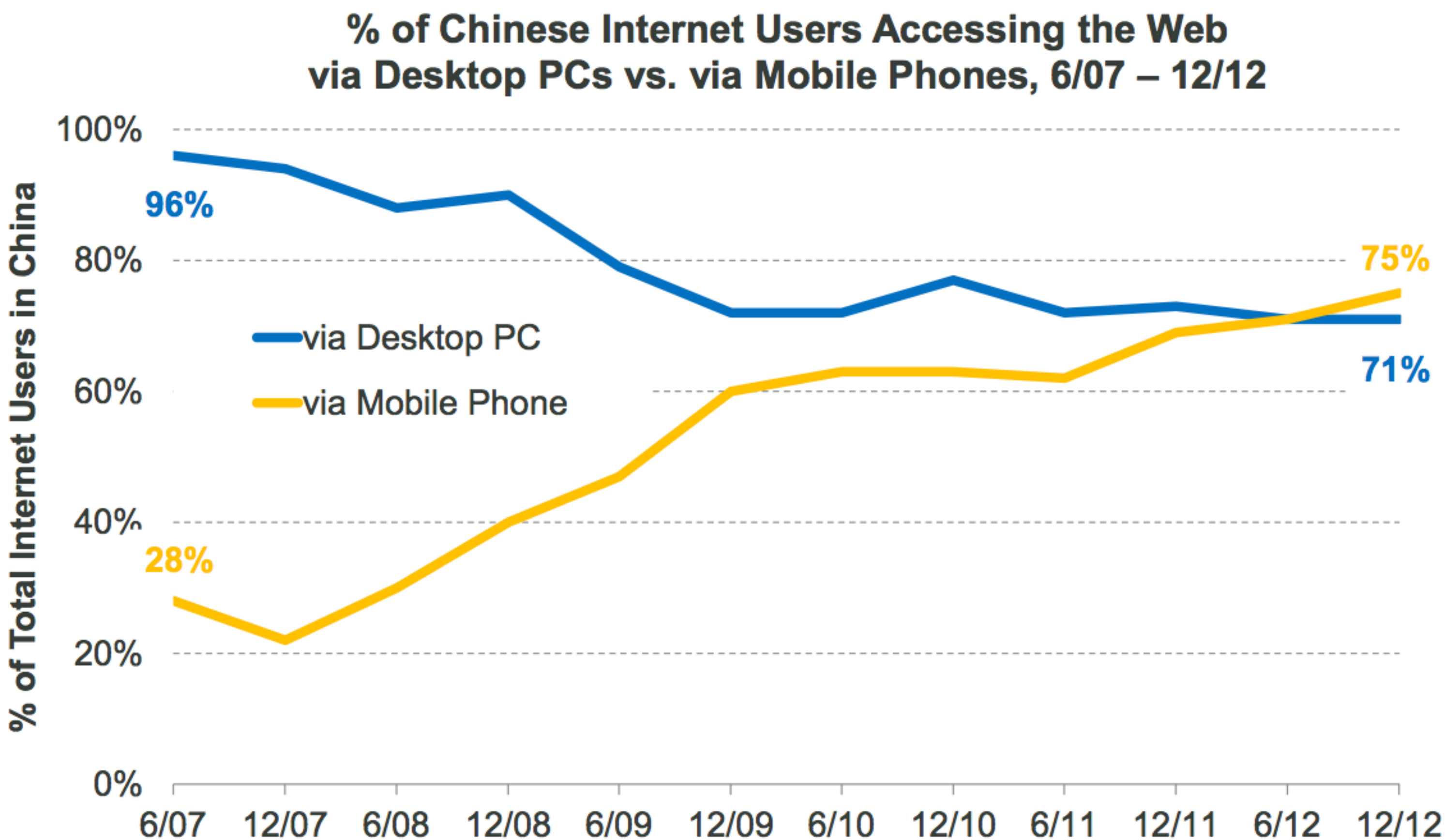
China – Mobile Internet Access Surpassed PC, Q2:12



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Source: CNNIC, 1/13.

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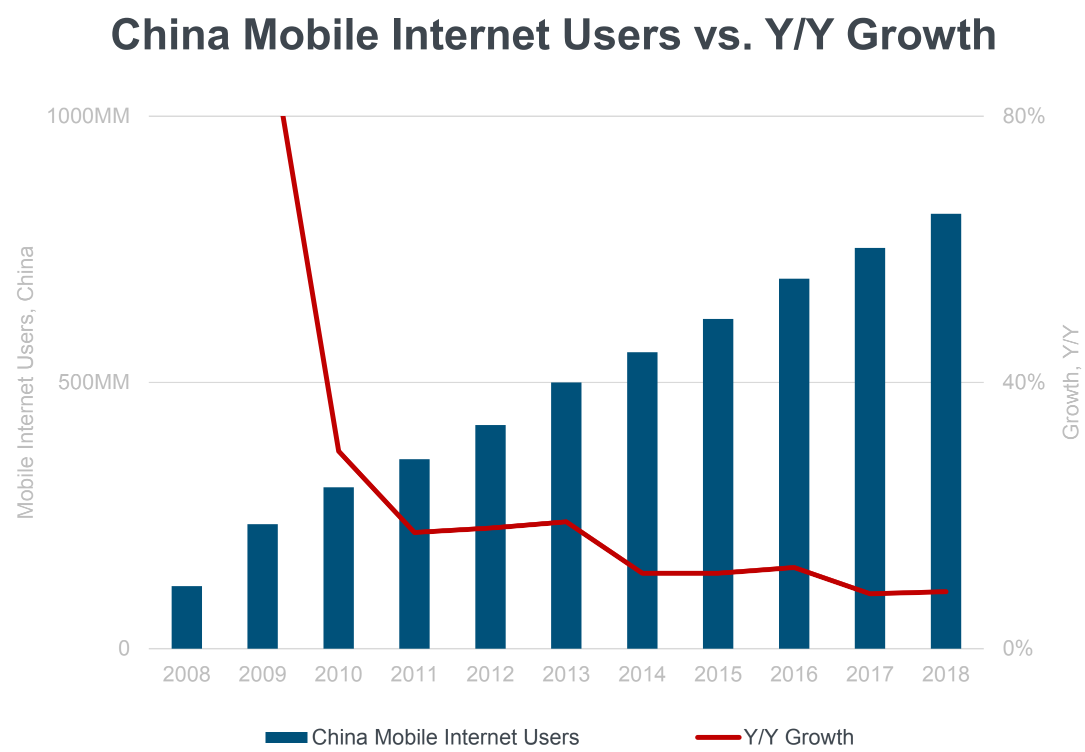
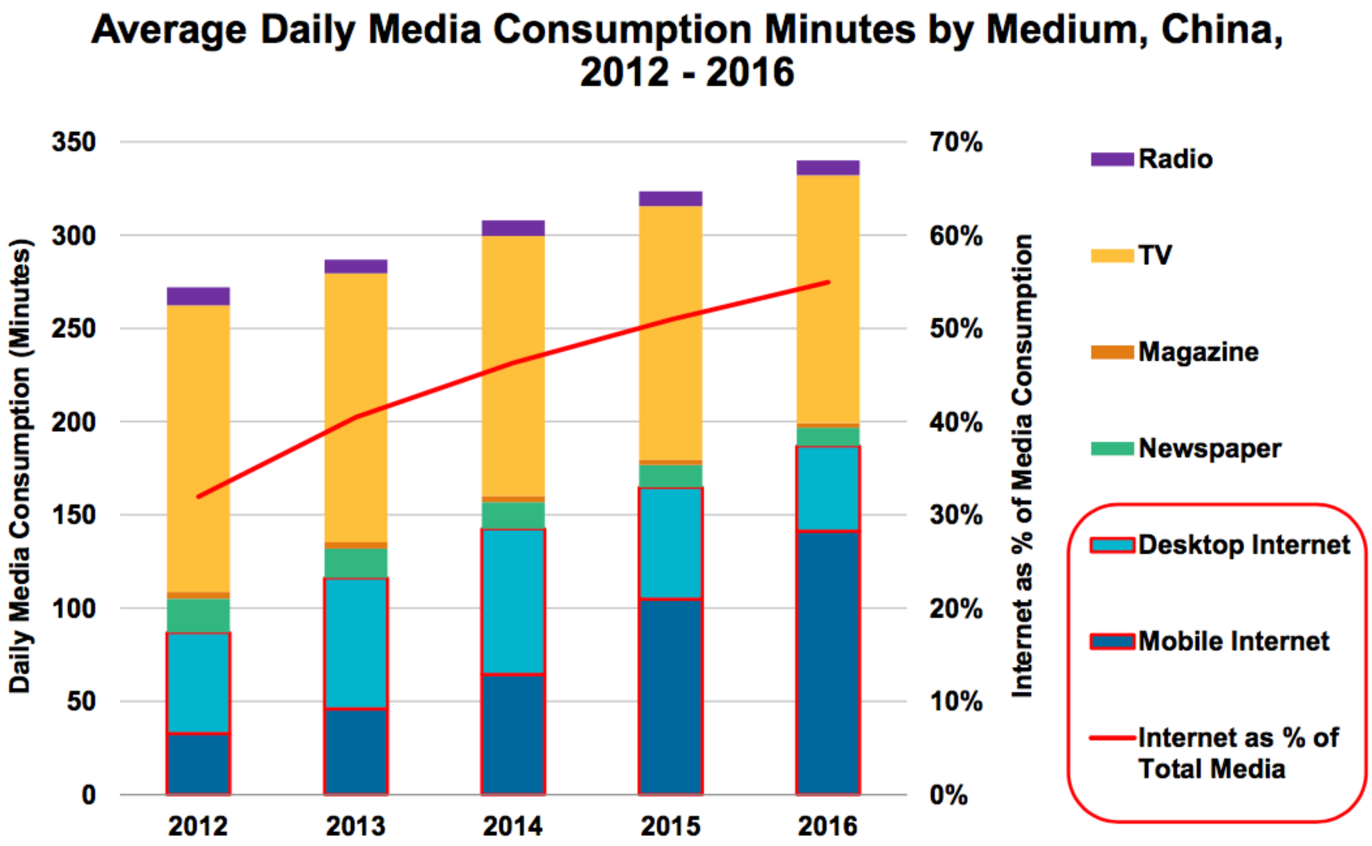
China Mobile Internet Users = 817MM…+9% vs. +8% Y/Y



Source: China Internet Network Information Center
Note: Mobile Internet User Data is as of Year-End.

China Media = Internet @ 55% of Time Spent   Mobile > TV (2016)



Source: Zenith Optimedia

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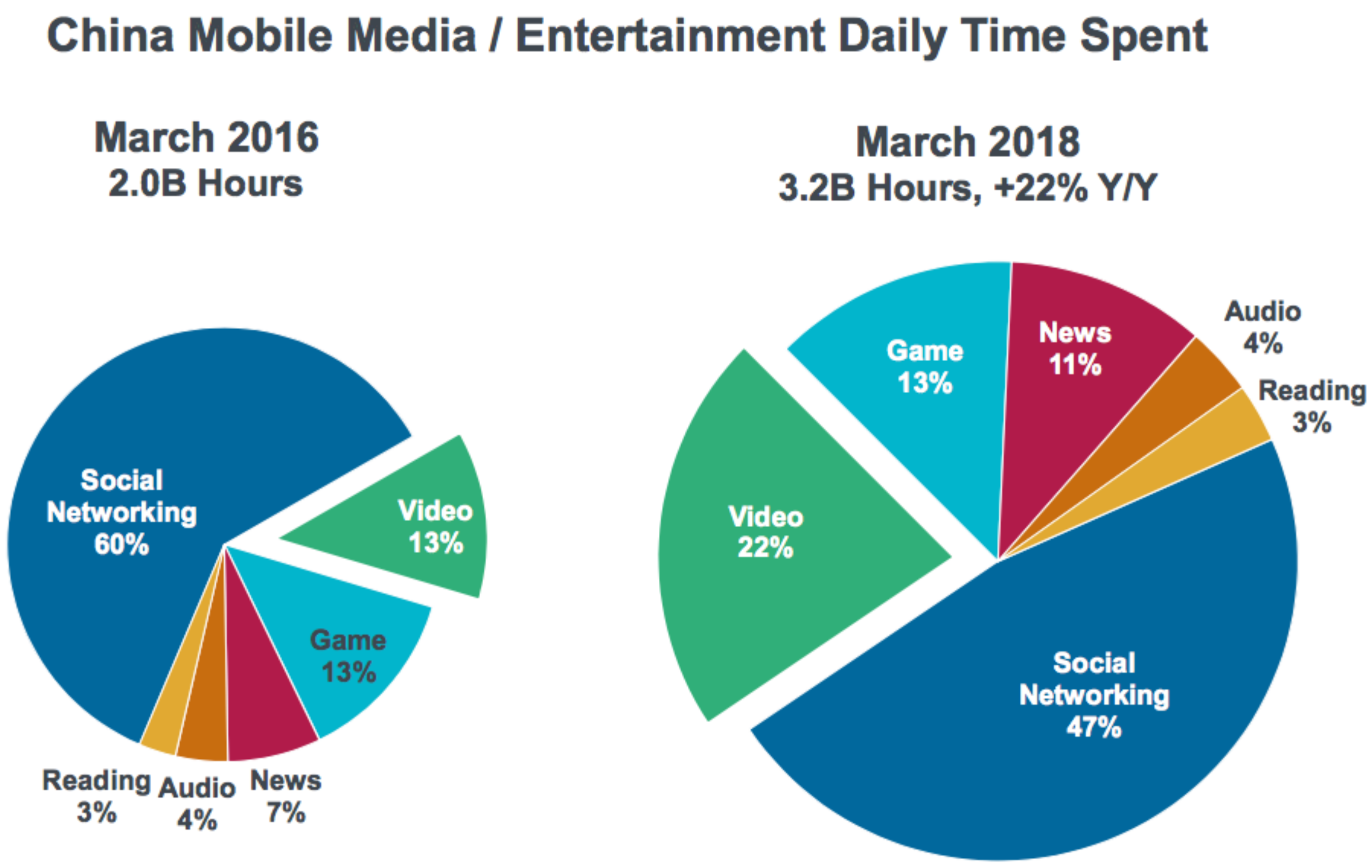
China Mobile Media / Entertainment Time Spent = +22% Y/Y…Mobile Video Growing Fastest



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2018
INTERNET TRENDS

Source: QuestMobile (3/18).

Hillhouse Capital

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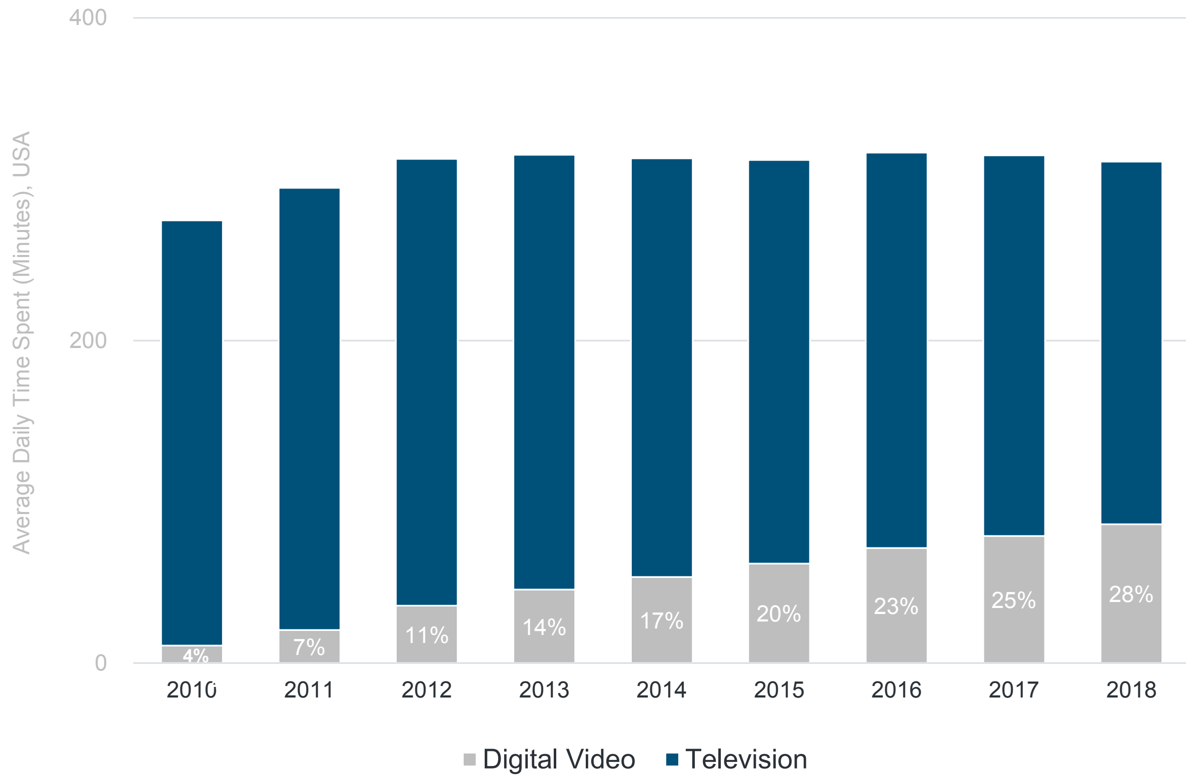
Video Time =
Digital +2x in Five Years @ 28% of Total (vs. TV)

Video Watching Daily Minutes – Digital vs. TV, USA

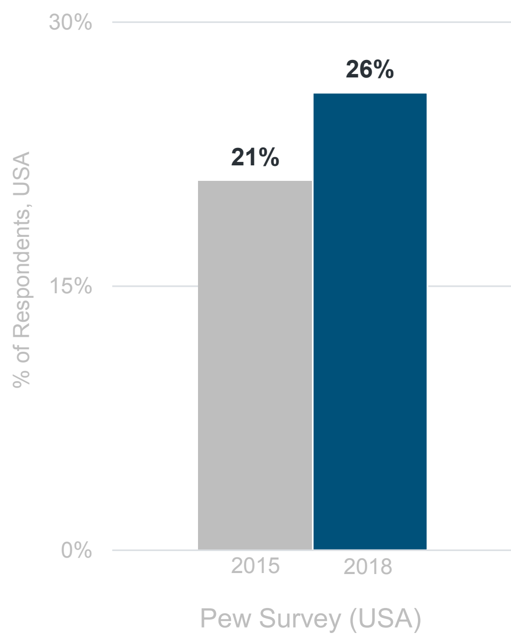


Source: eMarketer (10/18). Usage includes both home & work for consumers 18+. Non de-duplicated defined as time spent with each medium individually, regardless of multitasking.

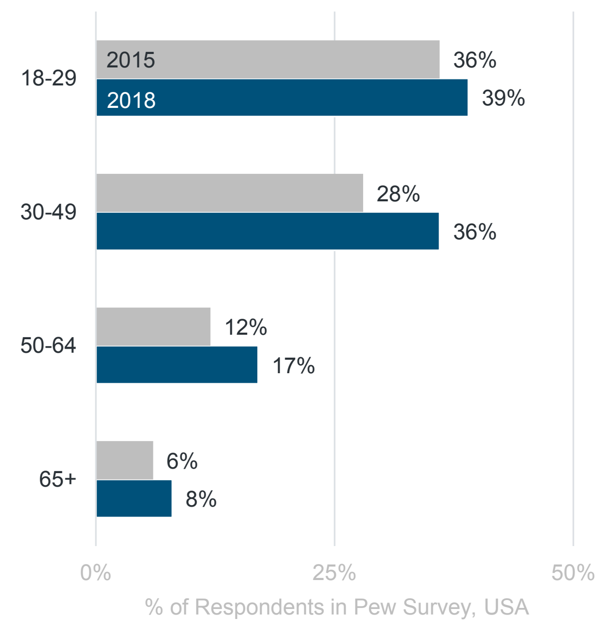
Adults 'Almost Constantly' Online = 26% vs. 21% Three Years Ago

% of Adults Online 'Almost Constantly'

Overall



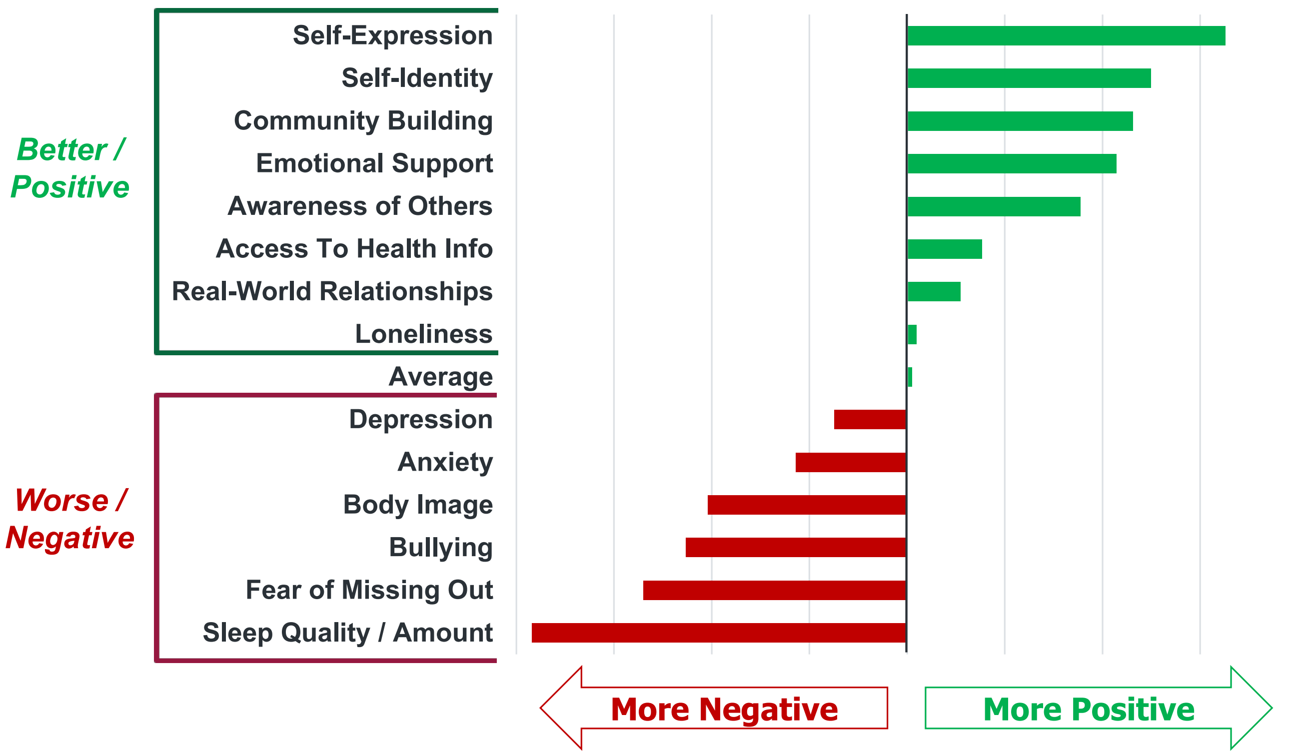
By Age Group



Source: Pew Research (7/15 + 1/18)

Social Media =
Positive & Negative

Do Social Media Platforms You Use Make These Health-Related Factors Better or Worse?

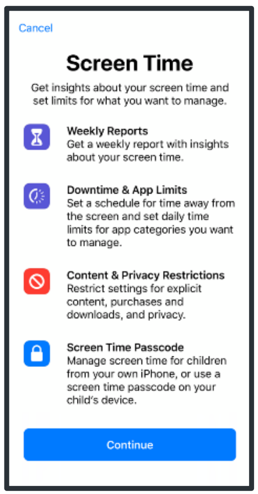


Source: Royal Society For Public Health Survey of 1,479 British teens in 'early-2017' Each teen was asked to rate 5 of the most popular social networks (YouTube, Facebook, Twitter, Snapchat & Instagram) on each dimension. Data presented = average of scores for each social network.

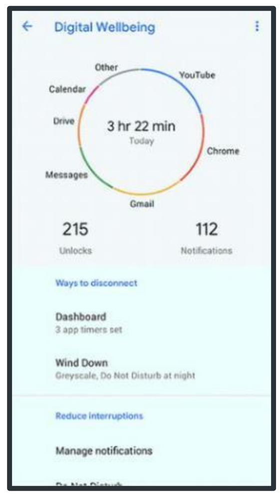
Digital Media =
Businesses Taking Action to Help Users Monitor Usage

2018
Major Platforms Launched Wellness / Time Tracking Features

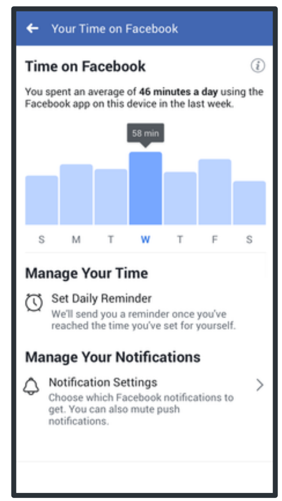
Apple
Screen Time



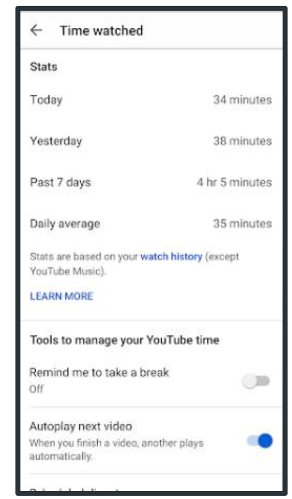
Google
Digital Wellbeing



Facebook
Your Time on Facebook



YouTube
Time Watched



Source: Apple, YouTube, Google, Facebook

Technology Cycles – Still Early Cycle on Smartphones + Tablets, Now Wearables Coming on Strong, Faster than Typical 10-Year Cycle

Technology Cycles Have Tended to Last Ten Years

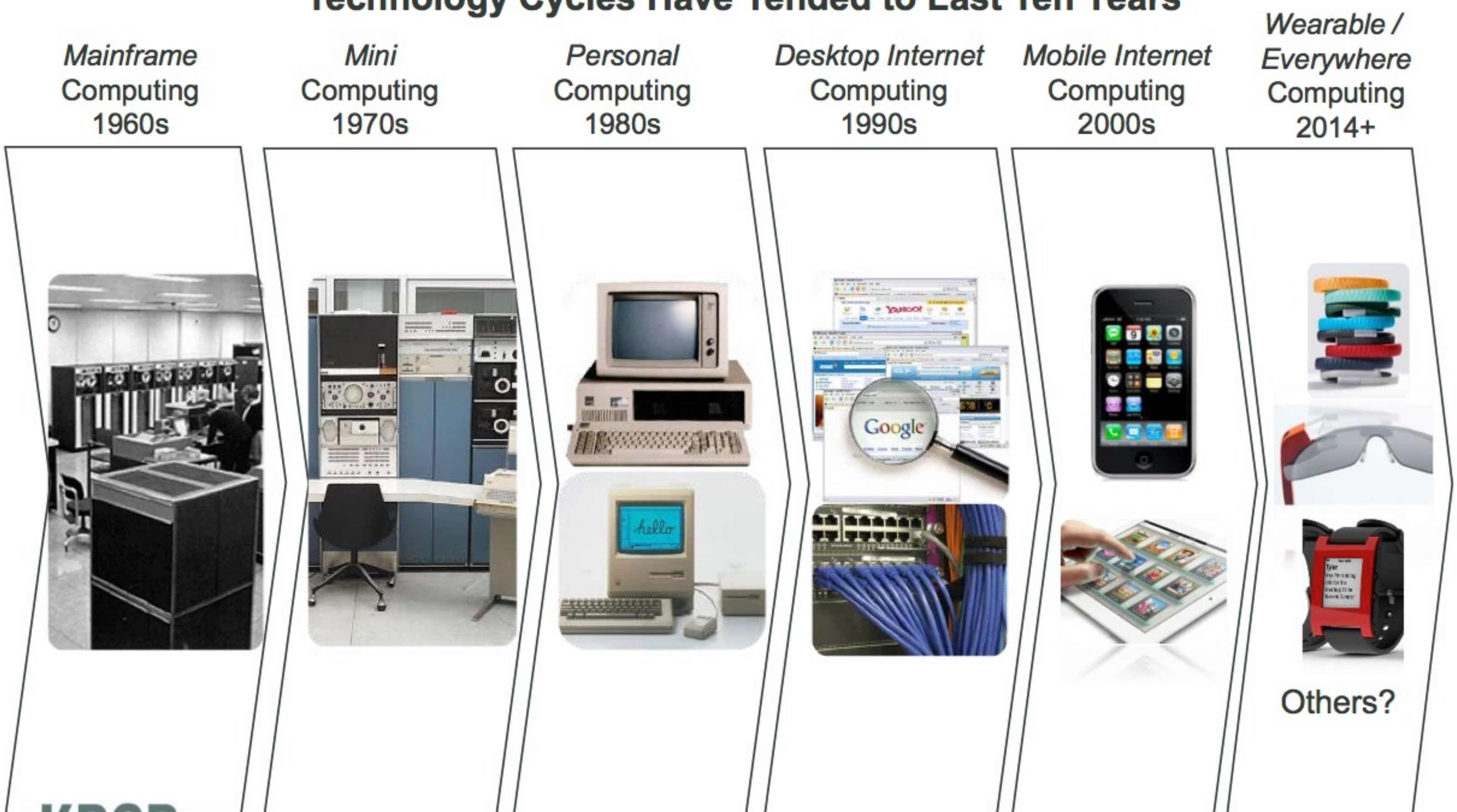


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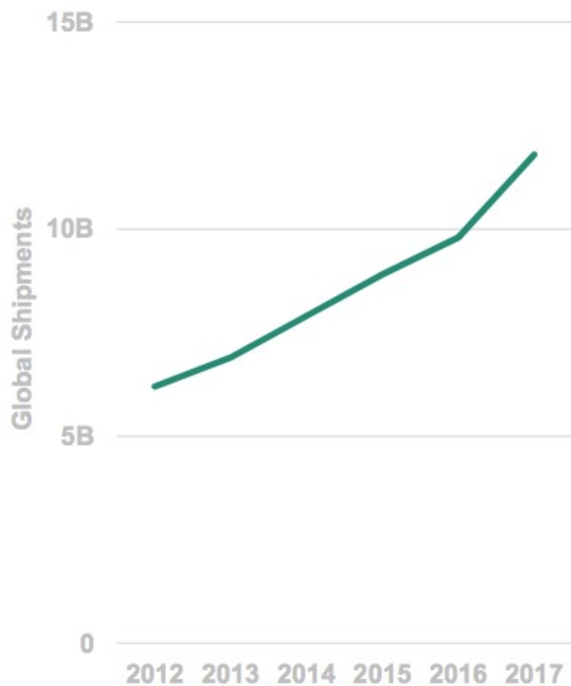
Image Source: Computersciencelab.com, Wikipedia, IBM, Apple, Google, NTT docomo, Google, Jawbone, Pebble.

...Data Gathering + Sharing + Optimization (2006 →) = Enabled by Sensor Pervasiveness...

MEMS Sensor / Actuator Shipments



Sensors + Data = In More Places

Visual Navigation
Google Maps

Shared Transportation
Mobike

Home Temperature
Nest

Predictive Maintenance
Samsara

Fitness Tracking
Motiv

Precision Cooking
Joule

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INTERNET TRENDS

Source: IC Insights (2018), Google Maps, Mobike, Nest, Samsara, Motiv, Joule. Note: MEMS sensors and actuators includes all MEMS-based sensors (e.g., Accelerometers, Gyroscopes, etc.), but does not include optical sensors, like CMOS image sensors, also includes actuators made using MEMs processes, per IC Insights.

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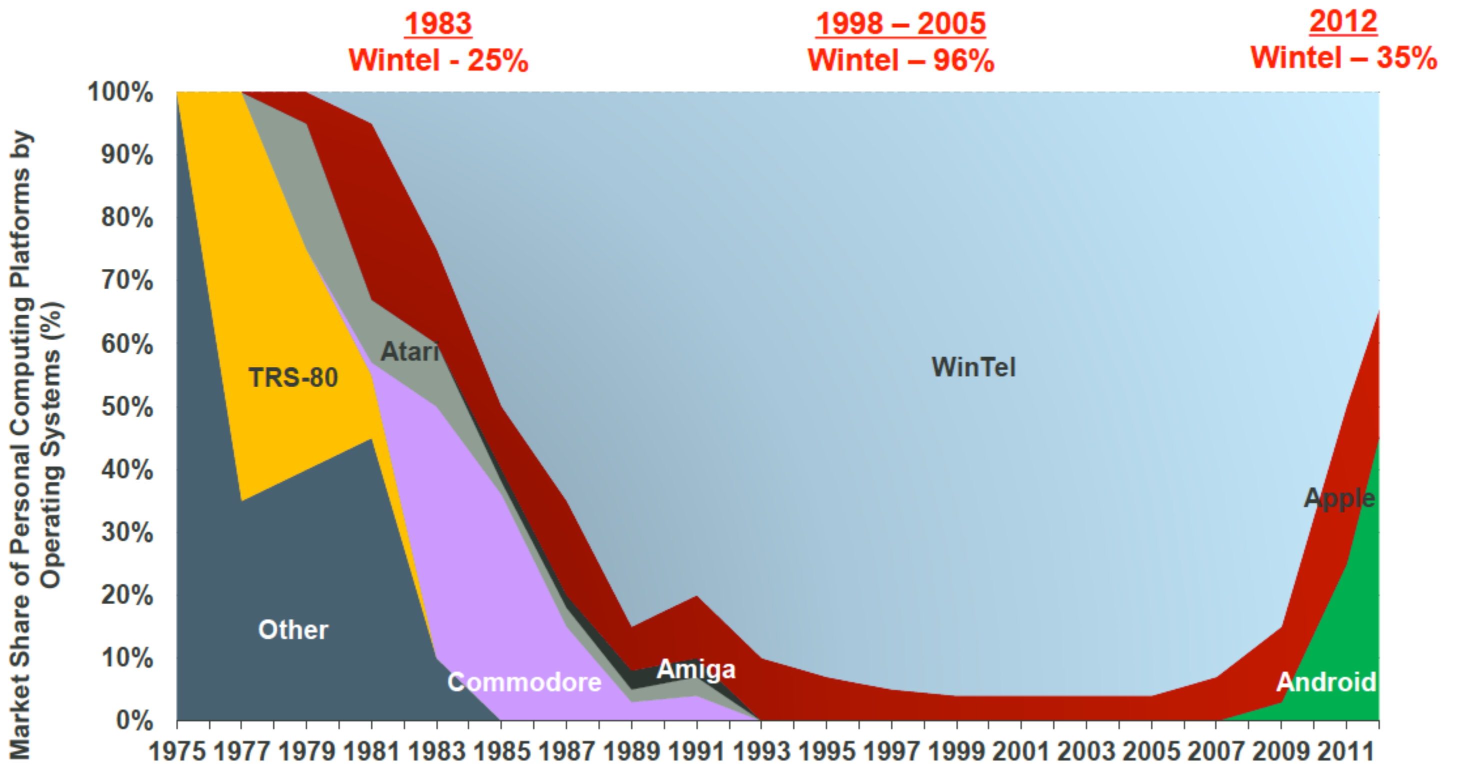
Re-Imagination of Computing Operating Systems - iOS + Android = 60% Share vs. 35% for Windows

Global Market Share of Personal Computing Platforms by Operating System Shipments, 1975 – 2012



KPCB

Source: Asymco.com (as of 2011), Public Filings, Morgan Stanley Research, Gartner for 2012 data.

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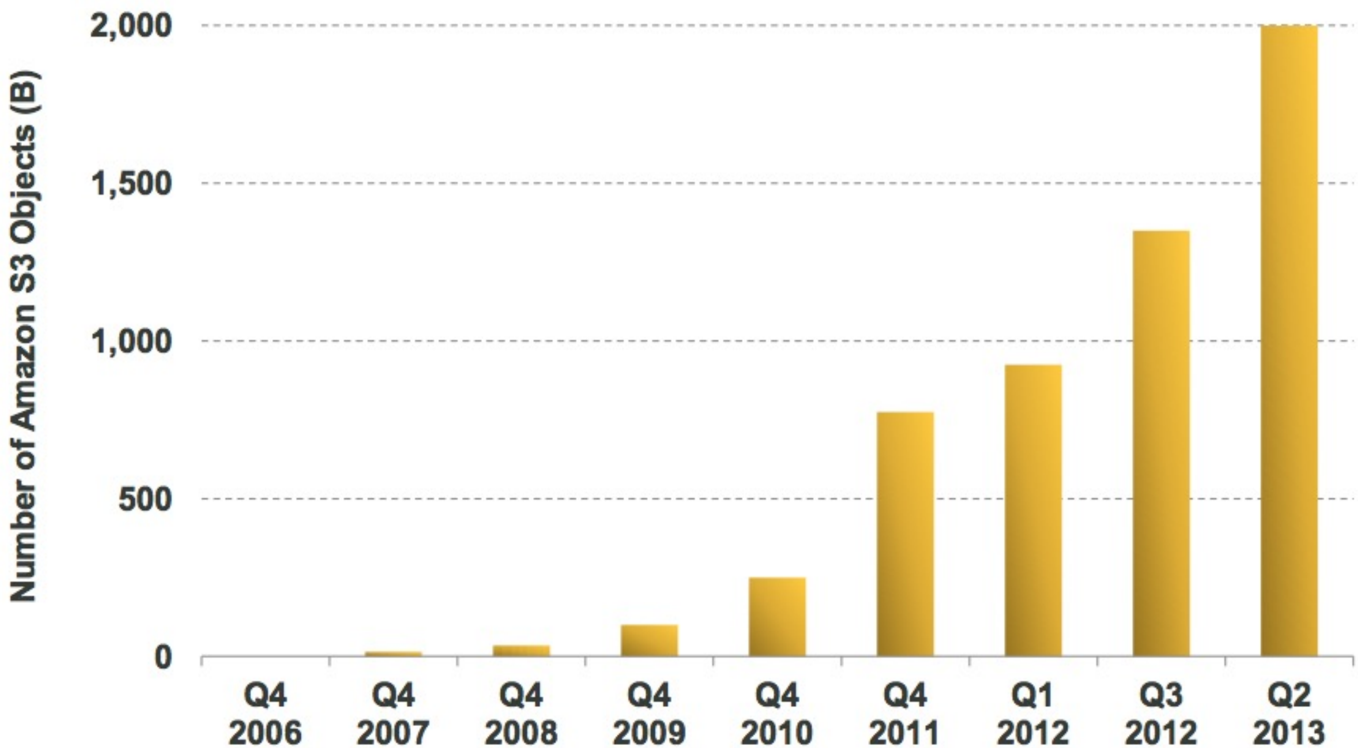
...While The Cloud Rises

Amazon Web Services (AWS) Leading Cloud Charge...

Objects Stored in Amazon S3* (B)



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*Note: S3 is AWS' storage product and used as proxy for AWS scale / growth .
Source: Company data.

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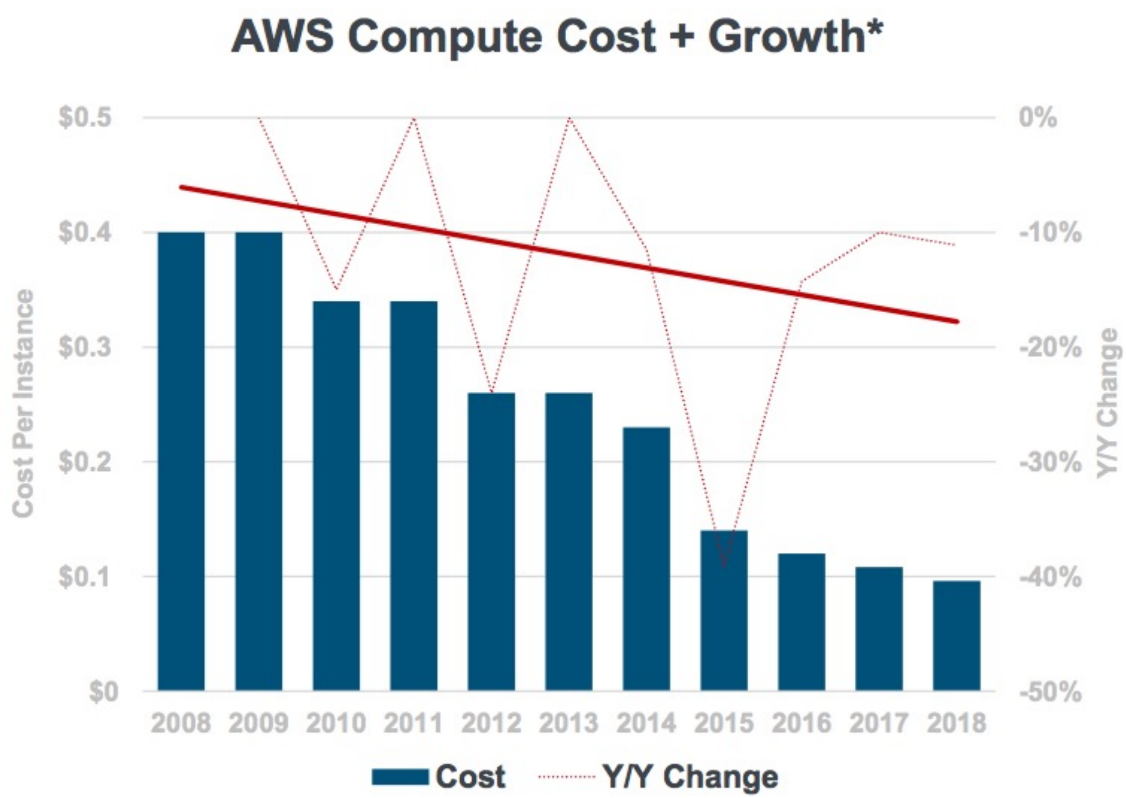
...Computing Big Bangs Volume Effects = Cloud Compute Cost Declines Continue -11% vs. -10% Y/Y...



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2018
INTERNET TRENDS

Source: The Internet Archive. *Cost data reflects price of 'current generation' m.large on-demand Linux instance in USA-East Virginia (m1.large = 2008-2013, m3.large = 2014-2015, m4.large = 2016-2017, m5.large = 2018). m.large chosen as a representative instance of general purpose compute; pricing does not account for increasing instance performance.

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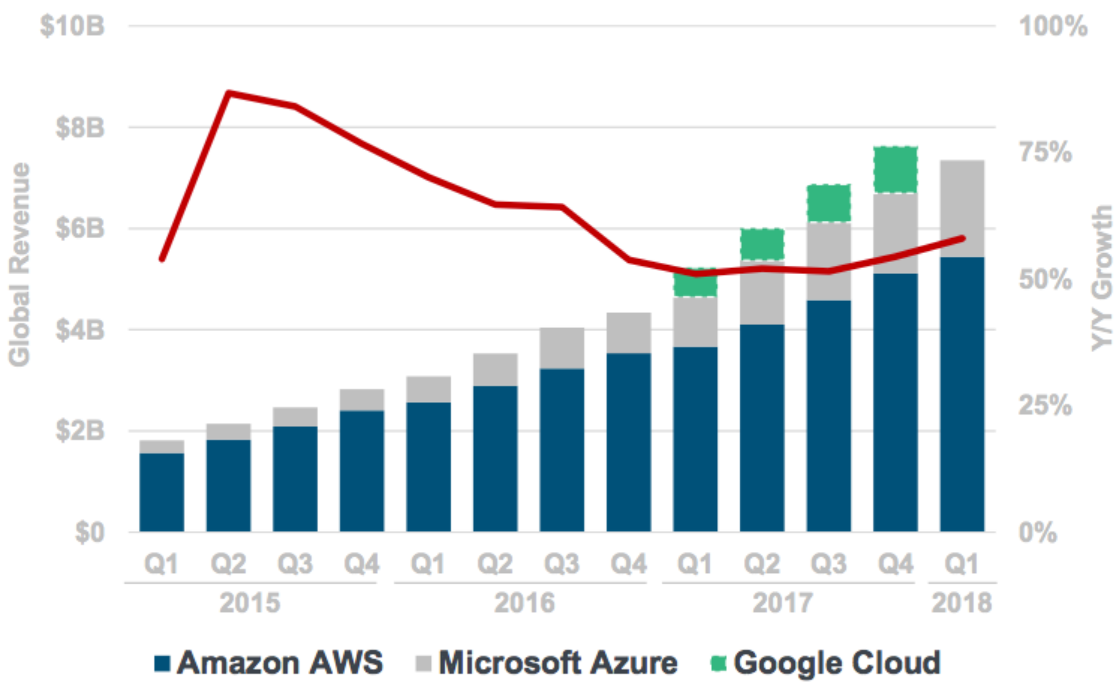
...Computing Big Bangs Volume Effects = Cloud Revenue Re-Accelerating +58% vs. +54% Q/Q

Cloud Service Revenue – Amazon + Microsoft + Google



KLEINER PERKINS 2018 INTERNET TRENDS

Source: Amazon AWS = Company filings, Microsoft Azure = Keith Weiss @ Morgan Stanley (4/18), Google Cloud = Brian Nowak @ Morgan Stanley (5/18). Note: Google Cloud revenue excluded in Y/Y growth rate calculation due to limited quarterly estimates.

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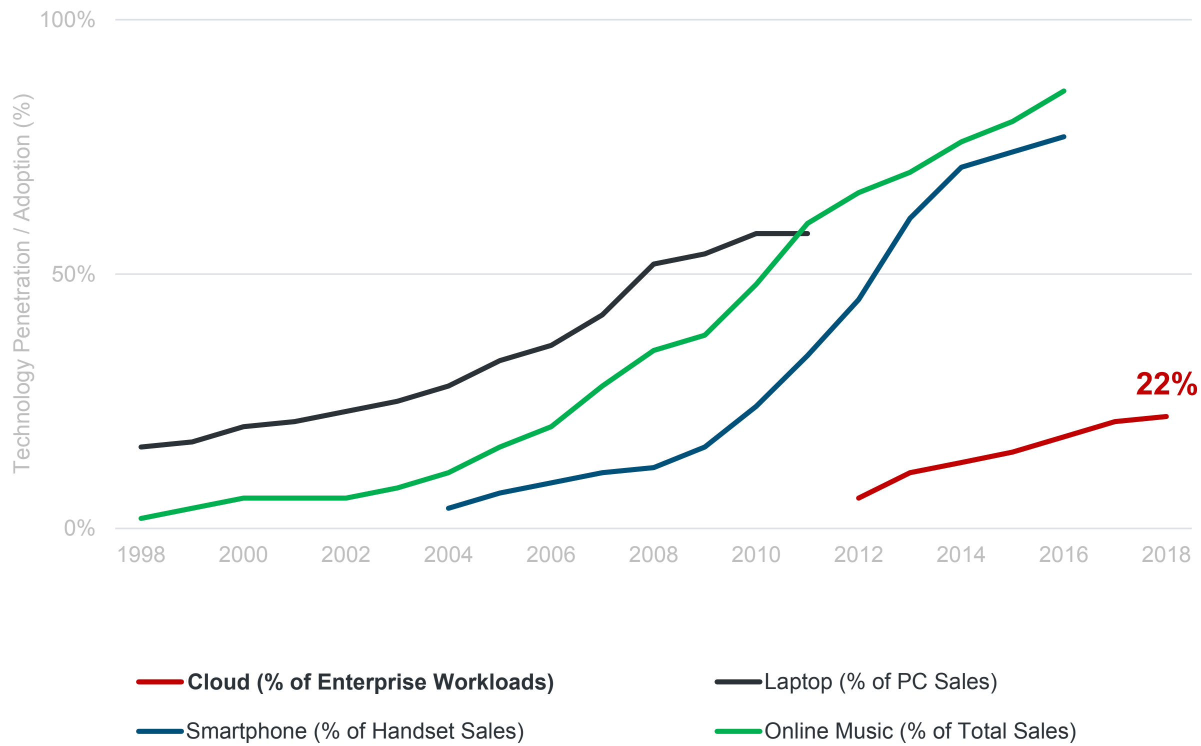
…Cloud Deployment = 22% of Workloads +2x vs. Five Years Ago

Technology Adoption Rates, Global per Morgan Stanley



Source: Morgan Stanley (5/18-3/19). Note: Estimate of Cloud workload penetration based on Morgan Stanley CIO survey in Q4 of each year.

Cloud Evolution / Tools = Paving Way for Innovation Across Infrastructure Landscape

New Methods of Software Delivery =

APIs / Browser Extensions creating new wave of capabilities (+ companies) for both companies and end users

Containers / Microservices =

Simplify software development process / improve consistency between testing & production environments / reduce complexity of managing & updating apps due to modular approach

Elastic Analytical Databases =

Likes of Google BigQuery / Snowflake / AWS Redshift Spectrum nearly infinitely scalable / usage based + have minimal maintenance requirements

Edge Computing =

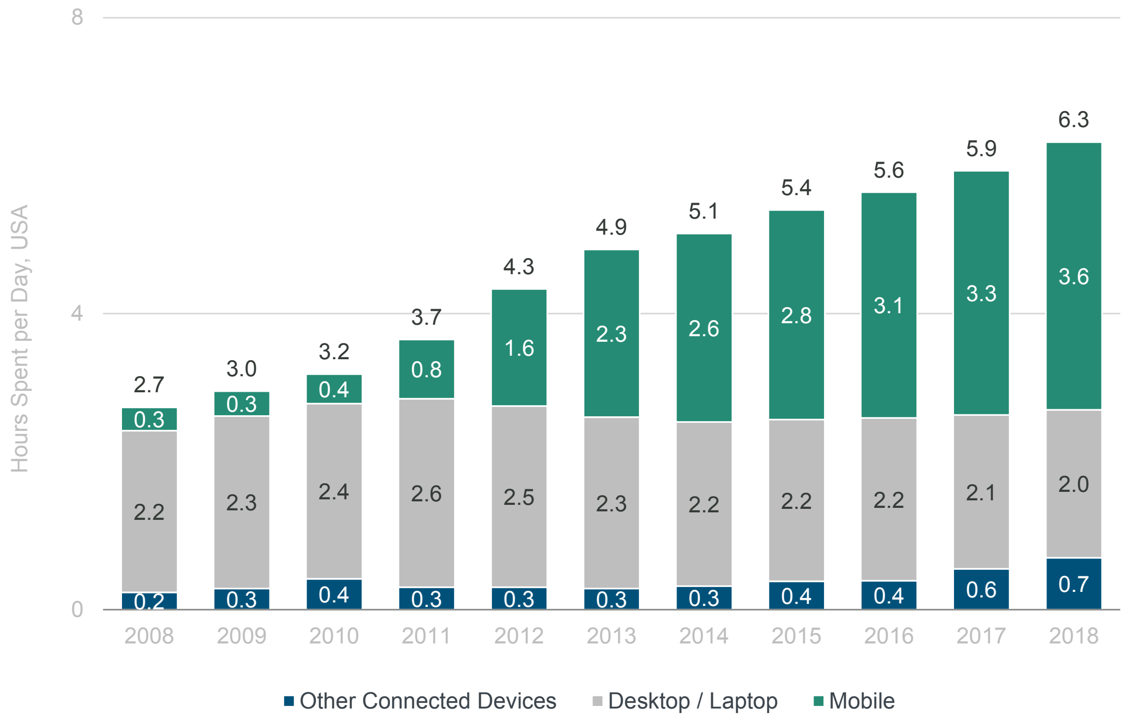
Pushing compute away from centralized nodes & closer to sources of data addresses many IT challenges when running data-centric workloads in cloud – reduces latency / can have security + compliance benefits

KLEINER PERKINS

Source: Lloyd Tabb, Looker Founder & CTO; Happiest Minds; Azuqua; TheServerSide; Forbes

Digital Media Usage = Accelerating +7% vs. +5% Y/Y

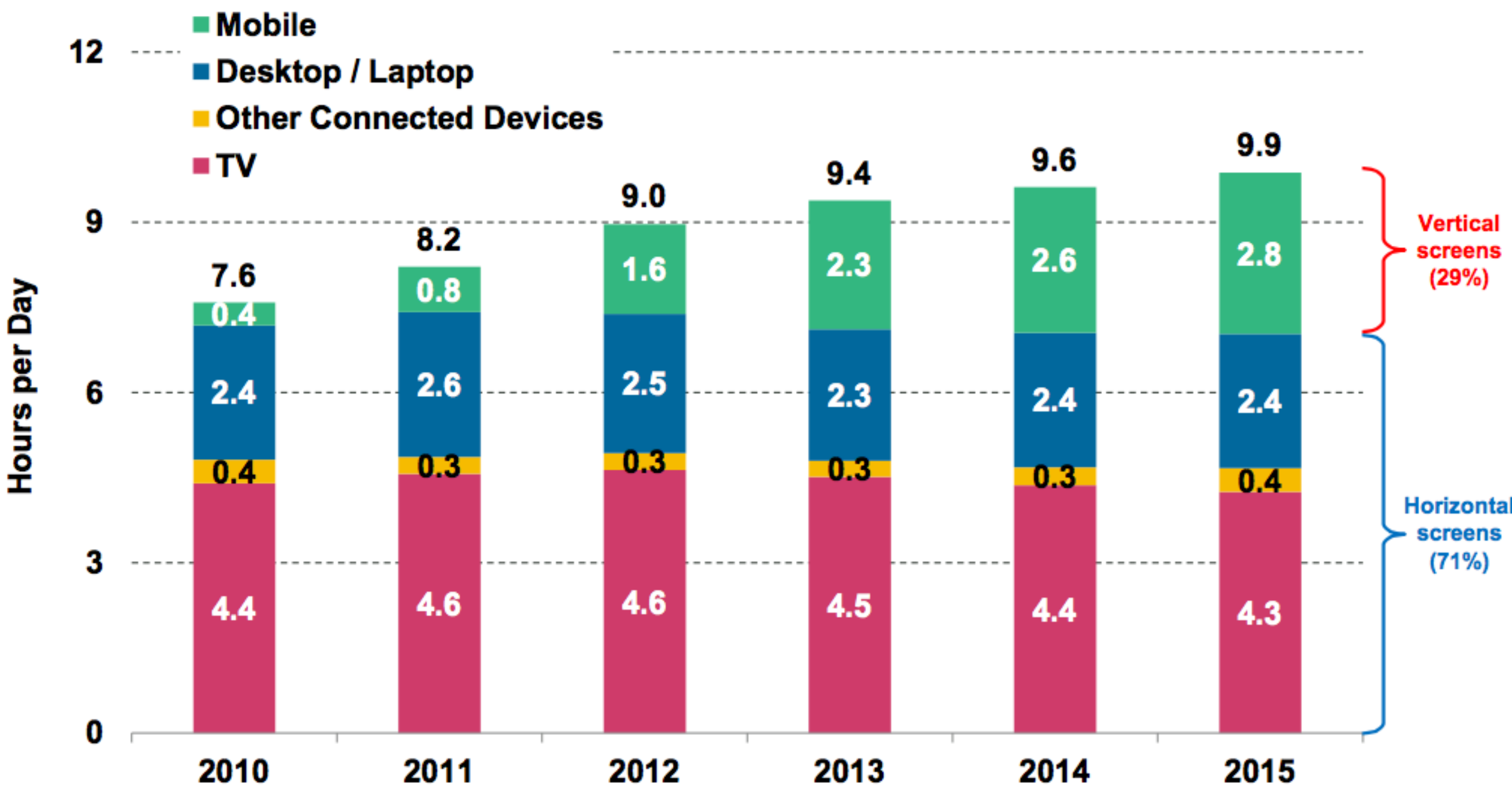
Daily Hours Spent with Digital Media per Adult User, USA



Source: eMarketer 9/14 (2008-2010), eMarketer 4/15 (2011-2013), eMarketer 4/17 (2014-2015), eMarketer 10/18 (2016-2018).
Note: Other connected devices include OTT & game consoles. Mobile includes smartphone & tablet. Usage includes both home & work for consumers 18+. Non deduped defined as time spent with each medium individually, regardless of multitasking.

...Vertical Viewing =
29% of View Time (Multi-Platform) vs. 5% Five Years Ago, USA...

Time Spent on Screens by Orientation (Hours / Day), USA, 2010 – 2015



@KPCB Source: eMarketer 4/15, Coatue analysis. Note: Other connected devices include OTT and game consoles. Mobile includes smartphone and tablet. Usage includes both home and work. Ages 18+; time spent with each medium includes all time spent with that medium, regardless of multitasking; for example, 1 hour of multitasking on desktop/laptop while watching TV is counted as 1 hour for TV and 1 hour for desktop/laptop. 24

Messaging Apps = Top Global Apps in Usage + Sessions

6+ of Top 10 most used apps globally = Messaging Apps

Top Apps by Usage

Rank	App
1	Facebook
2	WhatsApp
3	Messenger
4	Instagram
5	LINE
6	Viber
7	KakaoTalk
8	Clash of Clans
9	WeChat
10	Twitter

Top Apps By Number of Sessions

Rank	App	Sessions
1	KakaoTalk	55
2	WhatsApp	37
3	WeChat	29
4	VK	29
5	LINE	26
6	Viber	20
7	Facebook	20
8	Clash of Clans	16
9	Instagram	12
10	Messenger	8

Messaging Apps → significant app sessions

@KPCB Source: Quettra, Q1:15. Data ranked based on usage.
Quettra analyzes 75MM+ Android users spread out in more than 150 countries, collecting install and usage statistics of every application present on the device. Q1:15 data analyzed three months of data starting from 1/1/15. Data excludes Google apps and other commonly pre-installed apps to remove biases. Only apps with 10K+ installs worldwide and 100+ DAU are counted. 47

Messaging = Extensibility Expanding

Messaging
Tencent (2000 → 2018)

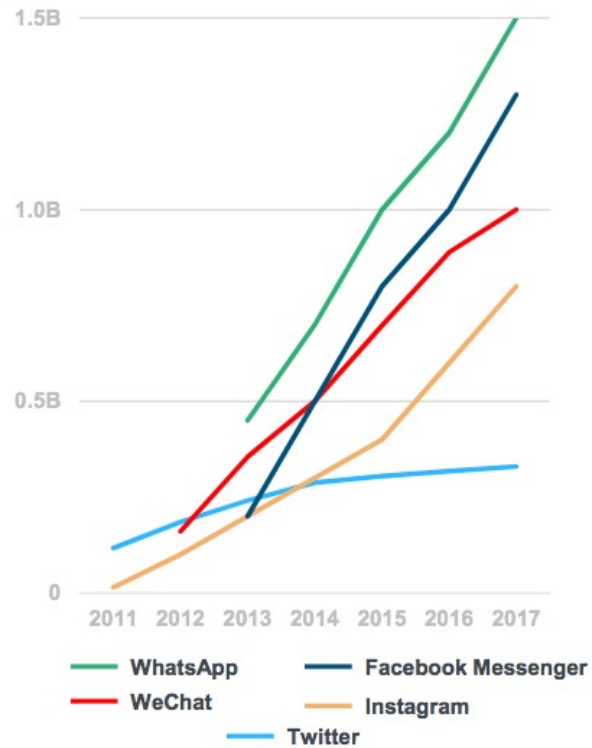
QQ

WeChat

I am on WeChat: marcopapa99

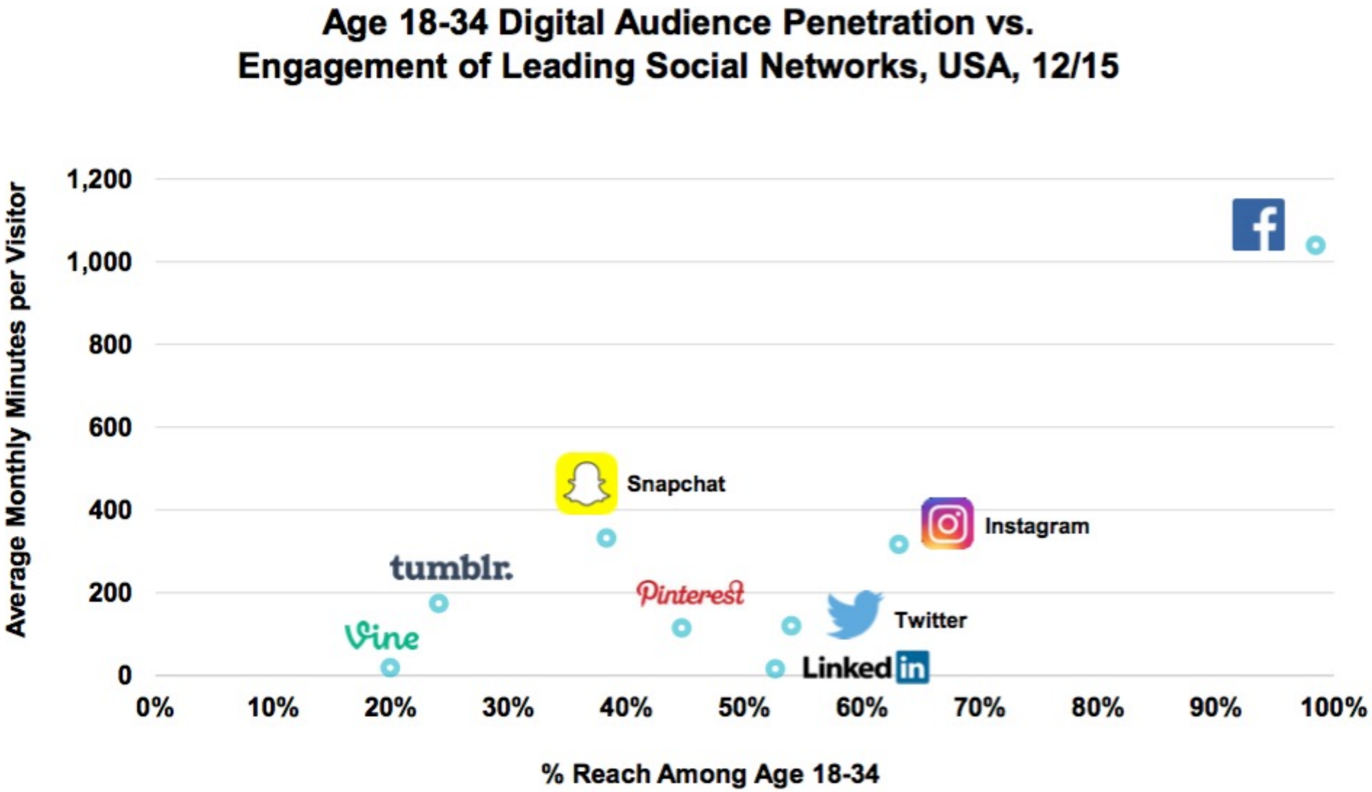
Messenger MAUs

MAU = Monthly Active Users



Source: Facebook, WhatsApp, Tencent, Instagram, Twitter, Morgan Stanley Research. Note: 2013 data for Instagram & Facebook Messenger are approximated from statements made in early 2014. Twitter users excludes SMS fast followers. MAUs (Monthly Active Users) are defined as users who log into a messenger on the web or through an application.

Millennial Social Network Engagement Leaders = Visual... Facebook / Snapchat / Instagram...



@KPCB Source: ComScore Media Metrix Multi-Platform, 12/15.

Asia-Based Messaging Leaders = Continue to Expand Uses / Services Beyond Social Messaging

	KakaoTalk	WeChat	LINE
Name	KakaoTalk	WeChat	LINE
Launch	March 2010	January 2011	June 2011
Primary Country	Korea	China	Japan
New Services Added 2015 -16*			
Banking / Financial Services	Kakao Bank (11/15)	WeBank (1/15)	Debit Card (2016)
Enterprise	✗	Enterprise WeChat (3/16)	✗
Online-To-Offline (O2O)	Kakao Hairshop (1H:16E) Kakao Driver (1H:16E)	✓	Grocery Delivery (2015)
TV	Kakao TV (6/15)	✓	Line Live & Line TV (2015)
Video Calls / Chat	(6/15)	✓	✓
Taxi Services	Kakao Taxi (3/15)	✓	✓
Previous Existing Services			
Messaging	✓	✓	✓
Group Messaging	✓	✓	✓
Voice Calls	Free VoIP calls (2012)	WeChat Phonebook (2014)	✓
Payments	KakaoPay (2014)	(2013)	Line Pay (2014)
Stickers	(2012)	Sticker shop (2013)	(2011)
Games	Game Center (2012)	(2014)	(2011)
Commerce	Kakao Page (2013)	Delivery support w / Yixun (2013)	Line Mall (2013)
Media	Kakao Topic (2014)	✓	✓
QR Codes	✓	QR code identity (2012)	✓
User Stories / Moments	Kakao Story (2012)	WeChat Moments	Line Home (2012)
Developer Platform	KakaoDevelopers	WeChat API	Line Partner (2012)

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Source: Company websites, press releases, Morgan Stanley Research.
*Blue shading denotes that at least one of the platforms listed has added new features since 2015. Some features for other platforms may have been added in prior years
Note: Enterprise denotes product made specifically for messaging or social networking within the enterprise, which is distinct from B2C messaging where businesses engage with current or potential customers.

Average Global Mobile User = ~33 Apps…12 Apps Used Daily… 80% of Time Spent in 3 Apps

Day in Life of a Mobile User, 2016

	Average # Apps Installed on Device*	Average Number of Apps Used Daily	Average Number of Apps Accounting for 80%+ of App Usage	Time Spent on Phone (per Day)	Most Commonly Used Apps
USA	37	12	3	5 Hours	Facebook Chrome YouTube
Worldwide	33	12	3	4 Hours	Facebook WhatsApp Chrome

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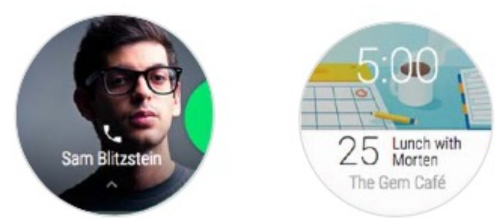
Source: SimilarWeb, 5/16.
*Apps installed does not include pre-installed apps. Most commonly used apps includes preloads.

Notifications = Growing Rapidly & Increasingly Interactive… Driving New Touch Points with Messaging Platforms + Other Apps

Direct Interaction on Notification Panel – without users interrupting what they're doing…

…More Up Close & Personal – as notifications appear on more & more mobile devices



@KPCB Images: Apple, Android Wear.

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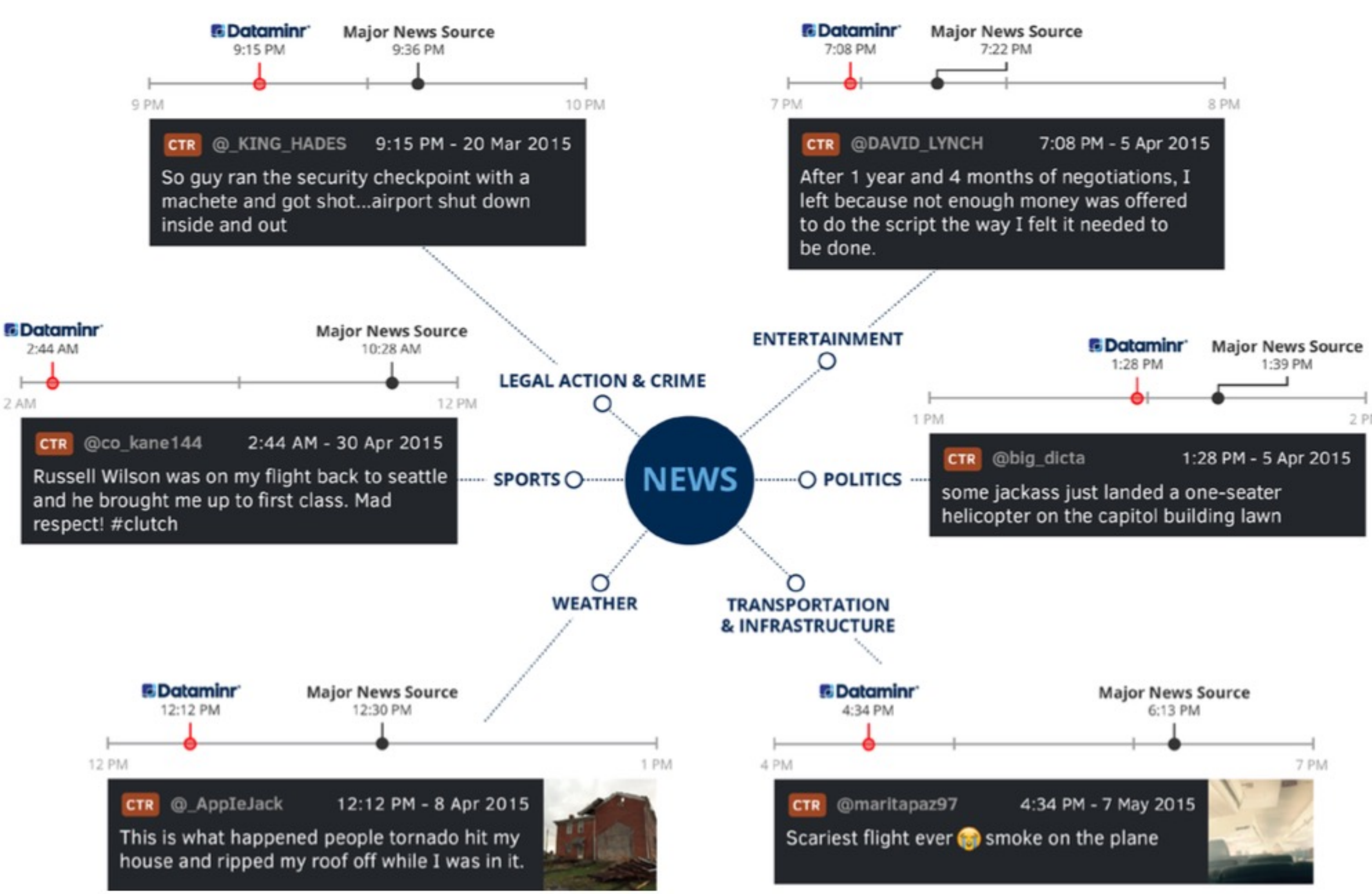
Users Increasingly First Source for News via Twitter / Dataminr



@KPCB Source: Dataminr, 5/15.

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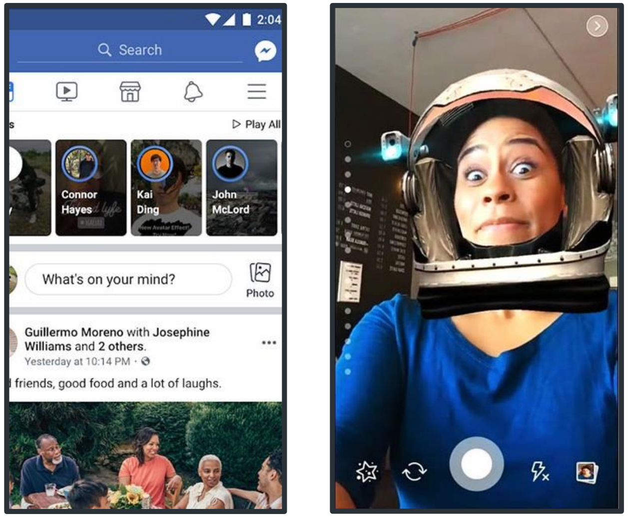
Video Evolution = Accelerating
Live (Linear) → On-Demand → Semi-Live → Real-Live

Live (Linear)	On-Demand	Semi-Live	Real-Live
Traditional TV 1926	***DVR / Streaming 1999***	***Snapchat Stories 2013***	***Periscope + Facebook Live 2015 / 2016***
Tune-In or Miss Out	Watch on Own Terms	Tune-In Within 24 Hours or Miss Out	Tune-In / Watch on Own Terms
Mass Concurrent Audience	Mass Disparate Audience	Mostly Personal Audience	Mass Audience, yet Personal
Real-Time Buzz	Anytime Buzz	Anytime Buzz	Real Time + Anytime Buzz

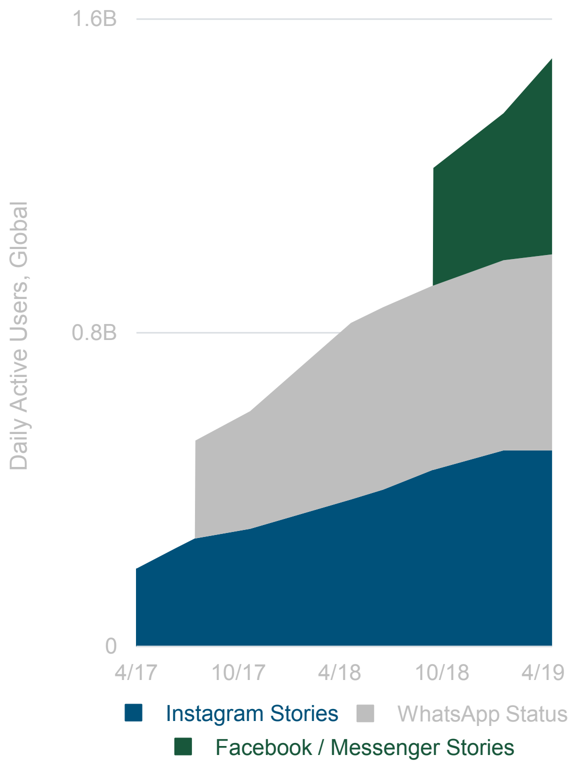
Images: Facebook, Twitter, Snapchat, Netflix, TiVopedia, BT.com
1926 - First television introduced by John Baird to members of the Royal Institution. 1999 - First DVR released by Tivo. 2013 – Snapchat Stories launched.

Video Time (Short-Form – Facebook Platform) = 1.5B DAUs + ~2x in One Year

Facebook Stories

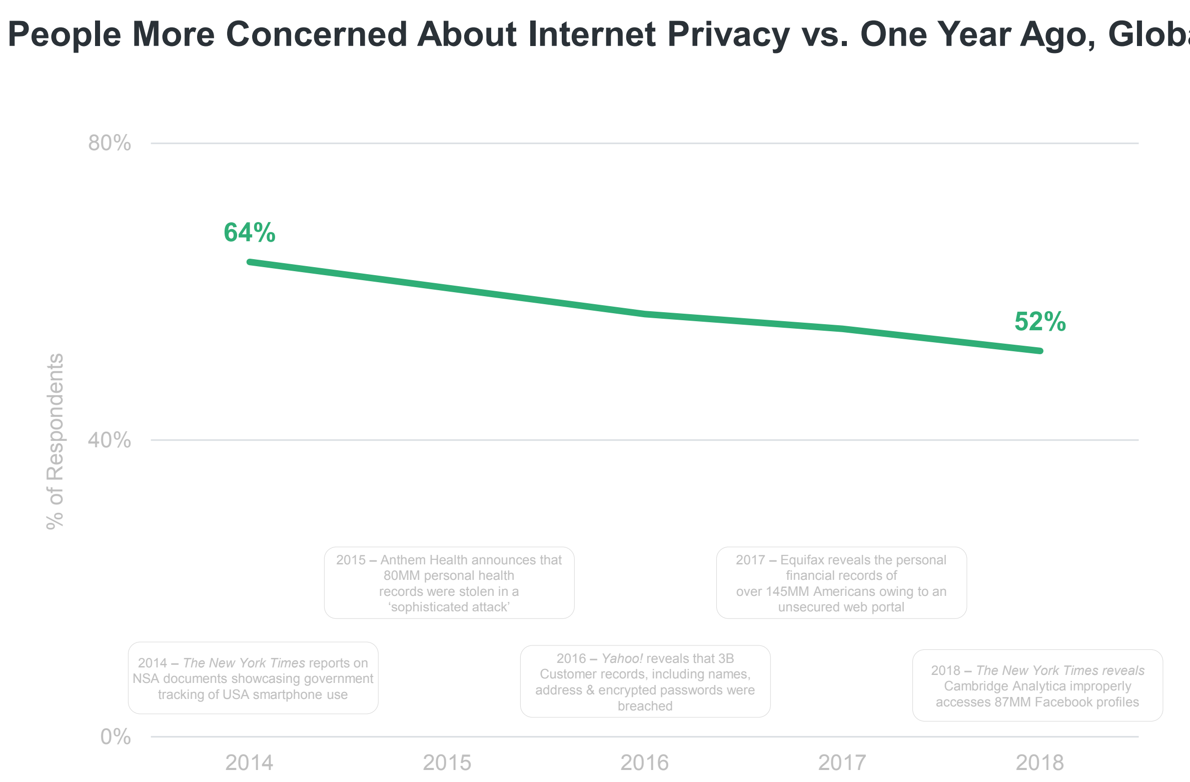


Daily Active Users, Global*



Source: Facebook (8/18,12/18). Daily Active Users (DAU) = Users that interact with a Story daily.
Vertical segments of chart = owing to first reported DAU metric for each product. *Not de-duped across platforms.

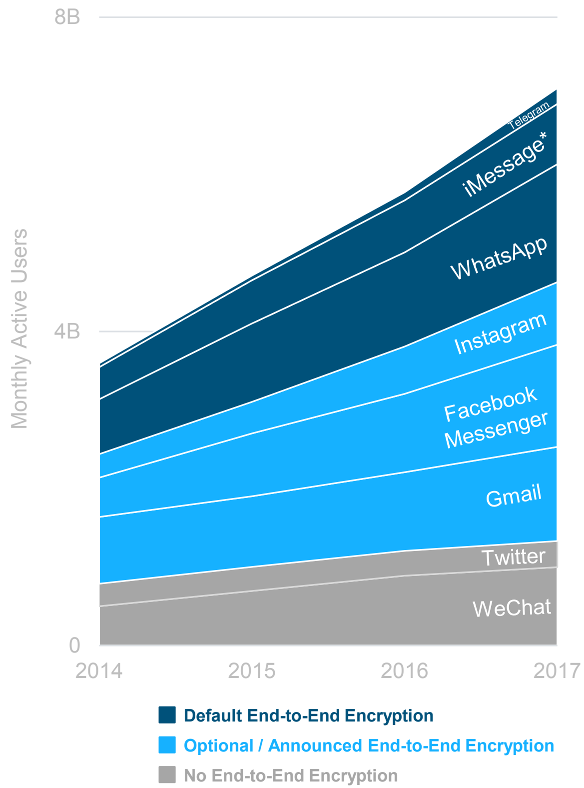
Digital Media =
Privacy Concerns High But Moderating



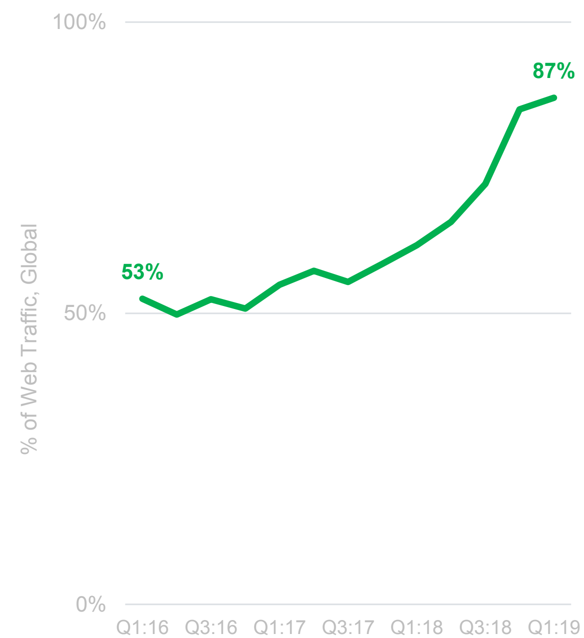
Source: CIGI-IPSOS Global Survey on Internet Security & Trust 2014, 2016, 2017, 2018. 2018 = 24,750 Internet users from 12/29/17-3/5/18. 2017 = 24,225, 2016 = 24,143, 2014 = 23,376.

Digital Media = Encrypted Messaging / Traffic Rising Rapidly

Select Messenger MAUs

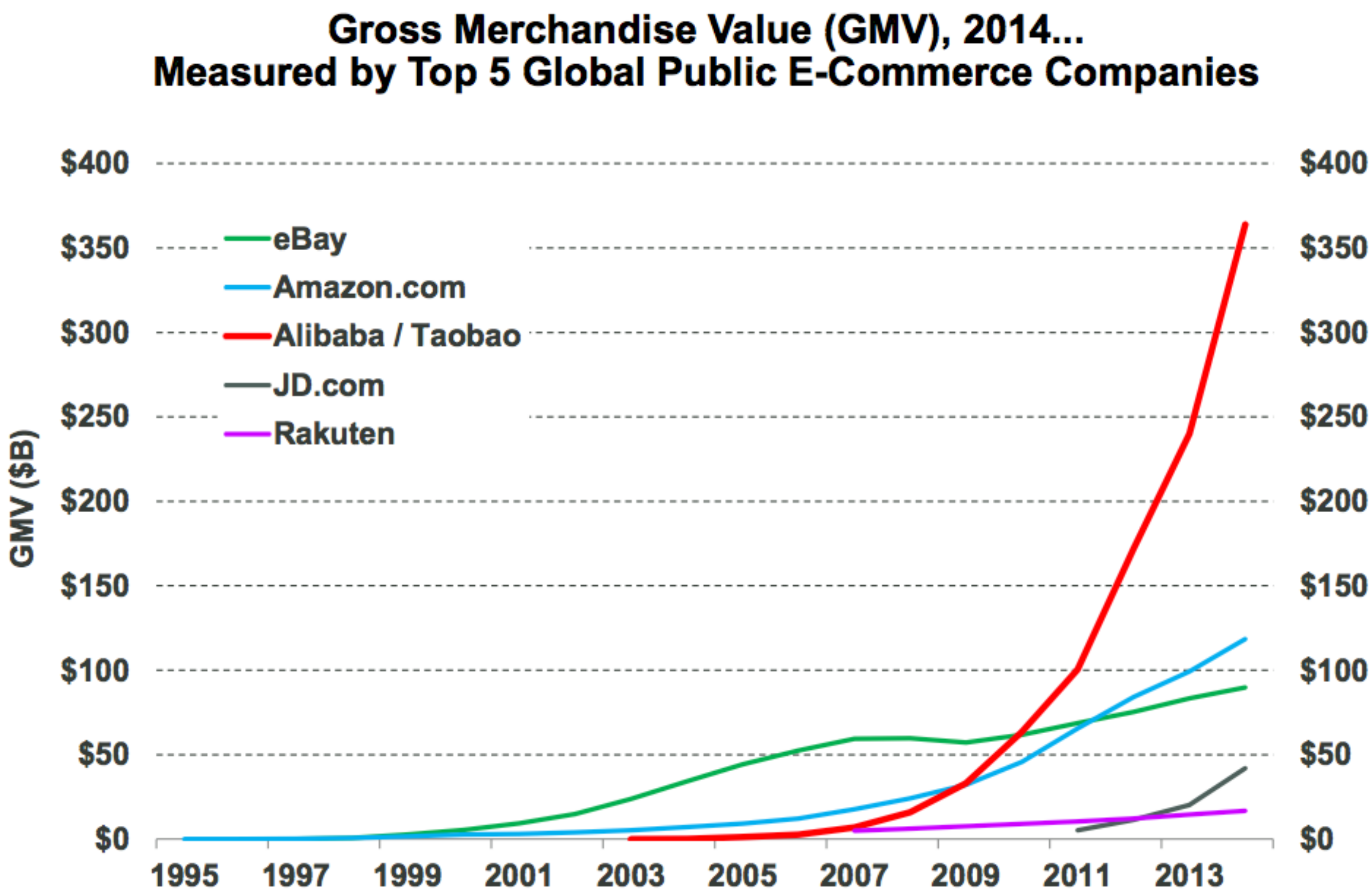


% of Web Traffic Encrypted



Source: Google, Tencent, Twitter, Facebook, Apple, Telegram releases & Morgan Stanley estimates. Note: *iMessage2 MAUs calculated by install base of Apple iPhones, as estimated by Credit Suisse (2014-2017). WhatsApp employs end-to-end encryption by default. Facebook Messenger has end-to-end encryption capabilities but users have to manually enable them. Instagram does not have end-to-end encryption but Facebook is planning to add that feature & make Facebook Messenger encrypted by default (1/19). All Gmail messages are encrypted at rest and in transit. Fortinet Q3:18 Quarterly Threat Landscape Report (11/18). HTTPS = Hyper Text Transfer Protocol Secure is the secure protocol over which data is sent between the browser and the website the user is connected to.

1st Generation 'Online Platforms / Marketplaces for *Products* Rising = Optimized for Desktop Internet + Traditional Shipping Delivery



@KPCB Source: Companies are eBay, Amazon, Alibaba, JD.com, and Rakuten. Per publicly available company data, Morgan Stanley Research.

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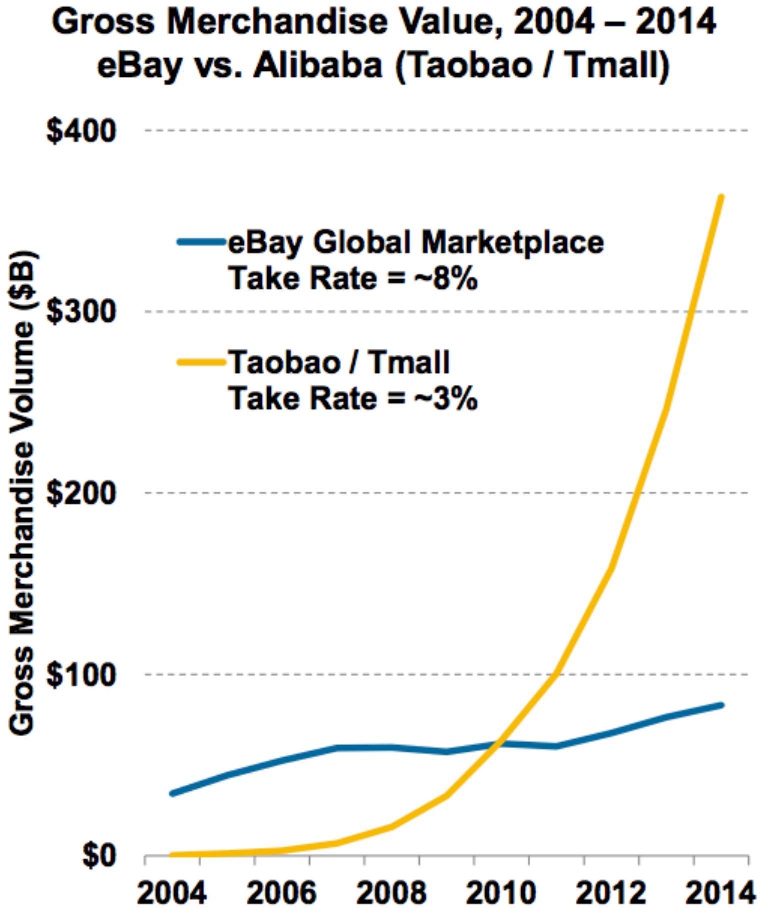
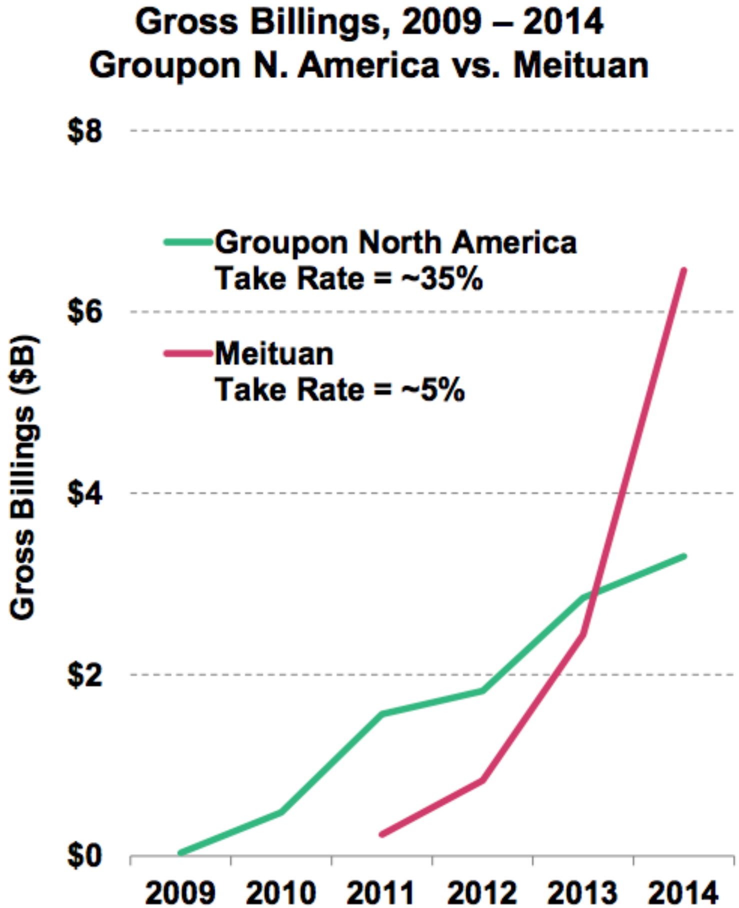
China E-Commerce = Low Take Rates* Helped China Marketplace Leaders Pass USA Peers





@KPCB

Source: Meituan gross billings data are estimates by Tuan800.com, eBay, Groupon, Alibaba GMV data per company.
Note: Take rate defined as net revenue divided by gross merchandise value or gross billings. eBay marketplace take rate excludes PayPal (~3%), eBay, Alibaba GMV data per company. Meituan take rate is estimate per media report.

Hillhouse Capital

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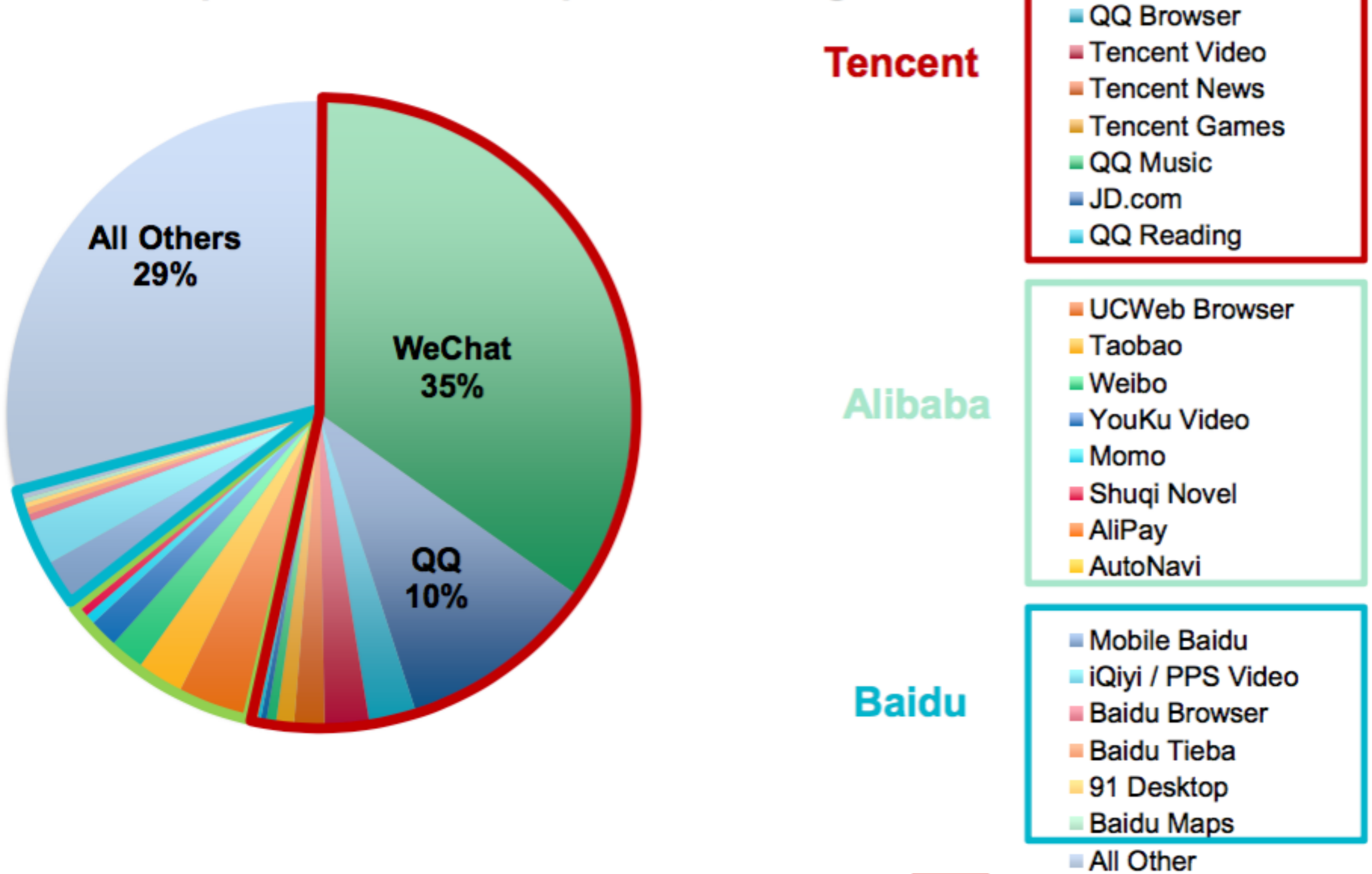
China Mobile Internet Usage Leaders...
Tencent + Alibaba + Baidu = 71% of Mobile Time Spent

Share of Mobile Time Spent, April 2016
Daily Mobile Time Spent = ~200 Minutes per User, Average



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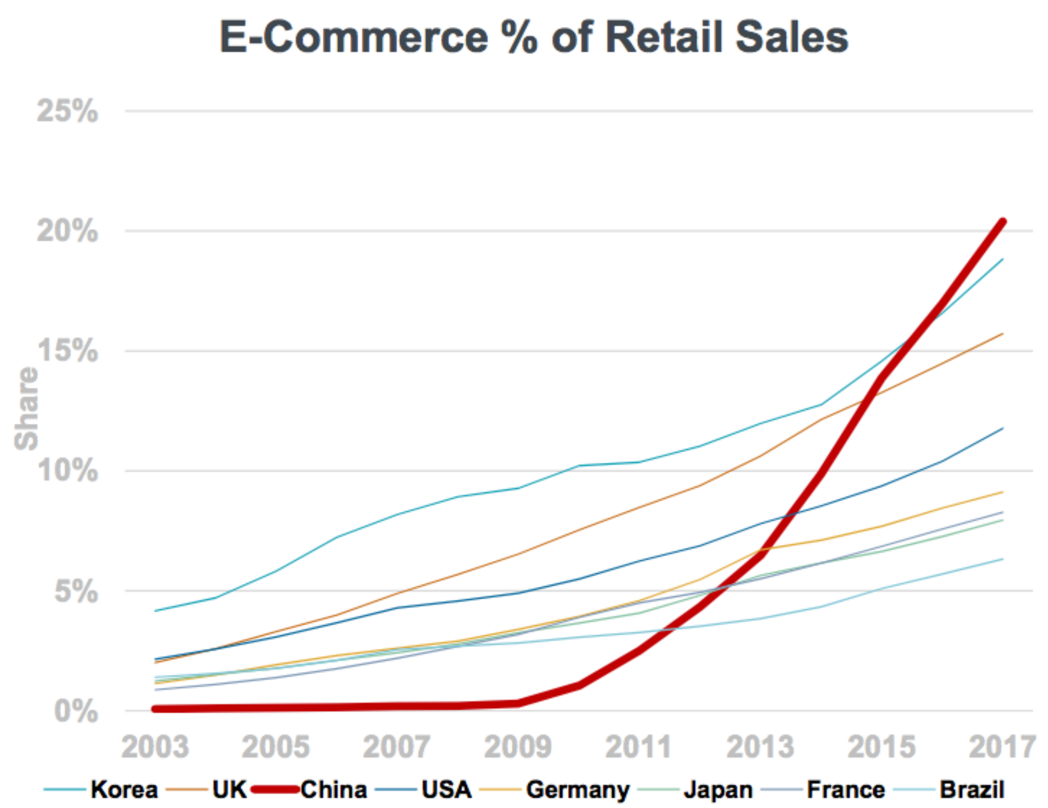
Note: Grouping of apps include strategic investments made by Tencent, Alibaba and Baidu. Only apps in top 50 by time spent share are called out. Source: QuestMobile, Trustdata, and Hillhouse estimates.

Hillhouse Capital | PAGE 167

Worldwide E-Commerce Share Gains Continue…
China @ 20% = Highest Penetration Rate + Fastest Growing



KLEINER PERKINS
2018
INTERNET TRENDS

Source: Euromonitor. Note data excludes certain consumer-to-consumer (C2C) transactions.

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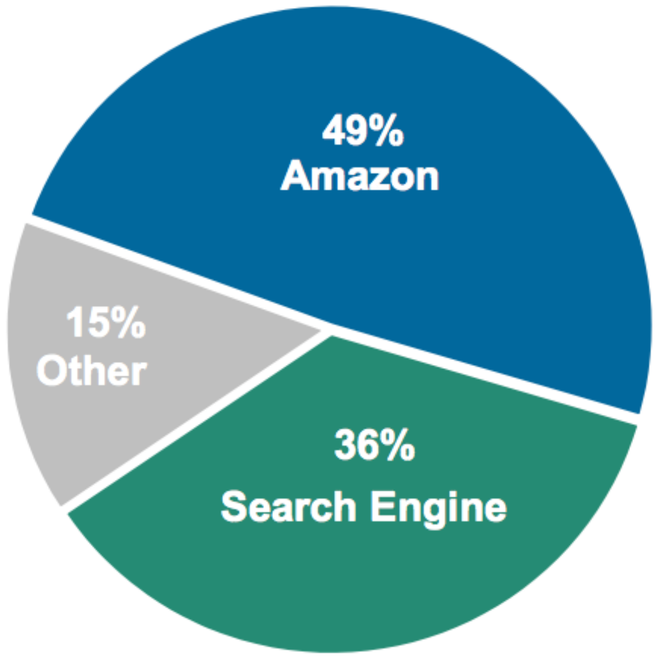
Product Finding = Often Starts @ Search (Amazon + Google...)

Where Do You Begin Your Product Search?



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2018
INTERNET TRENDS

Source: Survata (9/17). Note: n = 2,000 USA customers.

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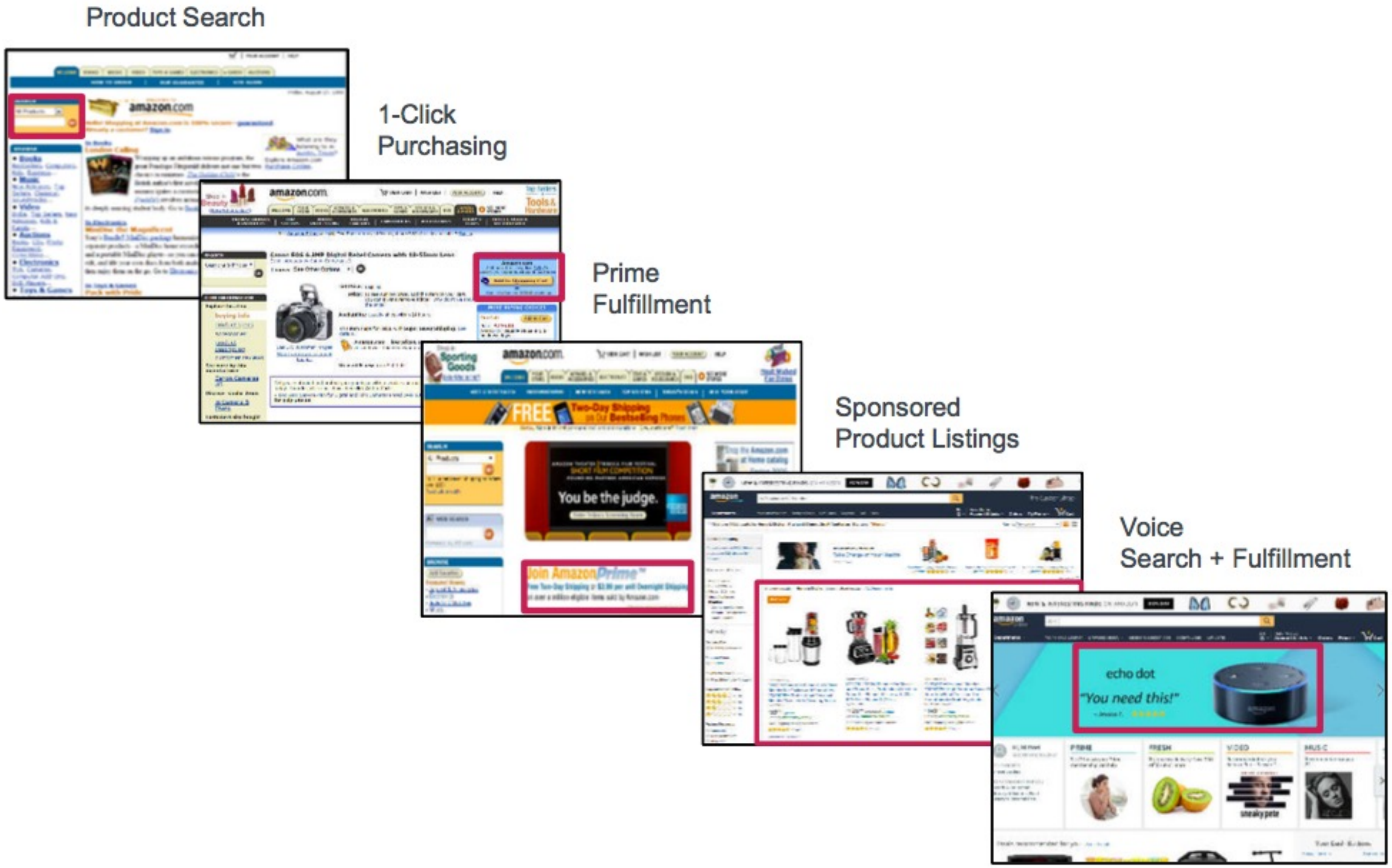
Product Finding (Amazon) = Started @ Search...Fulfilled by Amazon



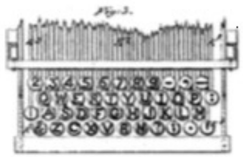
Source: The Internet Archive, Amazon.

Product Finding (Google) = Started @ Search…Fulfilled by Others

Source: The Internet Archive, Google.

Human-Computer Interaction (1830s – 2015), USA = Touch 1.0 → Touch 2.0 → Touch 3.0 → Voice

Punch Cards for Informatics
1832



QWERTY Keyboard
1872

Electromechanical Computer (Z3)
1941

Electronic Computer (ENIAC)
1943

Paper Tape Reader (Harvard Mark I)
1944

Mainframe Computers (IBM SSEC)
1948

Trackball
1952

Joystick
1967

Microcomputers (IBM Mark-8)
1974

Portable Computer (IBM 5100)
1975

Commercial Use of Window-Based GUI (Xerox Star)
1981

Commercial Use of Mouse (Apple Lisa)
1983

Commercial Use of Mobile Computing (PalmPilot)
1996

Touch + Camera - based Mobile Computing (iPhone 2G)
2007

Voice on Mobile (Siri)
2011

Voice on Connected / Ambient Devices (Amazon Echo)
2014

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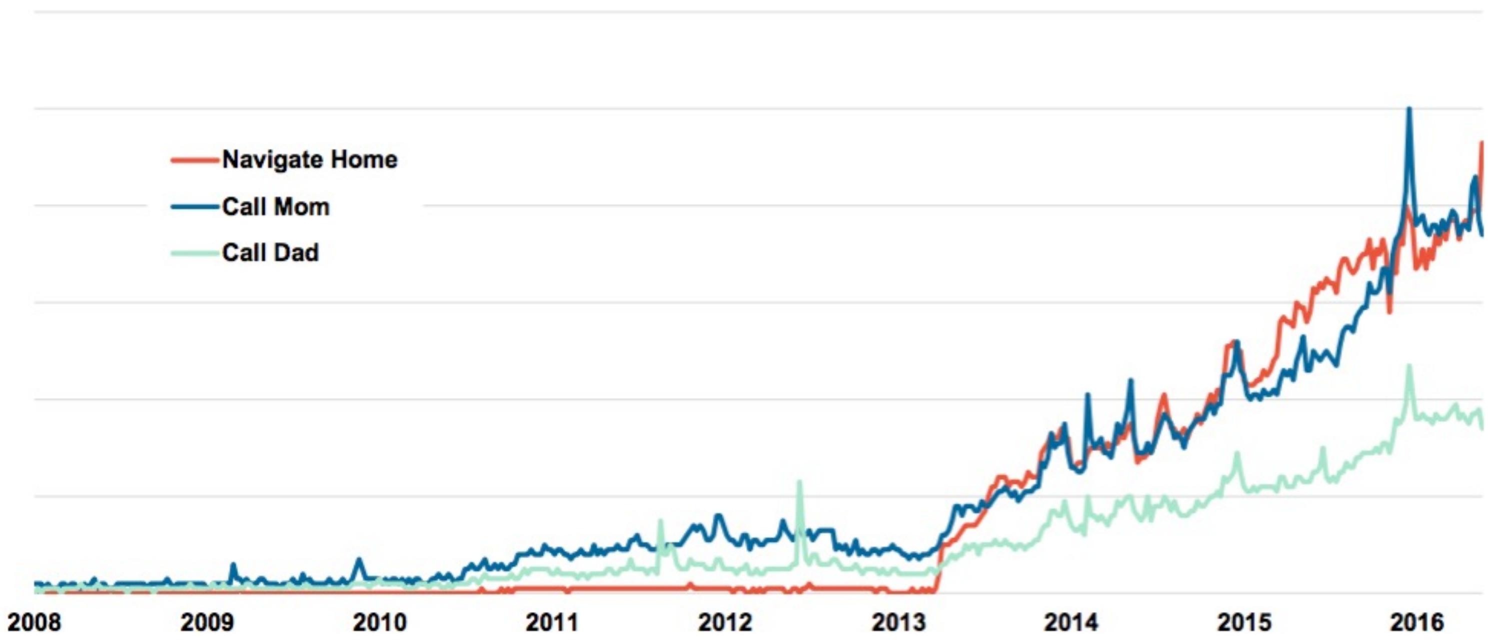
Source: University of Calgary, "History of Computer Interfaces" (Saul Greenberg)

Google Voice Search Queries = Up >35x Since 2008 & >7x Since 2010, per Google Trends

Google Trends imply queries associated with voice-related commands have risen >35x since 2008 after launch of iPhone & Google Voice Search

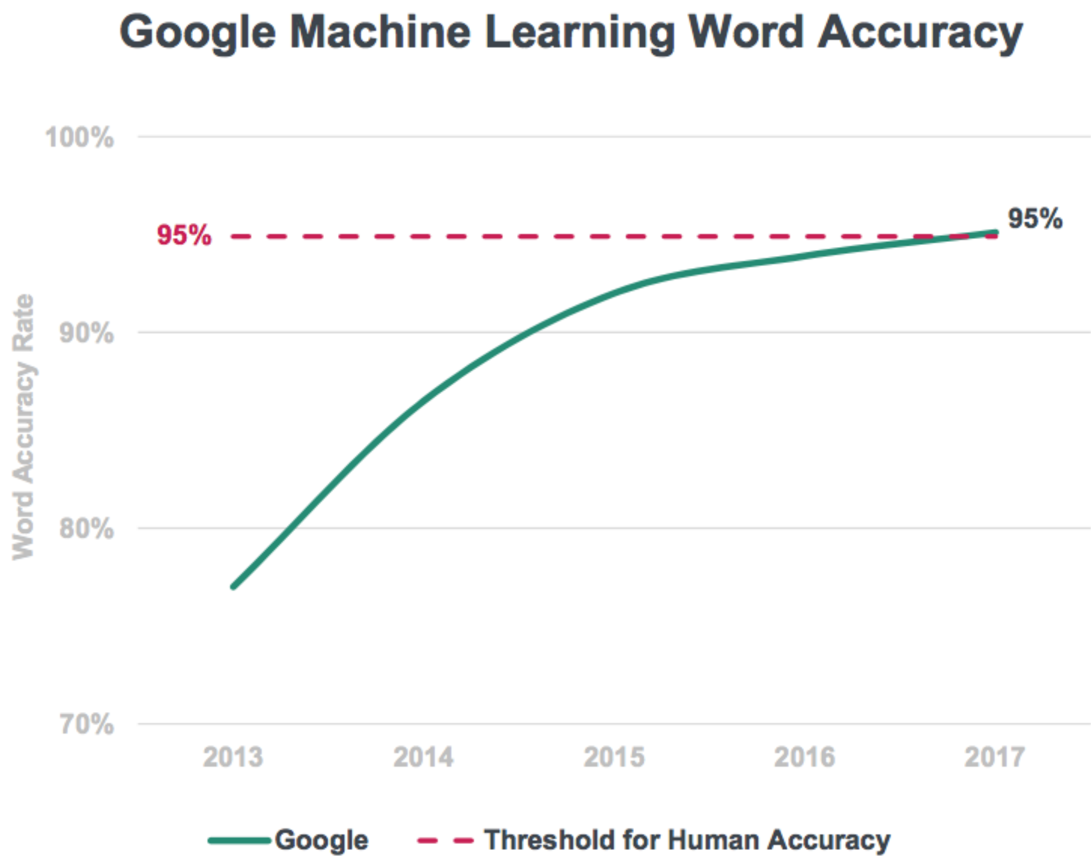
Google Trends, Worldwide, 2008 – 2016



@KPCB

Source: Google Trends
Note: Assume command-based queries are voice searches given lack of relevance for keyword-based search. Aggregate growth values determined using growth in Google Trends for three queries listed above.

Voice = Technology Lift Off…

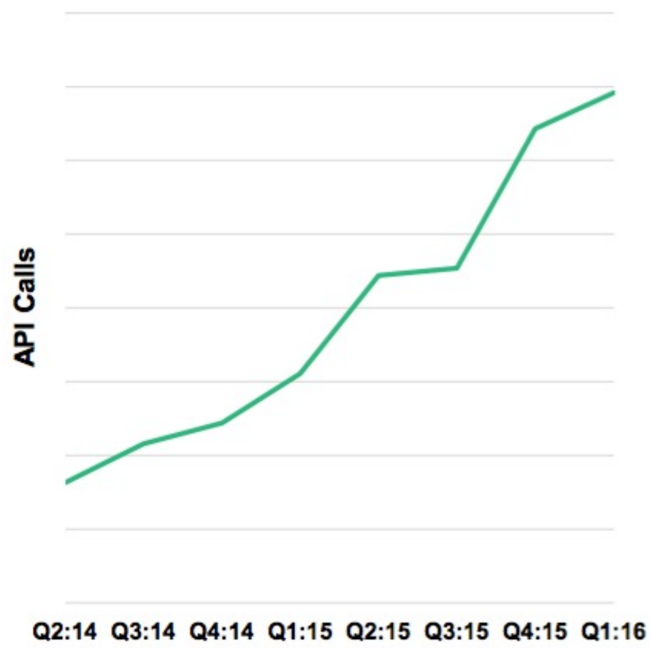


Source: Google (5/17). Note: Data as of 5/17/17 & refers to recognition accuracy for English language. Word error rate is evaluated using real world search data which is extremely diverse & more error prone than typical human dialogue.

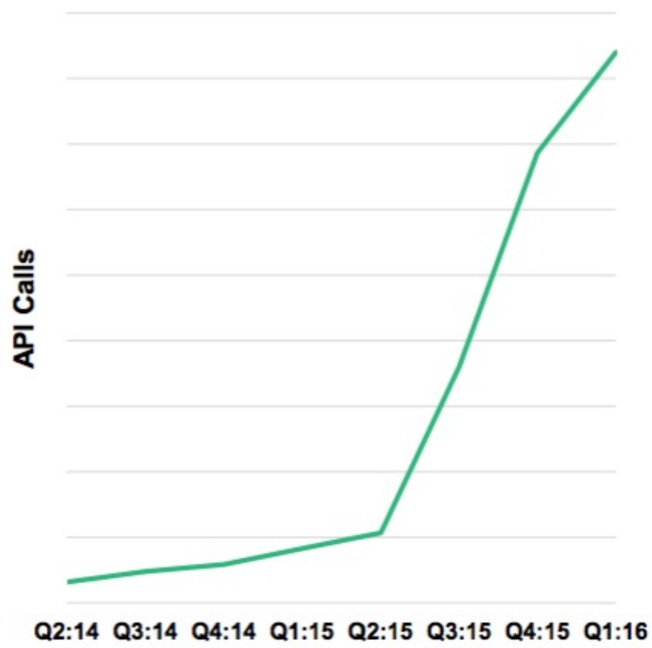
Baidu Voice = Input Growth >4x...Output >26x, Since Q2:14

Usage across all Baidu products growing rapidly...typing Chinese on small cellphone keyboard even more difficult than typing English...Text-to-Speech supplements speech recognition & key component of man-machine communications using voice

Baidu Speech Recognition Daily Usage by API Calls, Global, 2014 – 2016[1]



Baidu Text to Speech (TTS) Daily Usage by API Calls, Global, 2014 – 2016[2]



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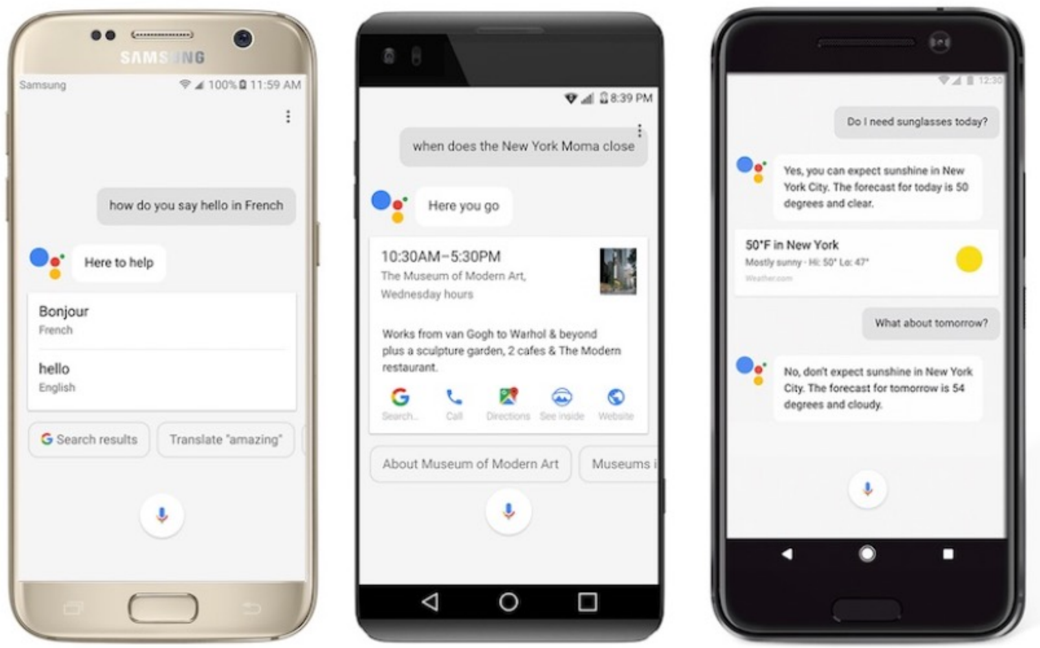
Source: Baidu
Note: (1) Data shown is growth of speech recognition at Baidu, as measured by the number of API calls to Baidu's speech recognition system across time, from multiple products. Most of these API calls were for Mandarin speech recognition. (2) Data shown is growth of TTS (text to speech) at Baidu, in terms of the total number of API calls to Baidu's TTS system across time, from multiple products. Most of these API calls were for Mandarin TTS.

Voice-Based *Mobile* Platform *Front-Ends* =
Voice Can Replace Typing

Google Assistant

Nearly 70% of Requests are Natural / Conversational Language, 5/17

20% of Mobile Queries Made via Voice, 5/16



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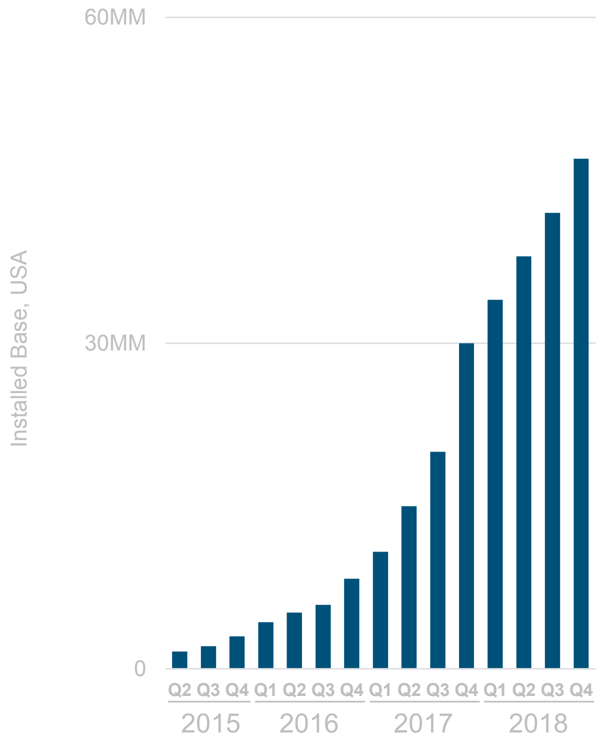
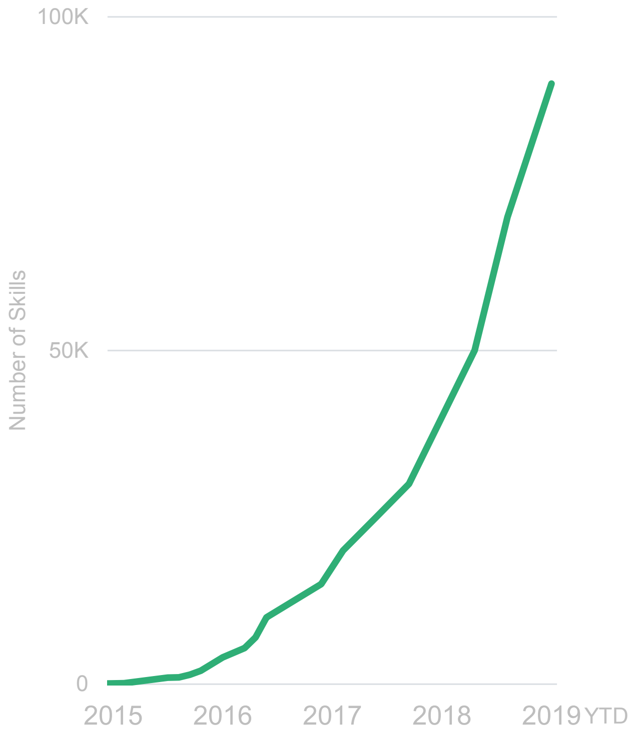
Source: Google I/O (5/16), Image: Macrumors (2/17)

…Voice =
47MM Amazon Echo Base + ~2x in One Year

Amazon Echo Installed Base

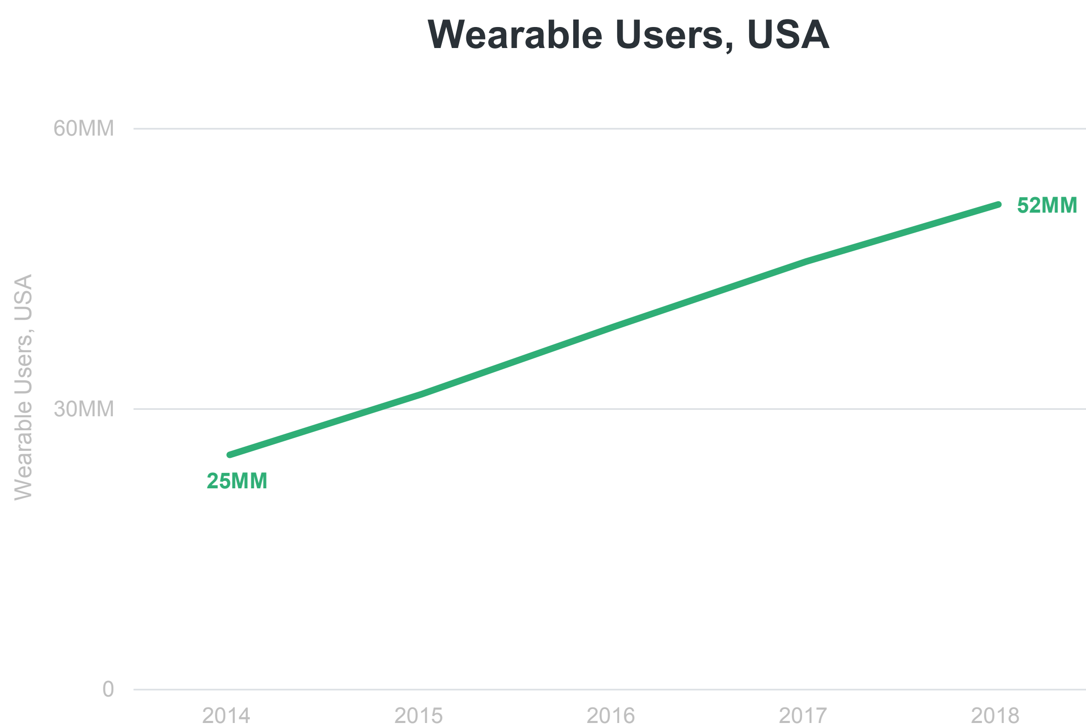


Amazon Echo Skills



Source: Consumer Intelligence Research Partners LLC (Echo install base, 2/18), Various media outlets including Geekwire, TechCrunch, & Wired (Echo skills, 4/19)

Wearables =
52MM Users + ~2x in Four Years



Source: eMarketer (11/18). Users = Monthly Active adults 18+ years old who wear accessories or clothing embedded with electronics, software, or sensors that have the ability to connect to the Internet (via built-in connectivity or tethering), which in turn collects & exchanges data with a manufacturer, operator, or other connected devices.

Is it a Car...Is it a Computer?...

Is it a Phone...Is it a Camera?

Is it a Car...Is it a Computer?

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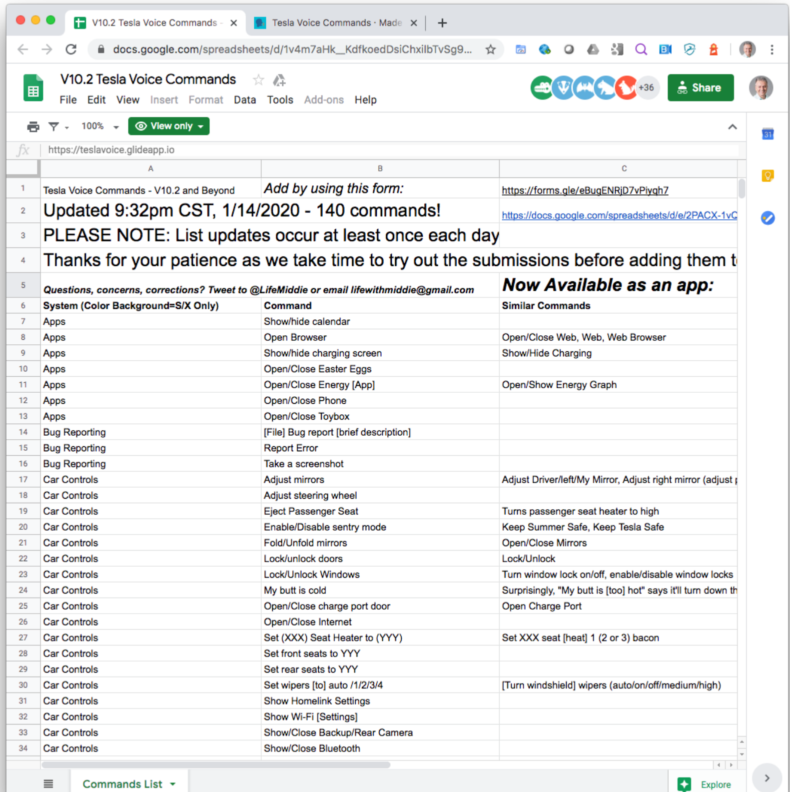
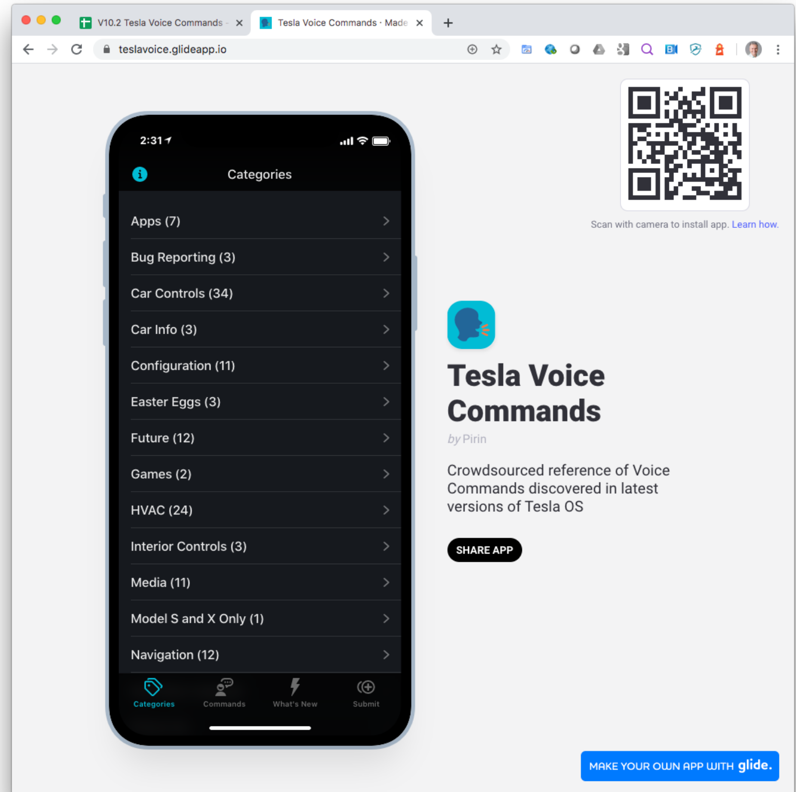
Source: Apple, Tesla

...One Can...
Lock / Monitor / Summon One's Tesla from One's Wrist

Tesla Voice Commands





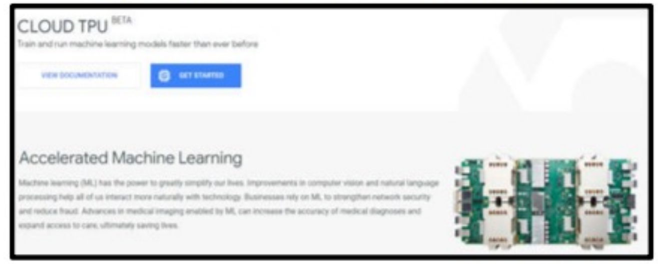
https://teslavoice.glideapp.io/

...Google = AI Platform Emerging from Google Cloud… Enabling Easier Data Processing / Collection for Others

Google Cloud AI Services / Infrastructure

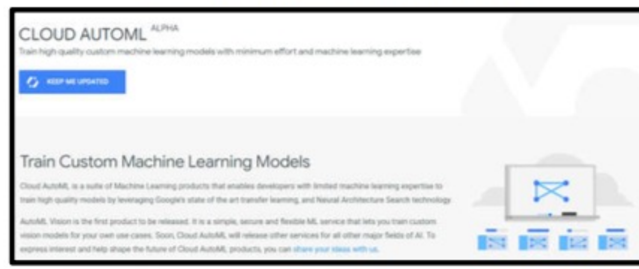
Google Cloud Vision API

AI Hardware – Tensor Processing Units



Dialogflow Conversational Platform

Cloud AutoML – Custom Models

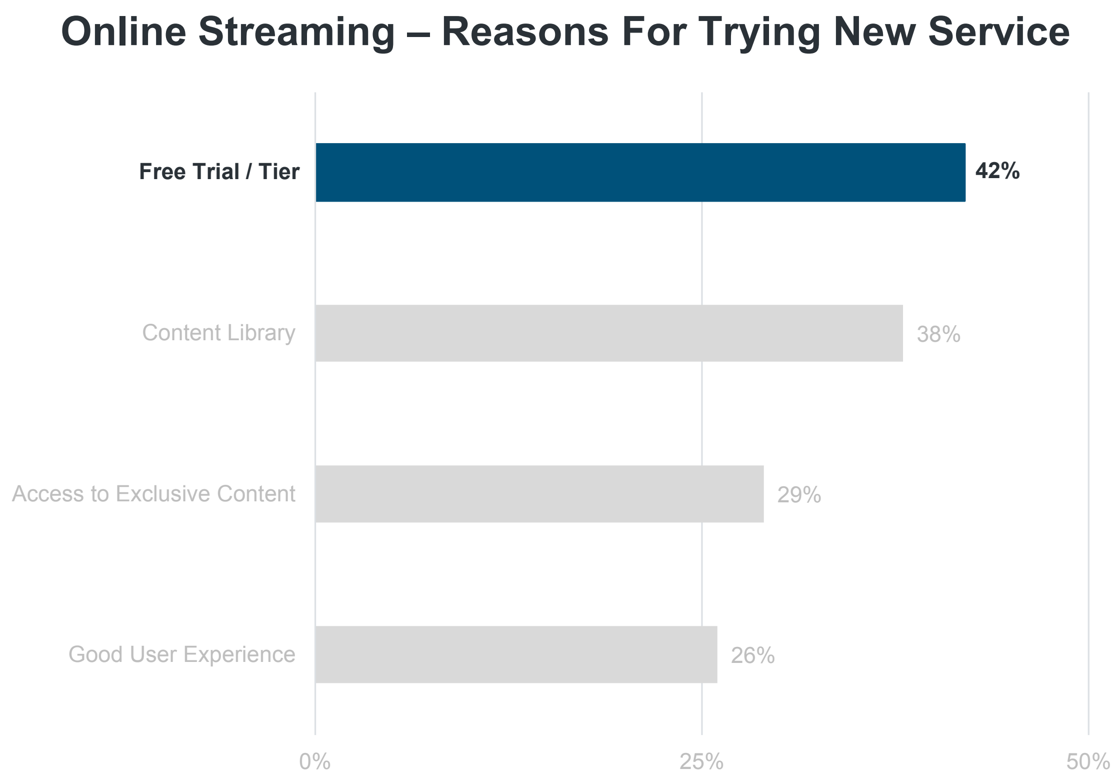


KLEINER PERKINS
2018
INTERNET TRENDS

Source: Google

200

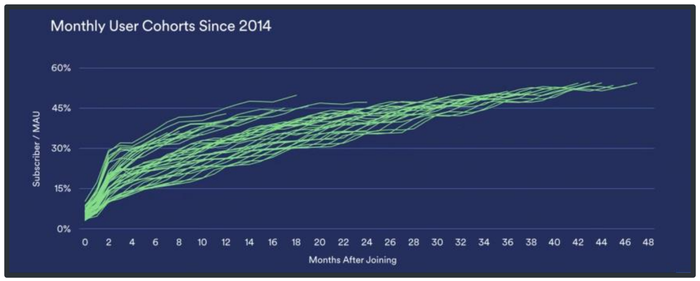
Effective + Efficient Marketing = Can Be Free Trial / Tier



Source: 2BrightCove / YouGov survey of 10,502 adults in USA, UK, France, Australia, Germany, Canada & UAE, 9/18.

Happy Customers…
Spotify = Free User Conversion to Paid Subscribers…

Free Ad-Supported Product…

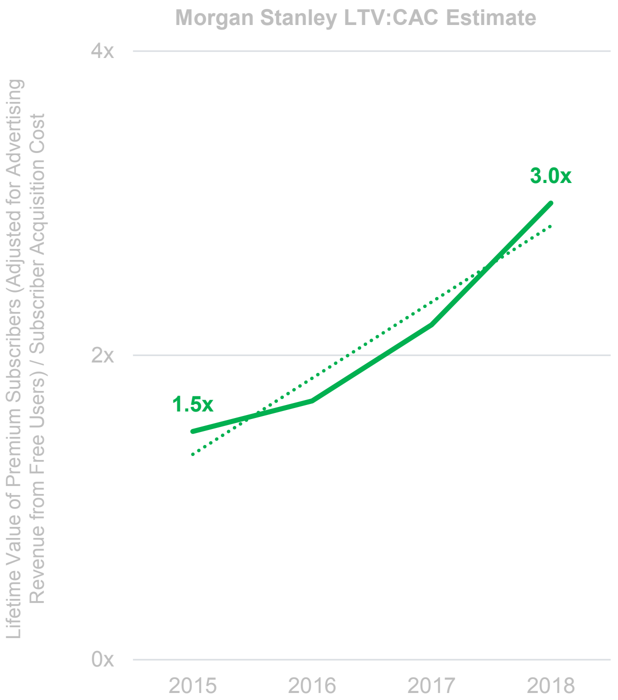


Our freemium model accounts for ~60% of our gross added premium subscribers... the ad-supported service is a subsidy program that offsets costs of new subscriber acquisition.

Developing a better user experience produces by far the most viral effect & impact when investing in growth*. Engagement drives conversion from free consumption to paid subscription.*

Barry McCarthy – CFO, Spotify, 3/18

…Rising LTV / Subscriber Acquisition Cost Ratio



Source: Spotify releases & Morgan Stanley estimates of LTV & SAC. Barry McCarthy @ Spotify Investor Day (3/18). Note: Morgan Stanley estimates LTV by calculating the value of a subscriber (including advertising revenue they generate before they upgrade to a paid user) accounting for gross margin & churn. Subscriber acquisition cost calculated by dividing sales & marketing spend in a relevant period by the number of new subscribers in the same period.

…Happy Customers…
Zoom = Free User Conversion to Paid Subscribers

Free to Join, When Paid User Hosts…

..we really want to get customers to test our product… It's really hard to get customers to try Zoom without a freemium product...

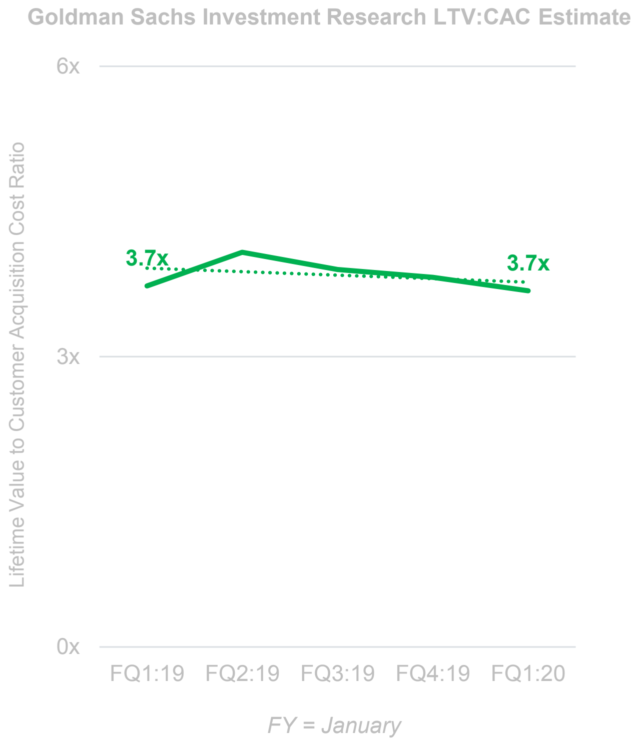
We make our freemium product work so well… If they like our product, very soon they are going to pay for the subscription.

The most important thing is to make sure the existing customer [is] happy rather than chasing after new prospects.

Our NPS is in the 67-69 range vs. our peers in the 20's… We do not want to spend money on [the] marketing side to generate leads.

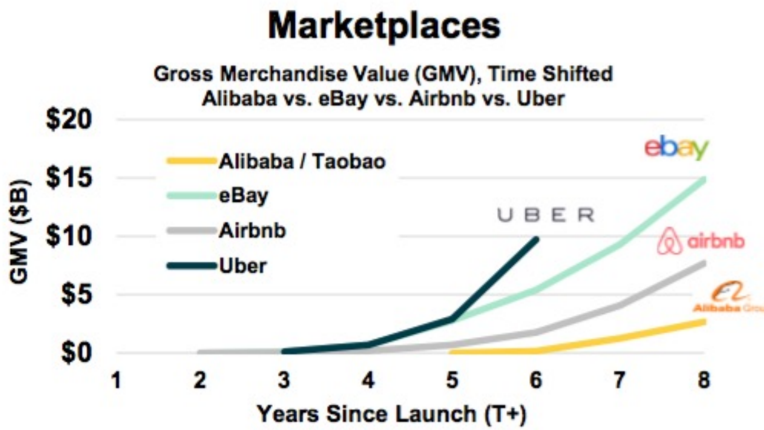
Eric Yuan – Founder / CEO, Zoom, 8/17

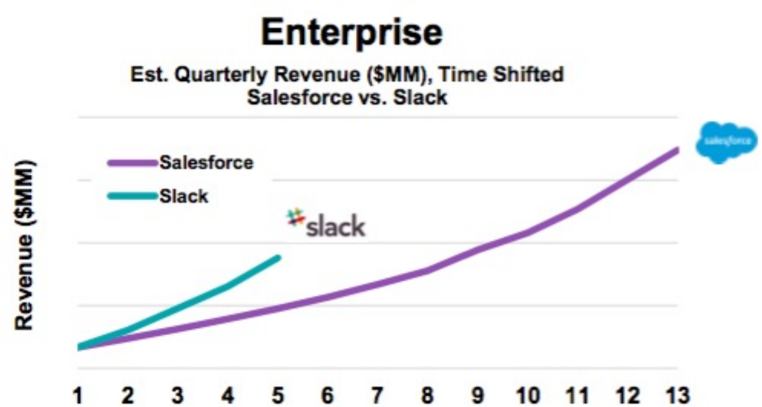
…High LTV / CAC Ratio*



Source: Interview with SaaStr (8/17). Zoom S-1. Goldman Sachs Investment Research estimates of LTV & CAC. Note: *Goldman Sachs Investment Research calculated LTV as the value of $1 of Annual Recurring Revenue, discounted for GAAP gross margin, Goldman Sachs Investment Research estimated Gross Churn & an 11% assumed discount rate. CAC is calculated as the amount of Sales & Marketing expense in a relevant period, divided by the gross new Annual Recurring Revenue added in the same period.

Current Generation of Internet Leaders = Growing Faster than Previous Generation





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Marketplaces Source: Company data, Morgan Stanley Research. eBay founded in 1995. Amazon founded in 1995. Alibaba.com founded in 1999 as B2B portal connecting Chinese manufacturers and overseas buyers. Uber launched 2009, gave first ride in 2010. Airbnb founded in 2008..
Commerce Source: Publicly available company data, Morgan Stanley Research. JD.com launched B2C shipments in 2004, founded 1998 as an online magneto-optical store. Amazon founded in 1995.
Enterprise Source: Slack. Graph starting point based on similar est. revenue figures. Salesforce quarterly revenue approximated from publicly disclosed annual GAAP revenues.

Global *Internet* Market Capitalization Leaders = USA Stable @ 18 of 30…China Stable @ 7 of 30

Rank 2019	Company	Region	Market Cap Value ($B) 6/7/19	6/7/16	% Change
1	Microsoft	USA	$1,007B	$410B	+146%
2	Amazon	USA	888	343	+159%
3	Apple	USA	875	540	+62%
4	Alphabet	USA	741	497	+49%
5	Facebook	USA	495	340	+46%
6	Alibaba	China	402	195	+106%
7	Tencent	China	398	206	+93%
8	Netflix	USA	158	43	+266%
9	Adobe	USA	136	50	+174%
10	PayPal	USA	134	46	+190%
11	Salesforce	USA	125	56	+123%
12	Booking.com	USA	77	67	+15%
13	Uber	USA	75	--	--
14	Recruit Holdings	Japan	52	20	+167%
15	ServiceNow	USA	51	12	+316%
16	Workday	USA	48	16	+197%
17	Meituan Dianping	China	44	--	--
18	JD.com	China	39	32	+22%
19	Baidu	China	38	60	(36%)
20	Activision Blizzard	USA	35	28	+25%
21	Shopify	Canada	34	2	+1,297%
22	NetEase	China	33	23	+44%
23	eBay	USA	33	28	+19%
24	Atlassian	Australia	32	5	+509%
25	MercadoLibre	Argentina	30	6	+388%
26	Twitter	USA	29	11	+173%
27	Square	USA	29	3	+808%
28	Electronic Arts	USA	29	23	+25%
29	Xiaomi	China	28	--	--
30	Spotify	Sweden	25	--	--
	Total		**$6,119**	**$3,064**	

BOND Internet Trends 2019

USA stable at 18 of 30 refers to the fact that 19 USA companies were in the top 30 using same parameters in 2018. China in 2018 = 8 of 30. Source: CapitalIQ data collected 6/7/19. Note: % change only applies to companies that were publicly traded on 6/7/16, private valuations not taken into account. Reflects a global list of publicly traded companies, some companies (e.g., Meituan Dianping) derive the majority of their revenue from one country. Companies selected are considered to be 'pureplay' Internet companies that do not derive a significant portion of revenues / profits from other business lines. Example companies that would be included otherwise = Reliance Industries (Refining), AT&T (Telecom), & Naspers (Television).

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Global Market Capitalization Leaders = USA Stable @ 23 of 30…Technology Stable @ 9 of 30

Rank 2019	Company	Sector	Region	Market Cap Value ($B) 6/7/19	6/7/16	% Change
1	Microsoft	Technology	USA	$1,007B	$410B	+146%
2	Amazon	Technology	USA	888	343	+159%
3	Apple	Technology	USA	875	540	+62%
4	Alphabet	Technology	USA	741	497	+49%
5	Berkshire Hathaway	Financial Services	USA	505	350	+44%
6	Facebook	Technology	USA	495	340	+46%
7	Alibaba	Technology	China	402	195	+106%
8	Tencent	Technology	China	398	206	+93%
9	Visa	Financial Services	USA	372	192	+94%
10	Johnson & Johnson	Healthcare	USA	368	318	+16%
11	JPMorgan	Financial Services	USA	354	239	+48%
12	Exxon Mobil	Energy	USA	316	371	(15%)
13	Nestlé	Food / Beverages	Switzerland	306	230	+33%
14	Walmart	Retail	USA	303	221	+37%
15	ICBC	Financial Services	China	285	224	+27%
16	Procter & Gamble	Home Goods	USA	273	220	+24%
17	Mastercard	Financial Services	USA	271	106	+156%
18	Bank of America	Financial Services	USA	262	149	+76%
19	Royal Dutch Shell	Energy	Netherlands	259	198	+31%
20	Samsung	Technology	South Korea	249	166	+50%
21	Disney	Media	USA	248	160	+55%
22	Cisco	Technology	USA	239	146	+64%
23	Pfizer	Pharmaceuticals	USA	238	212	+12%
24	AT&T	Telecom	USA	237	242	(2%)
25	Verizon	Telecom	USA	237	207	+15%
26	UnitedHealth	Healthcare	USA	235	131	+79%
27	Roche	Healthcare	Switzerland	233	224	+4%
28	Chevron	Energy	USA	231	191	+21%
29	Coca-Cola	Food / Beverages	USA	220	196	+12%
30	Home Depot	Retail	USA	217	161	+35%
	Total			**$11,264**	**$7,385**	

USA stable at 23 of 30 refers to the fact that 21 USA companies were in the top 30 using same parameters in 2018. Technology in 2018 = 10 of 30.
Source: CapitalIQ data collected 6/7/19. Reflects a global list of publicly traded companies, some companies (e.g., ICBC) derive the majority of their revenue from one country.

USA = 60% of Most Highly Valued *Tech* Companies Founded By… 1st or 2nd Generation Americans…1.9MM Employees, 2018

Immigrant Founders / Co-Founders of Top 25 USA Valued Public *Tech* Companies, Ranked by Market Capitalization

Rank	Company	Mkt Cap ($B)	LTM Rev ($B)	Employees (K)	Founder / Co-Founder (1st or 2nd Gen Immigrant)	Generation
1	Microsoft	$1,007B	$122B	131K	--	--
2	Amazon	888	242	648	Jeff Bezos	2nd, Cuba
3	Apple	875	258	132	Steve Jobs	2nd, Syria
4	Alphabet / Google	741	142	99	Sergey Brin	1st, Russia
5	Facebook	495	59	36	Eduardo Saverin	1st, Brazil
6	Cisco	239	51	74	--	--
7	Intel	206	71	107	--*	--
8	Oracle	182	40	137	Larry Ellison / Bob Miner	2nd, Russia / 2nd, Iran
9	Netflix	158	17	7	--	--
10	Adobe	136	10	21	--	--
11	PayPal	134	16	22	Max Levchin / Luke Nosek / Peter Thiel / Elon Musk***	1st, Ukraine / 1st, Poland / 1st, Germany / 1st, South Africa
12	Salesforce	125	14	35	--	--
13	IBM	118	79	351	Herman Hollerith	2nd, Germany
14	Texas Instruments	104	16	30	Cecil Green / J. Erik Jonsson	1st, UK / 2nd, Sweden
15	NVIDIA	89	11	13	Jensen Huang	1st, Taiwan
16	Qualcomm	84	21	35	Andrew Viterbi	1st, Italy
17	Booking.com	77	14	25	--	--
18	Uber	75	12	22	Garrett Camp	1st, Canada
19	Automatic Data Processing	73	14	57	Henry Taub	2nd, Poland
20	VMware	69	9	23	Edouard Bugnion	1st, Switzerland
21	Intuit	67	7	9	--	--
22	ServiceNow	51	3	8	--	--
23	Workday	48	3	11	Aneel Bhusri	2nd, India
24	Micron	38	30	36	--	--
25	Cognizant	36	16	282	Francisco D'Souza / Kumar Mahadeva	1st, India** / 1st, Sri Lanka

Source: CapIQ as of 6/7/19. *While Andy Grove (from Hungary) is not a co-founder of Intel, he joined as COO on the day it was incorporated. **Francisco D'Souza is a person of Indian origin born in Kenya. ***Max Levchin / Luke Nosek / Peter Thiel's startup Confinity merged with Elon Musk's startup X.com to form PayPal in 3/00.

USA = Many Highly Valued Private Tech Companies Founded By… 1st Generation Immigrants

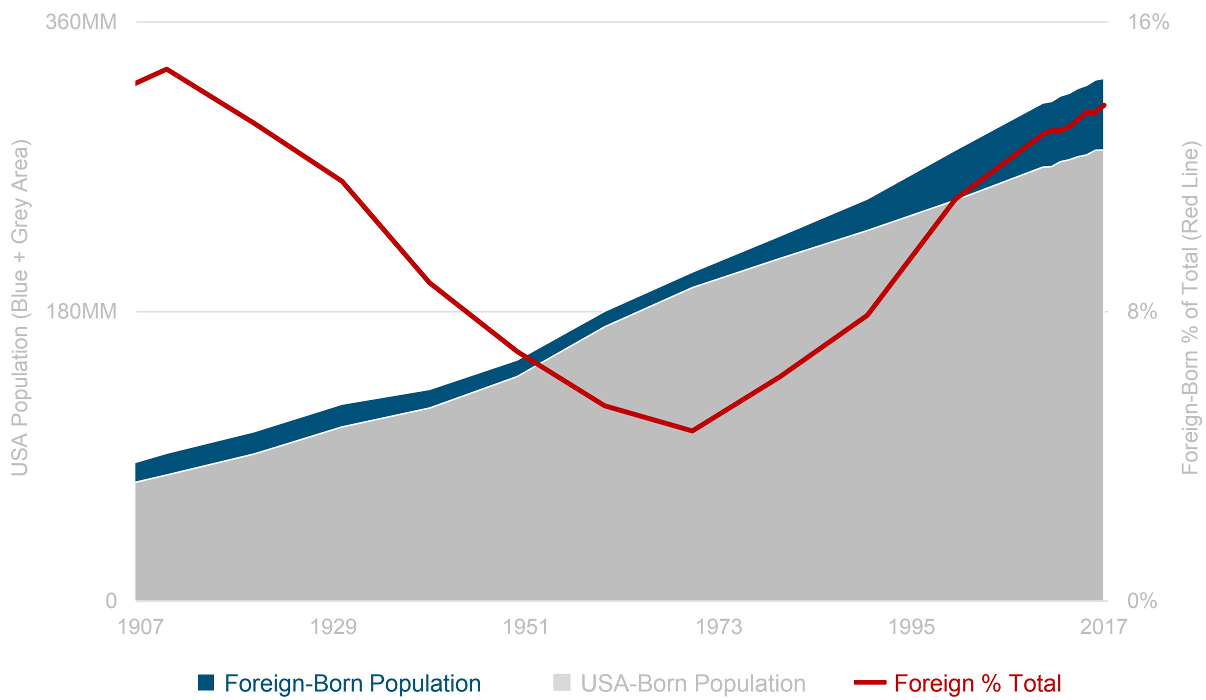
Company	Founder / Co-Founder	Country of Origin	Valuation ($B)
WeWork	Adam Neumann	Israel	$47B
SpaceX	Elon Musk	South Africa	31
Stripe	John Collison Patrick Collison	Ireland	23
Palantir	Peter Thiel	Germany	21
Epic Games	Mark Rein	Canada	15
DoorDash	Tony Xu	China	13
Wish	Peter Szulczewski Danny Zhang	Canada	9
Instacart	Apoorva Mehta	India	8
Slack	Stewart Butterfield Serguei Mourachov Cal Henderson	Canada Russia UK	7
UiPath*	Daniel Dines Marius Tirca	Romania	7
Tanium	David Hindawi	Iraq	7
Unity Technologies	David Helgason Nicholas Francis Joachim Ante	Iceland Denmark Germany	6
Robinhood	Baiju Bhatt Vlad Tenev	India Bulgaria	6
Compass	Ori Allon	Israel	4
Credit Karma	Kenneth Lin	China	4
Houzz	Adi Tatarko Alon Cohen	Israel	4
Snowflake	Marcin Zukowski Benoit Dageville Thierry Cruanes	Netherlands France France	4
Rubrik	Bipul Sinha Arvind Nithrakashyap Arvind Jain Soham Mazumdar	India India India India	3
Zoox	Tim Kentley-Klay	Australia	3
Oscar Health	Mario Schlosser	Germany	3

Company	Founder / Co-Founder	Country of Origin	Valuation ($B)
Crowdstrike	Dmitri Alperovitch	Russia	$3B
Affirm	Max Levchin	Ukraine	3
Databricks	Ali Ghodsi Matei Zaharia Ion Stoica	Sweden Romania Romania	3
Nuro	Jiajun Zhu Dave Ferguson	China New Zealand	3
Automation Anywhere	Ankur Kothari Mihir Shukla Neeti Metha Shukia Rushabh Parm	India India India India	3
Confluent	Jun Rao Neha Narkhede	China India	3
Roblox	David Baszucki	Canada	2
Medallia	Borge Hald	Norway	2
Lime	Toby Sun Brad Bao	China	2
Zume Pizza	Alex Garden	Canada	2
Gusto	Tomer London	Israel	2
Lemonade	Shai Wininger Daniel Schreiber	Israel	2
LegalZoom	Brian Lee	South Korea	2
Avant	Al Goldstein John Sun Paul Zhang	Uzbekistan China China	2
Apttus	Krik Krappe	UK	2
Postmates	Bastian Lehmann	Germany	2
Sprinklr	Ragy Thomas	India	2
Cloudflare	Michelle Zatlyn	Canada	2
Carta	Manu Kumar	India	2
ZocDoc	Oliver Kharraz	Germany	2
Warby Parker	Dave Gilboa	Sweden	2
Carbon3D	Alex Ermoshkin	Russia	2
Pony.ai	James Peng Tiancheng Lo	China	2
ServiceTitan	Ara Mahdessian Vahe Kuzoyan	Iran Armenia	2
Segment	Ilya Volodarsky	Russia	2
Quanergy	Tianyue Yu	China	2

Source for Valuation & Founders Backgrounds: Based on analysis by Pitchbook, Wall Street Journal, CB Insights, Forbes, Business Insider, & the National Foundation for American Policy. *UiPath is headquartered in New York, NY but was originally founded in Romania.

USA = 14% of Population Foreign-Born & Rising… Near All-Time High (1910) @ 15%

USA Population vs. Foreign-Born % of Total Population

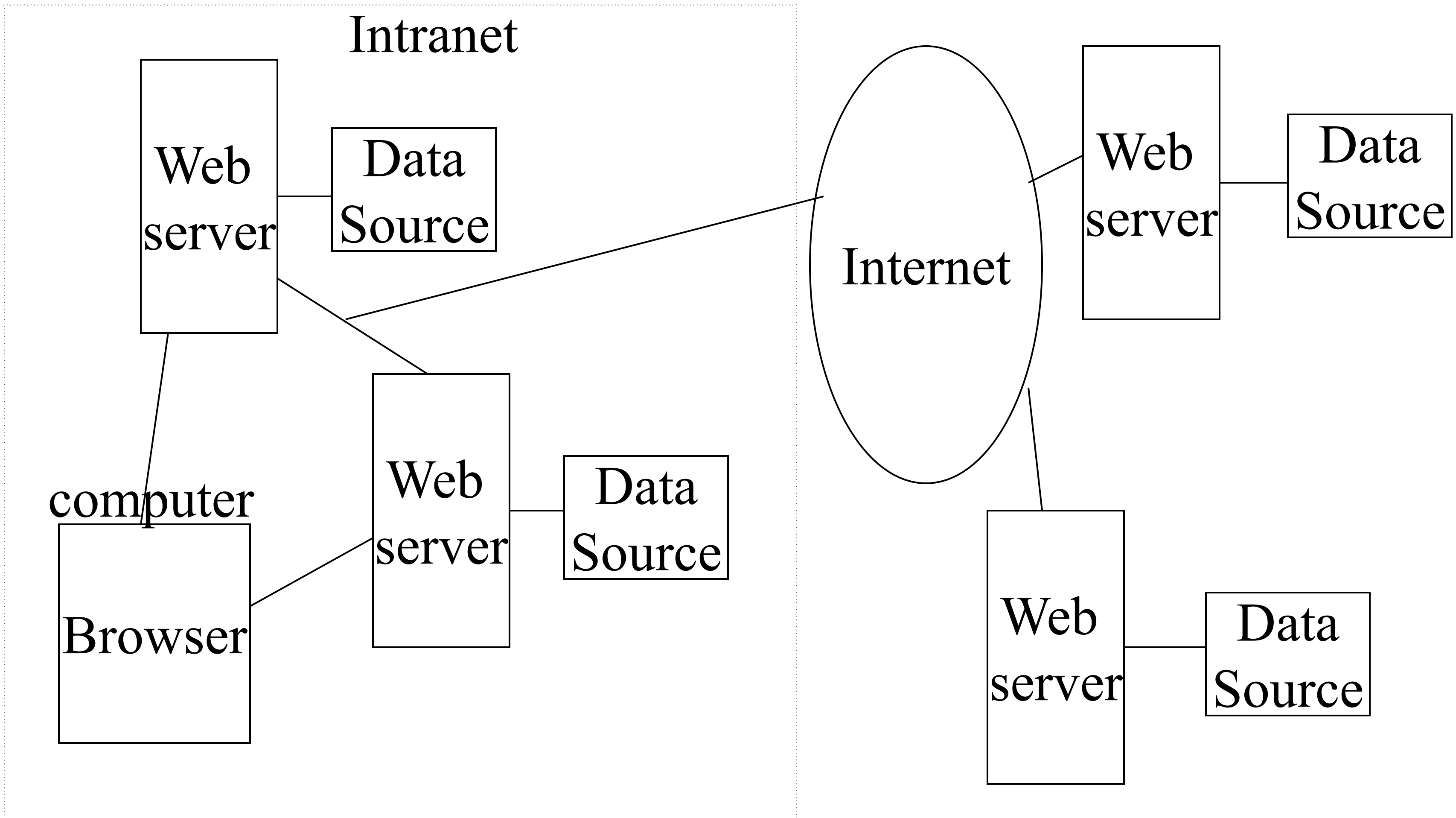


Source: Migration Policy Institute (MPI) tabulation of data from U.S. Census Bureau, 2010-2017 American Community Surveys (ACS) & 1970, 1990 & 2000 Decennial Census. All other data are from Campbell J. Gibson & Emily Lennon, "Historical Census Statistics on the Foreign-Born Population of the United States: 1850 to 1990" (Working Paper no. 29., U.S. Census Bureau, Washington, DC, 1999). Note: The term "immigrants" (also known as the foreign born) refers to people residing in the United States who were not U.S. citizens at birth. This population includes naturalized citizens, lawful permanent residents (LPRs), certain legal nonimmigrants (e.g., persons on student or work visas), those admitted under refugee or asylee status & persons illegally residing in the United States. Population growth figures from St. Louis FRED database.

The World Wide Web (WWW)

- **Material relevant to exams starts HERE with this slide.**

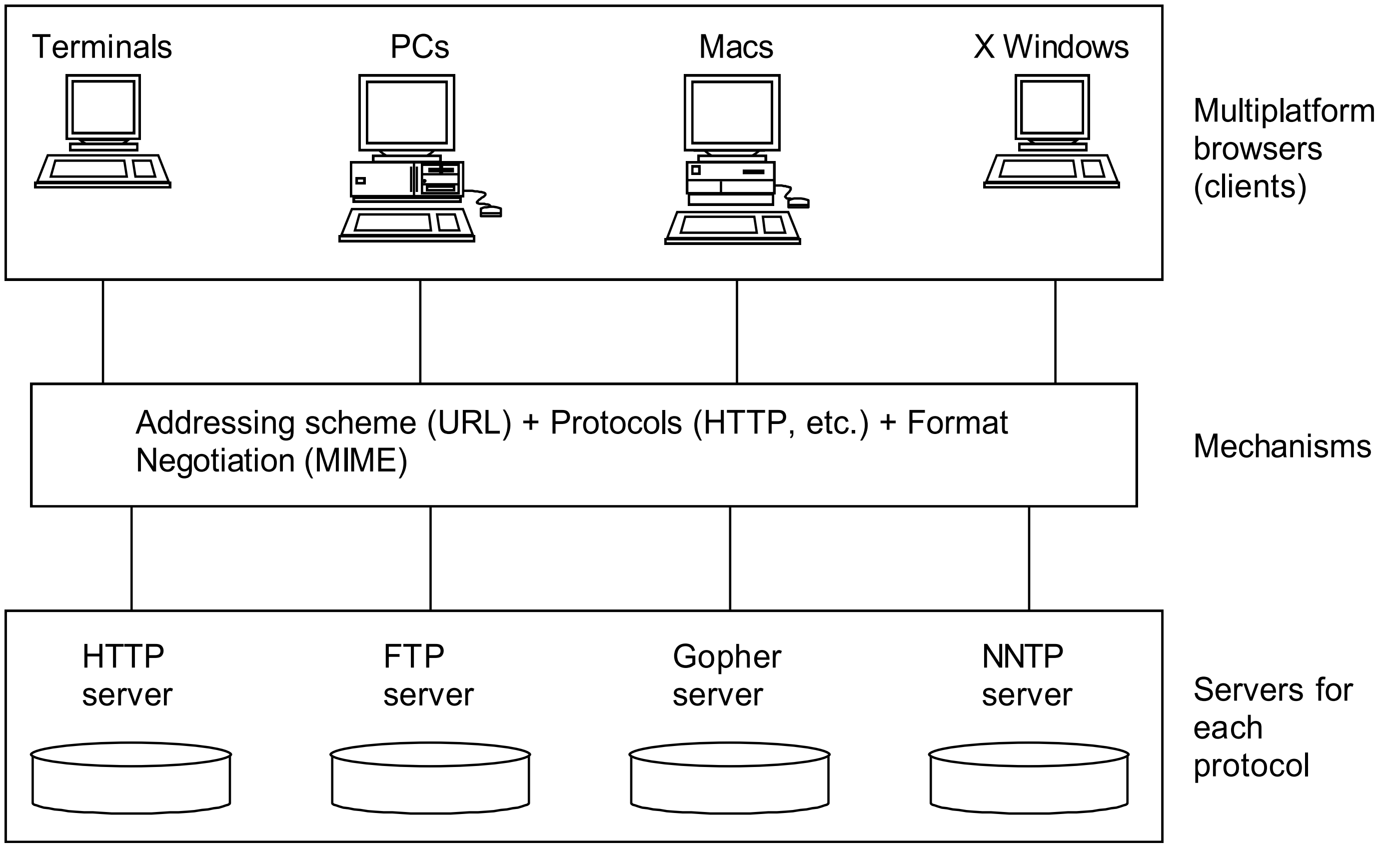
Graphical View of the WWW



Major Technology Components

- **Client/server architecture**
  - where client programs interact with web servers
- **Network protocol**
  - HTTP, Hypertext Transfer Protocol, is the language understood by browsers and web servers
  - designed to move quickly from document to document
- **Addressing system** (Uniform Resource Locators)
  - http://domain/directory/file.html
- **Markup Language**
  - every web server understands and every browser displays
  - includes support for HyperText and multimedia

Client/Server Architecture Model



The WWW Server

- Web browsers and Web servers communicate according to a protocol known as HTTP (HyperText Transfer Protocol)
  - The current HTTP protocol is version 2.0
- The Web server is a software system running on a machine often called the Web server, don't confuse them
- A web server can
  - receive and reply to HTTP requests
  - retrieve documents from specified directories
  - run programs in specified directories
  - handle limited forms of security
- A web server does not
  - know about the contents of a document, links in a document, images in a document or whether a particular file, e.g. a *.gif file, is in the correct format

Uniform Resource Locator (URL)

- A mechanism whereby an Internet resource can be specified in a single line of ASCII text
- See **RFC 1738:** http://www.faqs.org/rfcs/rfc1738.html

URL	**Refers to:**
file://pub/xt.ps	a PostScript file in directory pub on your local machine
ftp://usc.edu/docs/sweng.txt	file sweng.txt in directory docs on usc.edu, an anonymous ftp site
http://nunki.usc.edu/mydocs/book.doc	a file in directory mydocs on machine nunki.usc.edu, a WWW site
news:comp.compilers	the newsgroup computers.compilers
mailto:horowitz@usc.edu	an e-mail address

General Description of a URL

1. **Scheme** followed by a colon http:,ftp:, news:,mailto:,wais:,telnet:
2. **Double slash** (optional. Required for http, ftp) //
3. Internet **domain** name e.g., www.usc.edu
4. **Port** number (optional; e.g., www.usc.edu:8081)

   Standard or default port numbers:

   --- ftp is 21 gopher is 70

   --- telnet is 23 http is 80

   --- smtp is 25 nntp is 119

   --- imap is 143 secure nntp is 563

   --- pop3 is 110 secure pop3 is 995
5. **Path** e.g., /pub/docs

URL Character Set

- RFC 1738, Dec. 1994 defines the URL character set as
"...Only alphanumerics [0-9a-zA-Z], the special characters "$-_.+!*'()," **[not including the quotes]**, and reserved characters used for their reserved purposes may be used unencoded within a URL."
- However, HTML supports ISO-8859-1 (ISO-Latin) character set
  - HTML 4.x extends the character set to all of Unicode
- Therefore, in URLs an escape mechanism is used, % followed by two hex digits
- Characters that should be encoded include:
  %, /, ., .., #, ?, ;, :, $, +, @, &, =
- Here are some encoded values for so-called "unsafe" characters

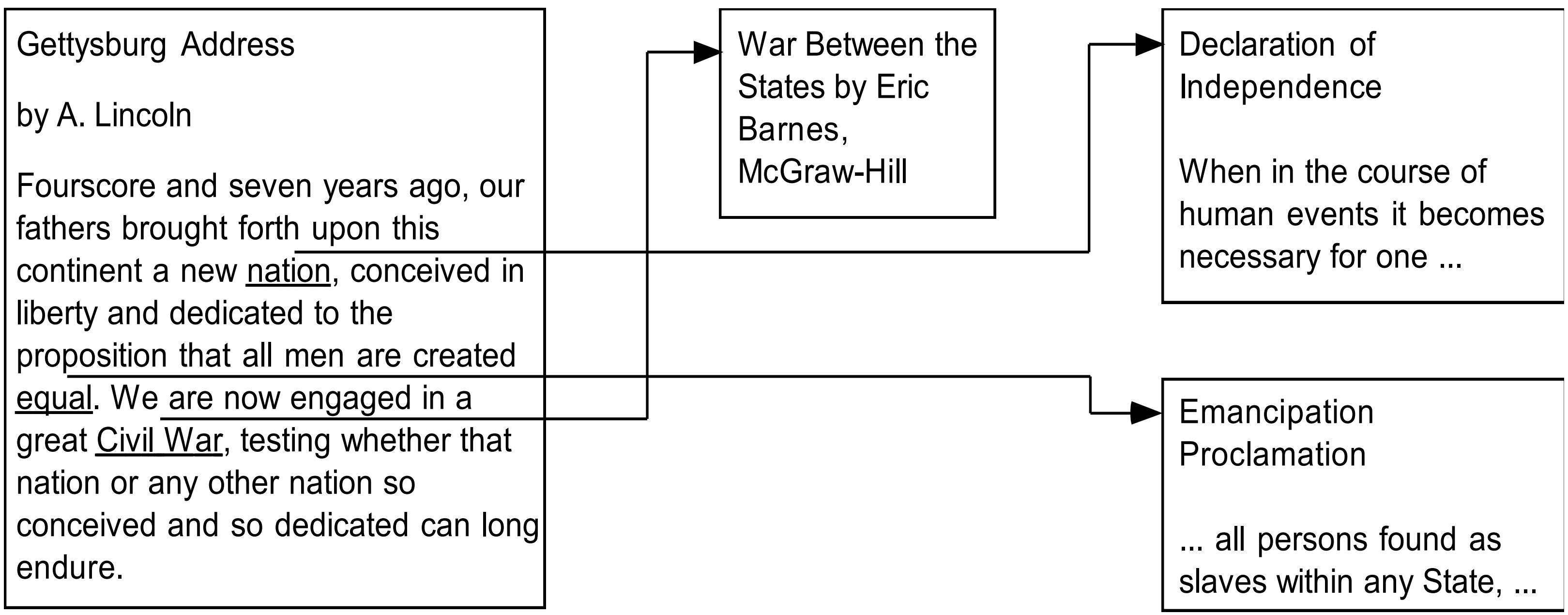
~	%7E	|	%7C
SPACE	%20	\	%5C
%	%25	^	%5E
&	%26	[	%5B
=	%3D	]	%5D
?	%3F	#	%23
{	%7B	>	%3E
}	%7D	<	%3C

Markup Languages

- HTML - hypertext markup language, specifies document layout and the specification of hypertext links to text, graphics and other types of objects
- Browsers display text and graphics using the markup as guidance
- However, HTML is *not* like a word processing program, e.g., Microsoft Word or WordPerfect, and *not* like a page description languages, e.g., postscript
  - as a result, translation into HTML can produce a result that does not look exactly like the original

What is HyperText?

- Regular text, with the additional feature of links to related documents
- As you read documents and follow links, you traverse a “web” of interconnections

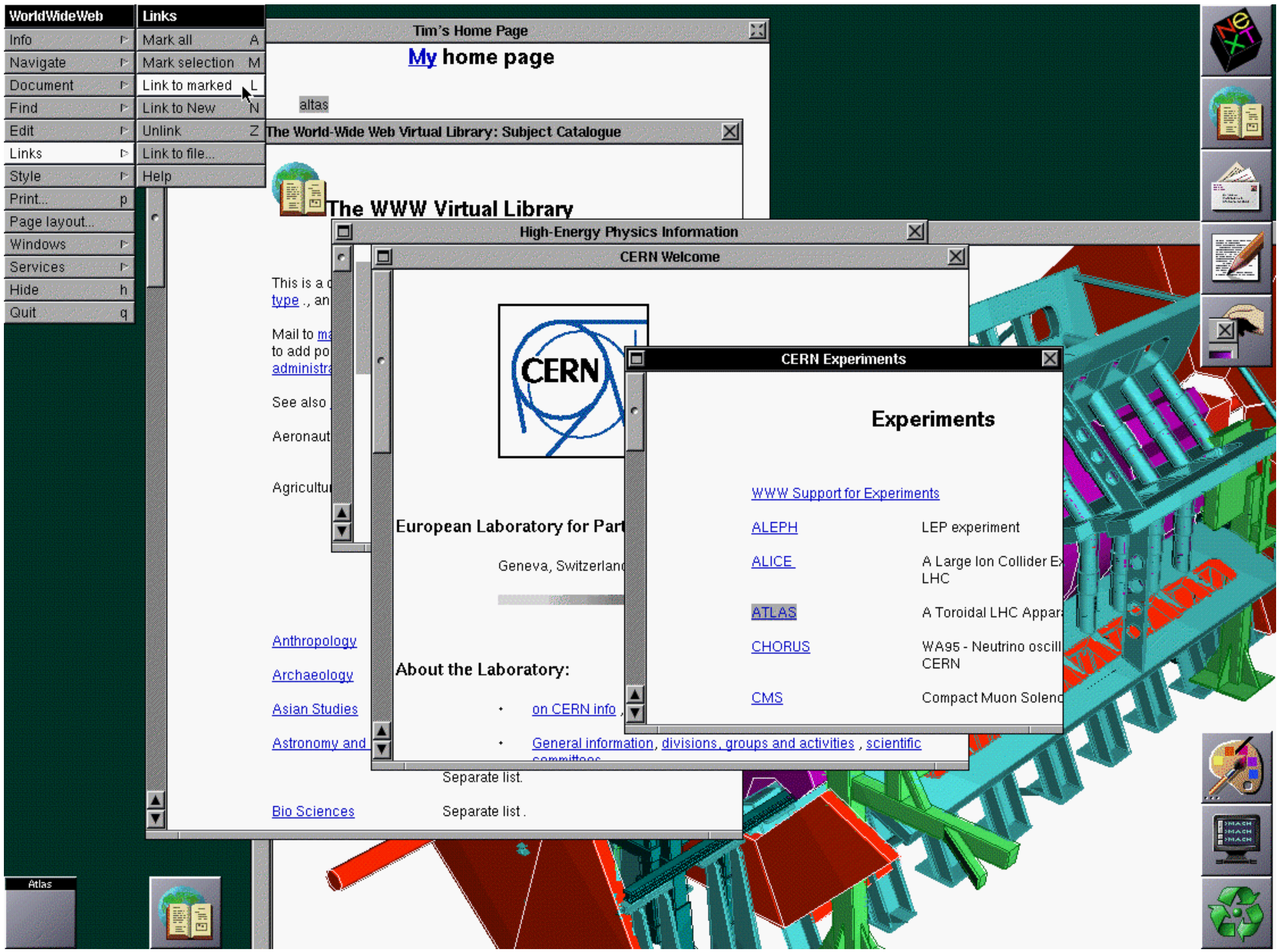


Early History of the WWW

- 1989-1990 Tim Berners-Lee conceives the WWW at CERN in Geneva
- 1990 Berners-Lee releases WWW prototype on NeXt computer
- 1992 Release of source code for line mode browser, lynx and HTTP
- 1993 Mosaic browser from NCSA is released
- 1993 WWW internet traffic now measures 1% of NSF backbone
- 12/94 Netscape Navigator 1.0 is released
  World Wide Web Consortium formed
- 1995 Microsoft Windows 95 and Internet Explorer 1.0 released
- 1995 Java is released
- 1998 Google is started
- 1999-2001 A burst of Internet start-up companies which flamed out because they were not profitable. Also known as he “Internet Bubble.”
- 2004 Firefox 1.0 is released
- 2005 YouTube is founded
- 2008 Google Chrome 1.0 is released

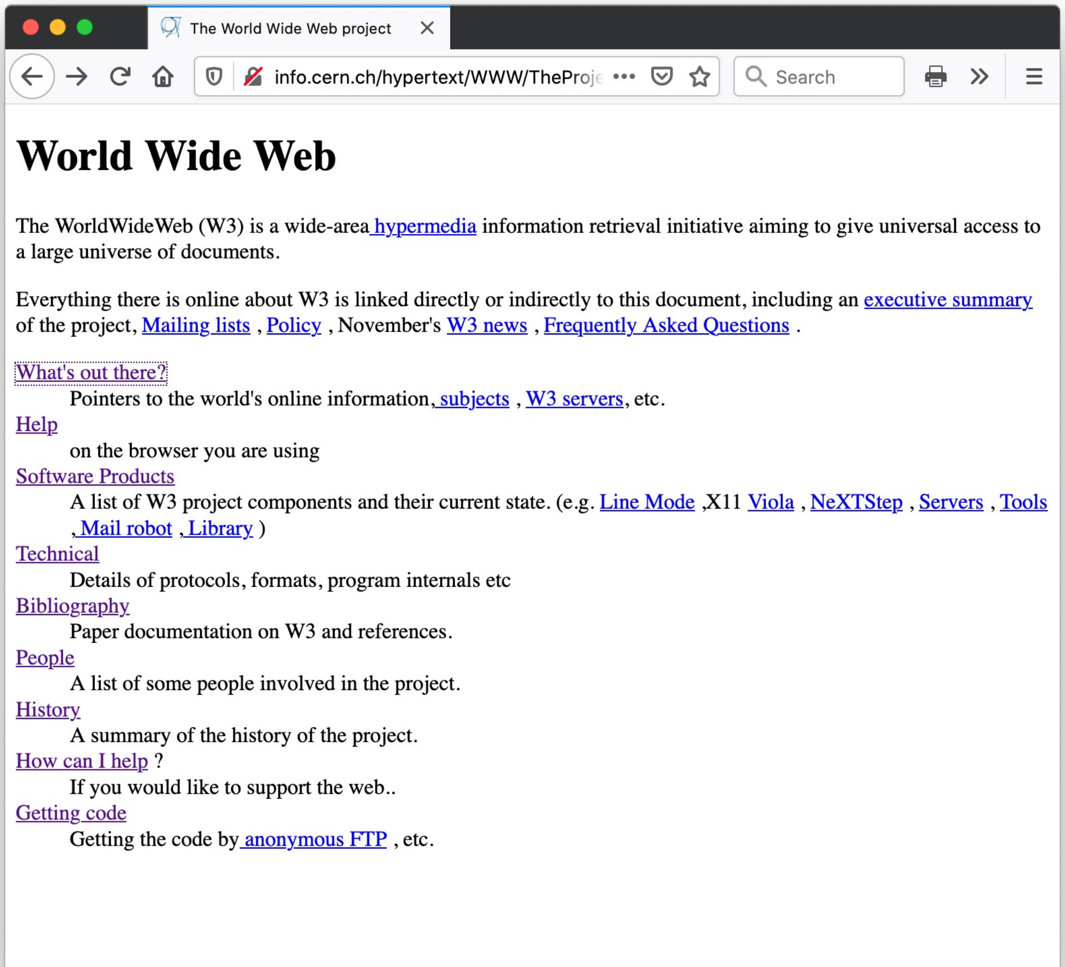
First Web Communication (Dec 1990)

See http://www.w3.org/History.html and tim Berners-Lee's presentation at the 10th anniversary, http://www.w3.org/2004/Talks/w3c10-HowItAllStarted/?n=1



Original WWW "The Project" site at CERN

http://info.cern.ch/hypertext/WWW/TheProject.html



London Olympics (July 2012)

See http://www.zdnet.com/article/web-inventor-tim-berners-lee-stars-in-olympics-opening-ceremony/

https://www.youtube.com/watch?v=KW6ivwDcOY4

Sir Tim Berners-Lee live-tweets during the 2012 Olympics opening ceremony, with a NeXT Cube by his side

WWW Consortium

- Founded in 1994, headed by Tim Berners-Lee, http://www.w3.org
- Goal: "to lead the World Wide Web to its full potential by developing common protocols that promote its evolution and ensure its interoperability."
- Many of the technologies guided by the WWW consortium will be discussed this semester:
  - HTML, Style Sheets, Document Object Model, international character sets, HTTP, XML, etc.